Lesson 1 — Extra Practice

The Cartesian Plane and Plotting Points

Part A — Quadrants, Axes, and Sign Patterns

1. Without plotting, state the quadrant each point lies in. **Careful:** some of these points lie on an axis, and a point on an axis is in *no* quadrant — for those, name the axis instead.

(a) $(-7, -2)$ (b) $(0, 9)$ (c) $(11, -4)$ (d) $(-6, 0)$

(e) $(-1, 8)$ (f) $(0, 0)$ (g) $(5, 5)$ (h) $(12, 0)$

2. Each description below tells you only the **signs** of the coordinates. Name the quadrant, or the part of the axis, that the point must lie on.

(a) $x > 0$ and $y < 0$ (b) $x < 0$ and $y < 0$ (c) $x = 0$ and $y < 0$

(d) $x < 0$ and $y = 0$ (e) $x > 0$ and $y > 0$ (f) $x = 0$ and $y = 0$

3. The point (a, b) lies in **Quadrant II**. That tells you the sign of a and the sign of b. Use those signs to decide which quadrant each point below lies in.

(a) (b, a) (b) $(-a, b)$ (c) $(a, -b)$ (d) $(-b, -a)$

4. Describe **all** of the points that satisfy each condition. Use words such as "axis", "quadrant", "horizontal line", or "vertical line" — not just one example point.

(a) Where are all the points whose x-coordinate is 0?

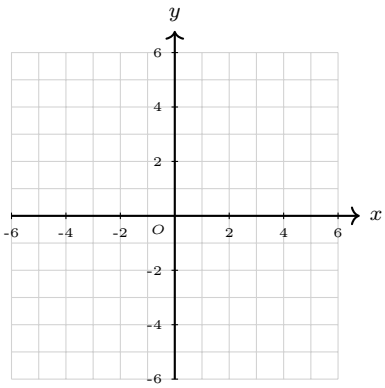
(b) Where are all the points whose y-coordinate is 0?

(c) Where are all the points with $y > 0$ *and* $x < 0$?

(d) Where are all the points with $y = -3$? Which quadrants does this set of points pass through?

Part B — Shapes on the Grid: Perimeter and Area

5. Plot $A(-4,3)$, $B(2,3)$, $C(2,-1)$, and $D(-4,-1)$ on the grid, then join $A \to B \to C \to D \to A$.

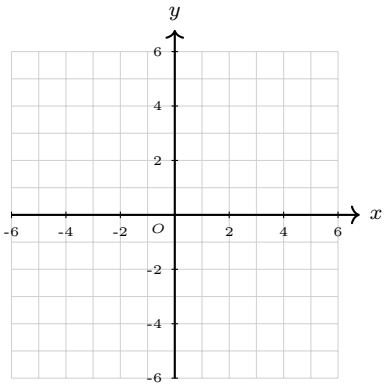


(a) What shape is $ABCD$?

(b) Find its perimeter by counting units.

(c) Find its area.

6. Plot $P(-3,-2)$, $Q(5,-2)$, and $R(5,4)$, then join $P \to Q \to R \to P$.

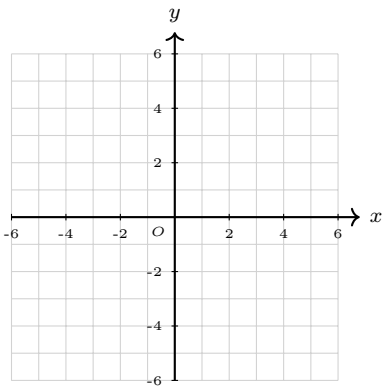


(a) Explain why $\angle Q$ is a right angle. *Hint:* look at which side is horizontal and which is vertical.

(b) Find the length of PQ and of QR by counting, then find the area of the triangle.

(c) Use the Pythagorean theorem (a Grade 8 review) to find PR, then give the perimeter.

7. Plot $W(-5,-3)$, $X(3,-3)$, $Y(1,2)$, and $Z(-3,2)$, then join $W \to X \to Y \to Z \to W$.



(a) Which two sides are parallel? How long is each of them?

(b) How far apart are those two parallel sides? (This is the height.)

(c) The area of a trapezoid is $\frac{1}{2}(b_1 + b_2)h$. Find the area.

Part C — Distance and Midpoint Without a Formula

8. Each pair of points below shares an x-coordinate or a y-coordinate, so the segment joining them is horizontal or vertical. Find its **length by counting**, and say whether the segment is horizontal or vertical.

(a) $(-6,2)$ and $(5,2)$ (b) $(3,-4)$ and $(3,6)$ (c) $(-2,-5)$ and $(-2,4)$ (d) $(-4,-6)$ and $(-4,5)$

9. The **midpoint** of a segment is the point exactly halfway along it. For a horizontal or vertical segment you can find it by *averaging* the two coordinates that differ and keeping the one that stays the same. Find the midpoint of each segment.
(a) $(-5, 4)$ to $(3, 4)$ (b) $(2, -6)$ to $(2, 4)$ (c) $(-3, -5)$ to $(6, -5)$ (d) $(-1, 7)$ to $(-1, -2)$

10. **Work backwards.** In each part you are given one endpoint A and the midpoint M. Find the other endpoint B.
(a) $A(-6, 3)$ and $M(-1, 3)$. Find B, then check that M really is halfway.
(b) $A(2, -5)$ and $M(2, -1)$. Find B, then state the length of the whole segment AB.

Part D — Reflections in the Axes and in the Origin

11. Reflecting a point is like folding the plane along a line. Consider the point $T(3, -4)$.
(a) Reflect T in the x-axis. Give the image, then state the sign rule in the form $(x, y) \to (____, ____)$.
(b) Reflect T in the y-axis. Give the image and the sign rule.
(c) Reflect T in the **origin** (a half-turn). Give the image and the sign rule.

12. Apply all three rules from question 11 to each point in the table below.

Point	$(-2, 5)$	$(6, 1)$	$(-4, -3)$	$(0, -6)$
Reflected in the x-axis				
Reflected in the y-axis				
Reflected in the origin				

One of the four points behaves differently under one of the reflections. Which point, which reflection, and why?

13. Triangle PQR has vertices $P(1,2)$, $Q(5,2)$, and $R(5,5)$. Plot it on the left grid. On the right grid, plot its image after a reflection in the x-axis and label the image P', Q', R'.

Original triangle PQR

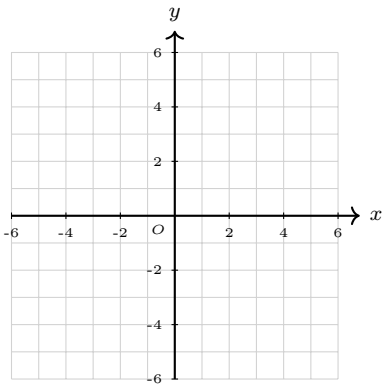
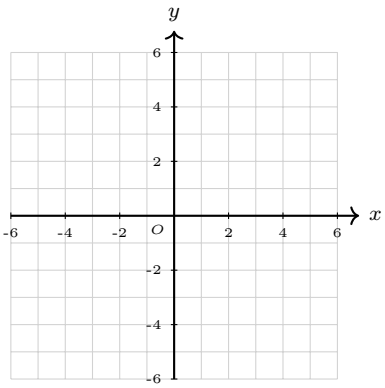


Image triangle $P'Q'R'$



(a) List the coordinates of P', Q', and R'.

(b) Which quadrant was PQR in? Which quadrant is $P'Q'R'$ in?

(c) Find the area of each triangle. What do you notice?

Part E — Working Backwards and Coordinate Puzzles

14. Three vertices of a rectangle are $(1,2)$, $(6,2)$, and $(6,7)$.

(a) Find the coordinates of the fourth vertex.

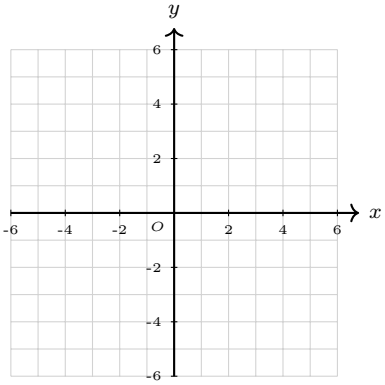
(b) Find the perimeter and the area of the rectangle.

(c) This rectangle deserves a more specific name. What is it, and how do the side lengths prove it?

15. **Parallelogram puzzle.** Three vertices of a parallelogram are $E(-4,-2)$, $F(0,-2)$, and $G(2,1)$. There is more than one point that could serve as the fourth vertex, because any one of the three given points may be the one "opposite" the missing one.

(a) Find the fourth vertex if EF and the missing side are the pair of parallel sides — that is, if the vertex opposite F is the missing one. Then find the other possible positions. Plot all of your parallelograms on the grid at right.

(b) How many different parallelograms are possible in total?



16. Three vertices of a square are $(-2,1)$, $(1,1)$, and $(1,-2)$.

(a) Find the fourth vertex.

(b) Find the side length, the perimeter, and the area.

(c) Which quadrants does this square pass through?

17. Describe, in words and with a sketch, where *every* point whose two coordinates are equal (that is, $x = y$) is located. Which quadrants do these points pass through, and which quadrants do they never enter?

18. **Stretch.** Consider every point (x, y) whose coordinates are both integers with $-2 \le x \le 2$ and $-2 \le y \le 2$. (a) How many such points are there? (b) How many of them lie in *no* quadrant? (c) How many lie in Quadrant III?

Answer Key. 1. (a) III (b) y-axis (c) IV (d) x-axis (e) II (f) the origin (g) I (h) x-axis. **2.** (a) IV (b) III (c) the y-axis, below the origin (d) the x-axis, left of the origin (e) I (f) the origin. **3.** Quadrant II means $a < 0$ and $b > 0$. (a) $(b, a) = (+, -)$: IV (b) $(-a, b) = (+, +)$: I (c) $(a, -b) = (-, -)$: III (d) $(-b, -a) = (-, +)$: II. **4.** (a) Every point on the y-axis (the whole vertical axis, including the origin). (b) Every point on the x-axis. (c) All of Quadrant II (left of the y-axis and above the x-axis), not including the axes themselves. (d) A horizontal line 3 units below the x-axis, through $(0, -3)$; it passes through Quadrants III and IV and crosses the y-axis between them. **5.** (a) A rectangle. (b) Width 6, height 4, so $P = 2(6 + 4) = 20$ units. (c) $A = 6 \times 4 = 24$ square units. **6.** (a) PQ is horizontal ($y = -2$ for both) and QR is vertical ($x = 5$ for both), and a horizontal segment meets a vertical segment at $90°$. (b) $PQ = 8$, $QR = 6$, $A = \frac{1}{2}(8)(6) = 24$ square units. (c) $PR = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$; perimeter $= 8 + 6 + 10 = 24$ units. **7.** (a) WX and ZY are both horizontal, so they are parallel: $WX = 8$ and $ZY = 4$. (b) Height $= 2 - (-3) = 5$. (c) $A = \frac{1}{2}(8 + 4)(5) = 30$ square units. **8.** (a) 11, horizontal (b) 10, vertical (c) 9, vertical (d) 11, vertical. **9.** (a) $(-1, 4)$ (b) $(2, -1)$ (c) $(1.5, -5)$ (d) $(-1, 2.5)$. **10.** (a) $B(4, 3)$; M is 5 units from each endpoint. (b) $B(2, 3)$; $AB = 8$ units. **11.** (a) $(3, 4)$; $(x, y) \to (x, -y)$ — keep x, change the sign of y. (b) $(-3, -4)$; $(x, y) \to (-x, y)$. (c) $(-3, 4)$; $(x, y) \to (-x, -y)$ — both signs change. **12.** x-axis: $(-2, -5)$, $(6, -1)$, $(-4, 3)$, $(0, 6)$. y-axis: $(2, 5)$, $(-6, 1)$, $(4, -3)$, $(0, -6)$. Origin: $(2, -5)$, $(-6, -1)$, $(4, 3)$, $(0, 6)$. The point $(0, -6)$ is unchanged by the reflection in the y-axis, because it already sits *on* the y-axis — the fold line passes through it. **13.** (a) $P'(1, -2)$, $Q'(5, -2)$, $R'(5, -5)$. (b) PQR is in Quadrant I; $P'Q'R'$ is in Quadrant IV. (c) Both have area $\frac{1}{2}(4)(3) = 6$ square units — a reflection changes position but not size or shape. **14.** (a) $(1, 7)$. (b) Sides 5 and 5, so $P = 20$ units and $A = 25$ square units. (c) It is a **square**: $(1, 2)$ to $(6, 2)$ is 5 units and $(6, 2)$ to $(6, 7)$ is also 5 units, so all four sides are equal. **15.** (a) Opposite F: $E + G - F = (-2, 1)$. The other two positions are $E + F - G = (-6, -5)$ (opposite G) and $F + G - E = (6, 1)$ (opposite E). (b) Three different parallelograms are possible. **16.** (a) $(-2, -2)$. (b) Side 3 units, $P = 12$ units, $A = 9$ square units. (c) All four quadrants — the square straddles both axes. **17.** They lie on a straight line through the origin that rises one unit for every unit to the right: $(-3, -3)$, $(-1, -1)$, $(0, 0)$, $(2, 2)$, $(4, 4)$, and so on. The line passes through Quadrant I (both coordinates positive) and Quadrant III (both negative) and through the origin; it never enters Quadrant II or Quadrant IV, because there x and y have opposite signs and so cannot be equal. **18.** (a) $5 \times 5 = 25$ points. (b) 9 of them lie on an axis: the five with $x = 0$ plus the five with $y = 0$, minus the origin counted twice. (c) 4 points — $(-1, -1)$, $(-1, -2)$, $(-2, -1)$, $(-2, -2)$; in fact each quadrant holds exactly 4 of the remaining 16.

Lesson 2 — Extra Practice

Growing Patterns and Types of Variables

Part A — Finding the Rule $t = mn + b$

A pattern that grows by a **constant amount** can be written as a rule $t = mn + b$, where n is the figure number and t is the number of tiles. The value m is the constant amount added each time, and b is what the rule would give at "Figure 0" — the starting value *before* the pattern begins. For example, a pattern starting at 7 and growing by 4 each time has $m = 4$ and $b = 7 - 4 = 3$, so $t = 4n + 3$.

1. A tile pattern gives the table below.

Figure Number, n	1	2	3	4	5
Number of Tiles, t	7	11	15	19	23

(a) Find m, the constant amount added each time.

(b) Work backwards one step from Figure 1 to find the value at Figure 0. This is b.

(c) Write the rule in the form $t = mn + b$, then test it on Figure 3.

(d) Use your rule to find the number of tiles in Figure 50.

(e) **Work backwards.** Which figure has 143 tiles?

(f) Why would counting up one figure at a time be a poor way to answer part (d)?

2. A second pattern gives this table.

Figure Number, n	1	2	3	4	5
Number of Tiles, t	2	7	12	17	22

(a) Write the rule $t = mn + b$.

(b) How many tiles are in Figure 20?

(c) **Which figure has 62 tiles?**

(d) Is 100 ever a value in this pattern? Show how you know.

3. The tile pattern below is made of three full rows plus one extra tile on top of the left-hand column.

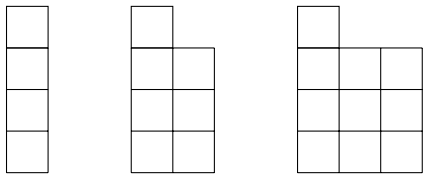


Figure 1 Figure 2 Figure 3

(a) Count the tiles in Figures 1, 2, and 3, then predict Figures 4 and 5 without drawing them.

(b) Write the rule $t = mn + b$. Explain where the 3 and the 1 come from in the picture.

(c) How many tiles are in Figure 50?

(d) **Work backwards:** which figure has 100 tiles?

(e) Which figure has 76 tiles?

(f) Explain why the extra tile on top does not change the constant difference m.

4. The table below follows the rule $t = 6n + 3$, but three entries have been erased. Fill them in.

Figure Number, n	1	2	3	4	5	6
Number of Tiles, t	9		21		33	

(a) Fill in the three missing values. Describe two different ways you could have found the entry at $n = 2$: one using the rule, and one using the constant difference.

(b) A student filled in $n = 4$ as 26. Which of the two checks above would have caught this error fastest, and how?

5. Reasoning about the two numbers in the rule $t = mn + b$.

(a) In a tile pattern, what does m tell you about the picture?

(b) What does b tell you? Why is b usually *not* the number of tiles in Figure 1?

(c) A pattern has $m = 0$. Describe what its figures look like, and whether it is still a growing pattern.

Part B — Patterns That Do Not Grow at a Constant Rate

6. Consider the pattern 1, 4, 9, 16, 25, … made from square arrays of tiles.

(a) Write the differences between consecutive terms. Are they constant?

(b) Write the next two terms.

(c) Describe the pattern in words, then write a rule for t in terms of n.

(d) How many tiles are in Figure 10?

(e) Explain why *no* rule of the form $t = mn + b$ can produce this pattern.

(f) What *is* constant about this pattern, if the first differences are not?

7. Consider the pattern 2, 6, 12, 20, 30, ...

(a) List the differences.

(b) Write the next two terms.

(c) Each term is a rectangle n tiles by $(n+1)$ tiles. Write the rule.

(d) Find Figure 10.

8. For each pattern below, state whether it grows by a **constant amount**. If it does, write the rule $t = mn + b$; if it does not, describe in words how it is being built.

(a) 5, 9, 13, 17, 21

(b) 1, 2, 4, 8, 16

(c) 20, 17, 14, 11, 8

(d) 1, 3, 6, 10, 15

Part C — Discrete or Continuous? (a preview of Lesson 5)

The values a variable is allowed to take can be **discrete** — separate, countable values with gaps between them, such as $1, 2, 3, \ldots$ tiles — or **continuous**, meaning any value in a range is possible, including fractions and decimals, with no gaps. Asking "could a value halfway between two of these make sense?" usually settles it.

9. Name the *independent* variable in each situation, then state whether it is discrete or continuous.

(a) The number of tiles in Figure n of a growing pattern.

(b) The depth of water in a bathtub while it fills.

(c) The total cost of n movie tickets at \$12 each.

(d) The distance a car has travelled after t hours.

(e) The number of buses needed for s students.

(f) The temperature of a cup of coffee as it cools.

10. Each situation below is genuinely arguable. Decide discrete or continuous and **justify your choice in a sentence** — the justification is the real answer here.

(a) The cost of filling a car, based on the number of litres of gasoline pumped.

(b) The height of a bean plant, measured and recorded once each morning for two weeks.

(c) The balance in a savings account that has interest added on the last day of each month. Is the balance discrete or continuous?

(d) The length of a candle, based on the number of hours it has been burning.

Part D — Independent and Dependent Variables

11. Identify the independent variable and the dependent variable in each situation.

(a) The battery percentage of a phone, based on the number of hours it has been charging.

(b) The price of a pizza, based on the number of extra toppings ordered.

(c) The distance a runner has covered, based on the number of minutes spent running.

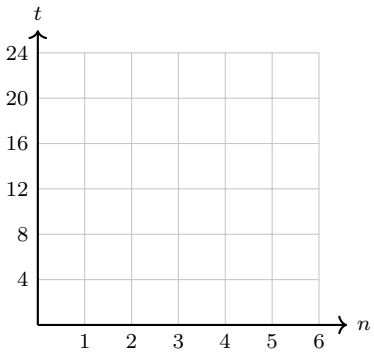
(d) The number of tiles in a figure, based on the figure number.

12. Two harder cases.

(a) Jordan says, "The number of hours I study depends on the mark I get on the test." Rewrite the statement so the variables are the right way round, and explain how you can tell.

(b) A table lists the shoe size and the height of 30 people. Neither one causes the other. Explain how you would still decide which variable to put on the horizontal axis.

13. A pattern follows the rule $t = 3n + 2$, where n is the figure number.



(a) Build a table of values for Figures 1 through 6.

(b) Plot the six ordered pairs (n, t) on the grid at left.

(c) Should you join the plotted points with a solid line? Explain using the word **discrete**.

Part E — Reasoning and Stretch

14. A tile pattern has the rule $t = 5n + 4$. Explain why n cannot be -3, and why it cannot be 2.5 either. What is the smallest value n is allowed to take?

15. In question 8(c) the pattern $20, 17, 14, 11, 8$ follows $t = -3n + 23$ and describes tiles being removed from a display. (a) What does the rule give for Figure 12? (b) Why is that answer impossible in the real situation? (c) What is the last figure that still makes sense, and how many tiles does it have?

16. Devon is working with the pattern $t = 4n + 1$.

(a) He claims Figure 12 has 50 tiles. Is he right? Show the calculation.

(b) Which figure has 101 tiles?

(c) Explain why 50 can *never* appear in this pattern, no matter how far it runs.

17. **Stretch — two patterns racing.** Pattern A starts with 3 tiles in Figure 1 and grows by 5 tiles each figure. Pattern B starts with 18 tiles in Figure 1 and grows by 2 tiles each figure.

Figure Number, n	1	2	3	4	5	6	7
Pattern A	3						
Pattern B	18						

(a) Complete the table.

(b) Write a rule for each pattern.

(c) At which figure number do the two patterns have the *same* number of tiles? How many tiles is that?

(d) After that figure, which pattern is ahead? Explain using the two values of m.

18. *(Multiple Choice)* A pattern follows the rule $t = 7n-4$. Which figure has 66 tiles?

A. Figure 8 B. Figure 9 C. Figure 10 D. Figure 11

19. *(Numerical Response)* A pattern begins with 9 tiles in Figure 1 and grows by 7 tiles each figure. The figure number that has 79 tiles is _____.

Record your answer from left to right:

Answer Key. 1. (a) $m = 4$ (b) Figure 0 would be $7 - 4 = 3$, so $b = 3$ (c) $t = 4n + 3$; check $n = 3$: $4(3) + 3 = 15$ ✓ (d) $t = 4(50) + 3 = 203$ tiles (e) $4n + 3 = 143 \Rightarrow 4n = 140 \Rightarrow n = 35$, Figure 35 (f) it would take 49 separate additions, with a chance of error at every step. **2.** (a) $m = 5$, Figure 0 is $2 - 5 = -3$, so $t = 5n - 3$ (b) $5(20) - 3 = 97$ (c) $5n - 3 = 62 \Rightarrow 5n = 65 \Rightarrow n = 13$, Figure 13 (d) No: $5n - 3 = 100$ gives $5n = 103$ and $n = 20.6$, which is not a whole figure number. **3.** (a) 4, 7, 10; Figure 4 has 13 and Figure 5 has 16 (b) $t = 3n + 1$; the 3 is the three full rows added with each new column, and the 1 is the single extra tile on top, which is there in every figure (c) $3(50) + 1 = 151$ tiles (d) $3n + 1 = 100 \Rightarrow n = 33$ (e) $3n + 1 = 76 \Rightarrow n = 25$ (f) the extra tile is added once and never again, so it shifts every value up by 1 but changes no difference between consecutive figures. **4.** (a) $n = 2$ gives 15, $n = 4$ gives 27, $n = 6$ gives 39. Using the rule: $6(2) + 3 = 15$. Using the constant difference: $m = 6$, so add 6 to the previous entry, $9 + 6 = 15$. (b) The constant difference: $21 + 6 = 27$, not 26 — a single addition exposes it immediately, and the differences $21 - 15 = 6$ but $26 - 21 = 5$ are visibly unequal. **5.** (a) m is the number of tiles added to move from one figure to the next — the size of the repeating "chunk" of the picture. (b) b is the fixed part of every figure, the piece that never grows; equivalently the value the rule gives at Figure 0. It is not Figure 1's count because Figure 1 already contains one copy of the growing chunk, so Figure 1 is $m + b$. (c) Every figure would be identical — no tiles added — so it is a constant pattern, not a growing one. **6.** (a) $3, 5, 7, 9$ — not constant, they grow by 2 each time (b) 36, 49 (c) each figure is a square array of tiles, n by n, so $t = n^2$ (d) $10^2 = 100$ tiles (e) a rule $t = mn + b$ always adds the same amount m each step, but here the step keeps increasing (f) the *second* differences are constant: $5 - 3 = 2$, $7 - 5 = 2$, $9 - 7 = 2$. **7.** (a) $4, 6, 8, 10$ — increasing, not constant (b) 42, 56 (c) $t = n(n+1)$ (d) $10 \times 11 = 110$ tiles. **8.** (a) Constant, $m = 4$ and Figure 0 is 1, so $t = 4n + 1$. (b) Not constant — each term is *double* the one before (differences $1, 2, 4, 8$). (c) Constant, $m = -3$ and Figure 0 is 23, so $t = -3n + 23$; the pattern shrinks by 3 each figure. (d) Not constant — the differences are $2, 3, 4, 5$, each one larger than the last (these are the triangular numbers, the running totals $1, 1 + 2, 1 + 2 + 3, \ldots$). **9.** (a) figure number n — discrete (b) time elapsed — continuous (c) number of tickets n — discrete (d) time t in hours — continuous (e) number of students s — discrete (f) time elapsed — continuous. **10.** (a) Continuous: a pump can deliver 12.73 L, so any amount in the range is possible — rounding on the display does not change what is physically possible. (b) Either answer is defensible; the reason is what matters. The plant's height itself is continuous, but the *recorded data* is discrete — one measurement per day and nothing in between. (c) Discrete: the balance jumps on the last day of the month and then stays flat, so the graph is a set of steps rather than a smooth rise. (d) Continuous: the candle shortens smoothly, so a length halfway between two readings genuinely occurs. **11.** (a) Ind: hours charging; Dep: battery percentage. (b) Ind: number of extra toppings; Dep: price. (c) Ind: minutes spent running; Dep: distance covered. (d) Ind: figure number n; Dep: number of tiles t. **12.** (a) "The mark I get on the test depends on the number of hours I study." The study hours come first and are the thing Jordan controls, so they are independent; the mark is the result, so it is dependent. (b) Put whichever one you are using as the input on the horizontal axis; the question you are asking decides it, not correlation. **13.** (a) t: $5, 8, 11, 14, 17, 20$. (b) Plot $(1, 5), (2, 8), (3, 11), (4, 14), (5, 17), (6, 20)$. (c) No. The figure number is discrete — there is no Figure 2.5 — so only the separate points have meaning and they should be left unjoined. **14.** A figure number counts figures, so it must be a positive whole number: there is no Figure -3 and no half a figure, so 2.5 is not allowed either. The smallest value is $n = 1$. **15.** (a) $-3(12) + 23 = -13$. (b) A display cannot hold a negative number of tiles. (c) $-3(7) + 23 = 2$, and $-3(8) + 23 = -1$, so Figure 7 is the last one that makes sense, with 2 tiles. **16.** (a) No: $4(12) + 1 = 49$, not 50. (b) $4n + 1 = 101 \Rightarrow 4n = 100 \Rightarrow n = 25$, Figure 25. (c) $4n$ is always even, so $4n + 1$ is always odd; 50 is even, so it can never occur (solving directly, $n = 12.25$, not a whole figure number). **17.** (a) A: $3, 8, 13, 18, 23, 28, 33$; B: $18, 20, 22, 24, 26, 28, 30$. (b) A: $t = 5n - 2$; B: $t = 2n + 16$. (c) At Figure 6, where both have 28 tiles. (d) Pattern A. Its m is 5 against B's 2, so A gains 3 more tiles per figure; once the two are level at Figure 6, A pulls ahead by 3 tiles for every further figure (33 versus 30 at Figure 7). **18.** C ($7n - 4 = 66 \Rightarrow 7n = 70 \Rightarrow n = 10$). **19.** 11 (the rule is $t = 7n + 2$, and $7n + 2 = 79 \Rightarrow n = 11$).

Lesson 3 — Extra Practice

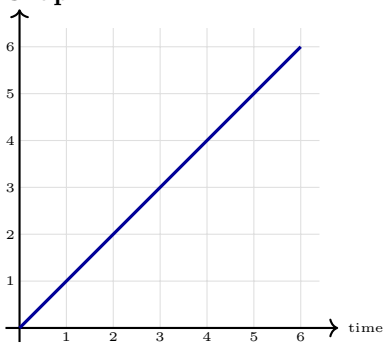
Describing Relationships Between Two Quantities

Note: some questions in this set look ahead — reading the *shape* of a graph as a story is Lesson 11, and steady versus changing rates come back in Unit 6. Treat them as a preview.

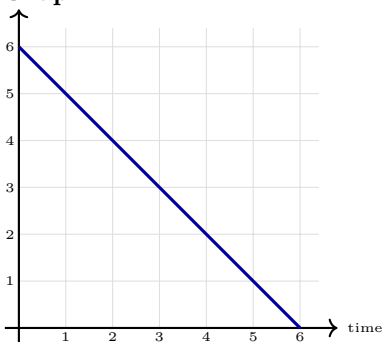
Part A — Shapes, Variables, and What a Relationship Means

1. Each graph below shows the general **shape** of a relationship between two quantities. In every graph, time runs along the horizontal axis and the quantity runs up the vertical axis.

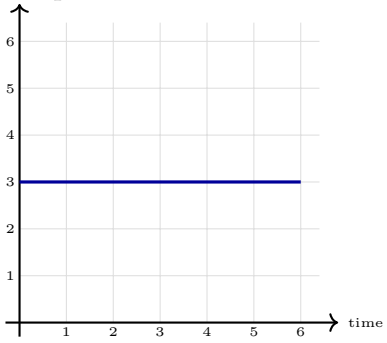
Graph A



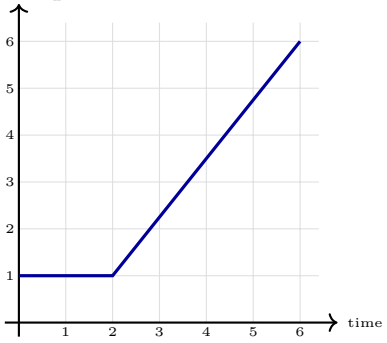
Graph B



Graph C



Graph D



Write the letter of the graph that best matches each situation.

(a) Water is pumped into an empty tank at a steady rate. The quantity is the volume of water in the tank. _____

(b) A candle burns down steadily from its full height until nothing is left. The quantity is the height of the candle. _____

(c) A car is parked at a mall 3 km from home for the whole afternoon. The quantity is the car's distance from home. _____

(d) A hiker rests at a lookout 1 km along the trail, then walks steadily onward. The quantity is the hiker's distance from the trailhead. _____

2. For each pair of quantities, state the **independent** variable and the **dependent** variable, then name the quantity that belongs on the **horizontal** axis.

(a) the mass of a parcel and the cost of shipping it

(b) the battery percentage of a phone and the number of minutes it has been charging

(c) the distance a taxi drives and the fare the rider pays

(d) the height of a plant and the number of days since its seed was planted

3. For each situation, state whether the graph would be **increasing**, **decreasing**, **constant**, or **decreasing then constant**.

(a) the money left on a bus pass as rides are taken

(b) the temperature of a pot of water that is already boiling and stays on the burner

(c) the height of a young tree over ten years

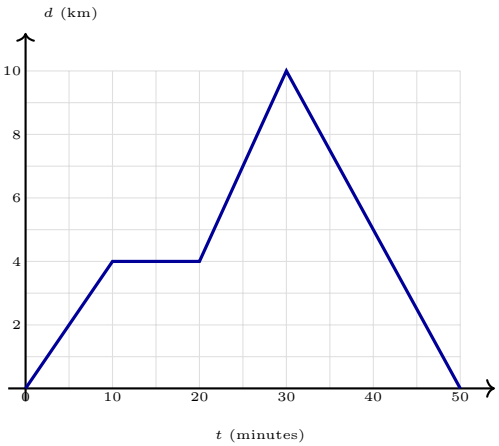
(d) the number of unopened cans in a case as you drink one a day for four days, then stop for a week

4. Across one summer, the number of ice cream cones sold at a beach and the number of sunburns treated at the first-aid station rise and fall together. A student concludes, "Eating ice cream causes sunburns." Explain what is wrong with this reasoning, and name the quantity that is really behind both.

5. Two quantities can be strongly related without either one causing the other. Describe your **own** example of such a pair (not the ice cream one), and explain in one or two sentences why neither quantity causes the other.

Part B — Reading and Telling Stories with Graphs

6. The graph below shows Jamal's distance from home, d kilometres, after t minutes of cycling.



(a) During which time interval was Jamal stopped? How do you know?

(b) How far from home was Jamal after 25 minutes?

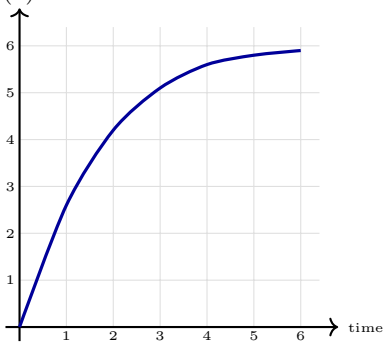
(c) During which interval was Jamal travelling **fastest**? Support your answer by calculating a speed, in kilometres per minute, for each of the three moving sections.

(d) During which interval was Jamal returning home? Explain what tells you this.

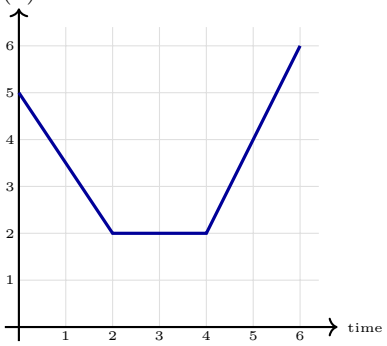
(e) What total distance did Jamal cycle? (This is not the same as his final distance from home.)

7. Each graph below has no labels on its axes except "time." For each one, write a short, plausible **story** that the graph could be describing. Say what quantity is on the vertical axis, and make sure your story explains every section of the graph.

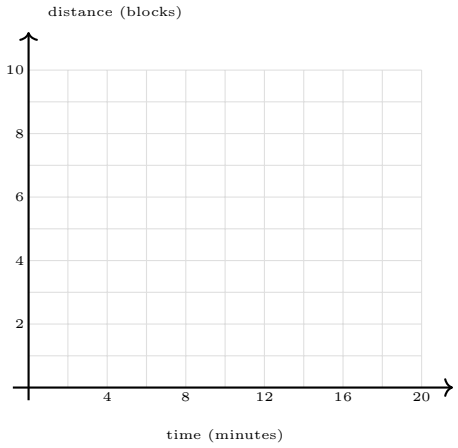
(a)



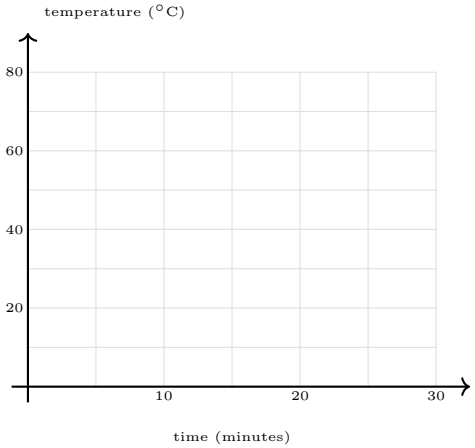
(b)



8. Maya leaves home and walks 3 blocks in 5 minutes. She waits 4 minutes for a friend, and then the two of them walk 5 more blocks in the next 8 minutes. Sketch a graph of Maya's distance from home against time on the grid below. Label each of the three sections.



9. A mug of tea is poured at 80°C and left on a counter in a room that stays at 20°C. Sketch a graph of the tea's temperature against time on the grid below, and explain in one sentence why your graph levels off instead of continuing to fall.



Part C — Tables, Ordered Pairs, and Linear or Non-Linear

10. A relation is **linear** when equal steps in the input always produce equal steps in the output — that is, when the **first differences** of the outputs are constant. Decide whether each table is linear or non-linear. Show the first differences.

(a)

x	1	2	3	4	5
y	4	7	10	13	16

(b)

x	1	2	3	4	5
y	1	4	9	16	25

(c)

x	0	2	4	6	8
y	20	17	14	11	8

(d)

x	0	1	2	3	4
y	2	6	18	54	162

11. **Careful** — in these tables the inputs do **not** go up in equal steps, so comparing the output differences alone will fool you. For each table, compare the *rate* $\dfrac{\text{change in } y}{\text{change in } x}$ across each gap, then state whether the relation is linear.

(a)

x	1	3	4	7
y	5	11	14	23

(b)

x	0	1	3	6
y	2	3	7	16

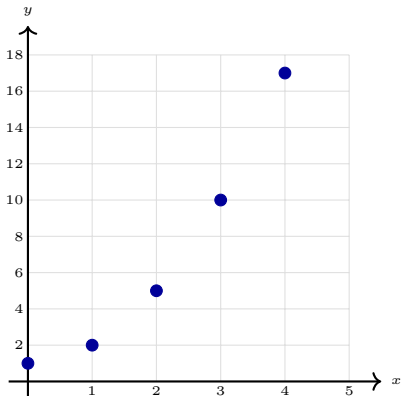
12. The table below shows a linear relation with two outputs missing.

x	2	4	6	8	10
y	7	___	19	___	31

(a) Explain how the given outputs tell you the step size, then fill in the two missing values.

(b) Write the completed relation as a set of ordered pairs (x, y).

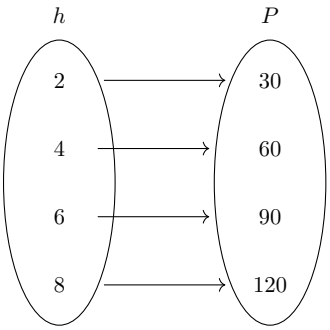
13. The points of a relation have been plotted below.



(a) Write the relation as a set of ordered pairs and as a table of values.

(b) Is the relation linear or non-linear? Justify using first differences.

14. A mapping diagram for a summer job is shown below: each input is a number of hours worked, h, and each output is the pay earned, P dollars.



(a) Write the relation as a set of ordered pairs (h, P).

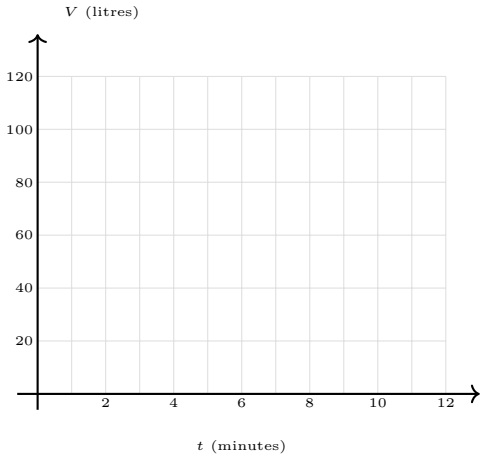
(b) State the independent and dependent variables, and explain how the words of the situation — not the diagram — tell you which is which.

(c) Is this relation linear? Justify your answer using first differences.

(d) Determine the hourly rate of pay, and use it to predict the pay for 11 hours of work.

Part D — Stretch

15. An empty 120 L tank is filled at a steady 20 L per minute for 5 minutes. The valve is then closed and the tank is left untouched for 4 minutes. Finally the tank is drained at a steady 40 L per minute until it is empty.



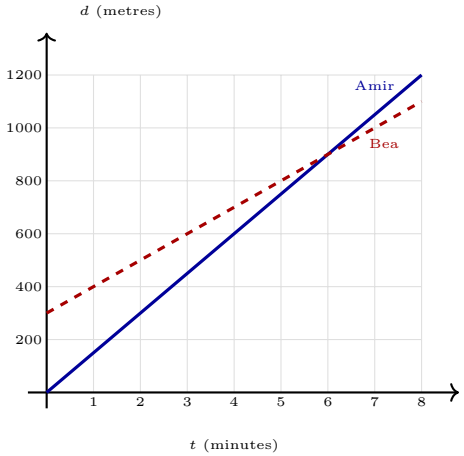
(a) How much water is in the tank at $t = 5$ minutes? Does the tank ever get full?

(b) At what time is the tank empty again? Show your calculation.

(c) Sketch the complete graph of volume against time on the grid at left, and label the three sections.

(d) Which section of your graph is steepest, and what does that steepness mean in this situation?

16. The graph below shows two runners on a straight 1200 m track. Amir's line starts at the origin; Bea's line starts higher because she is given a head start.



(a) How big is Bea's head start?

(b) Determine each runner's speed in metres per minute. Who is faster?

(c) At what time does Amir catch Bea, and how far along the track are they at that moment?

(d) Who is ahead at 4 minutes, and by how much? Explain how the graph shows this at a glance.

17. A swimming pool holds 500 L and is drained at a steady 25 L per minute. Two of the three tables below cannot describe this situation.

Table 1

t	0	2	4	6
V	500	450	400	350

Table 2

t	0	2	4	6
V	500	550	600	650

Table 3

t	0	2	4	6
V	500	400	300	200

(a) Which table matches the situation? Justify your choice with a rate.

(b) Describe a situation that each of the other two tables *could* describe, including the rate in your description.

18. A Ferris wheel turns at a steady speed. Explain why a graph of a rider's height above the ground against time cannot be an increasing line, and describe the shape it must have instead.

19. On a distance–time graph, can a section ever be a perfectly **vertical** segment? Explain what such a segment would be claiming about the moving object, and why that is impossible.

Answer Key. 1. (a) A (b) B (c) C (d) D. **2.** Independent (horizontal axis) / dependent: (a) parcel mass / shipping cost (b) charging time / battery percentage (c) distance driven / fare (d) days since planting / plant height. (The order a pair is listed in doesn't tell you which is which — ask which one depends on the other.) **3.** (a) decreasing (b) constant (c) increasing (d) decreasing then constant. **4.** Rising together is not the same as one causing the other. Hot, sunny weather is the hidden third quantity, driving both ice cream sales and time spent in the sun. **5.** Answers vary. Sample: running-shoe sales and marathon finishers in a city — both rise because more people are training; buying shoes does not make anyone finish a race. **6.** (a) 10–20 min; the graph is horizontal, so his distance from home is not changing (b) 7 km, since $4 + 5(0.6) = 7$ (c) the 20–30 min section: the rates are $4 \div 10 = 0.4$, $6 \div 10 = 0.6$ and $10 \div 20 = 0.5$ km/min (d) 30–50 min; the graph slopes downward, so his distance from home shrinks, reaching 0 (home) at 50 min (e) 10 km out + 10 km back = 20 km. **7.** Answers vary. (a) rises quickly, then levels off — e.g. fans arriving until a stadium is nearly full. (b) a steady fall, a flat stretch, then a steeper rise — e.g. a bathtub draining, sitting while the plug is replaced, then refilling faster. **8.** A line from $(0, 0)$ to $(5, 3)$; a horizontal segment from $(5, 3)$ to $(9, 3)$; a line from $(9, 3)$ to $(17, 8)$. **9.** A curve from 80°C, falling steeply then more slowly, levelling off at 20°C: the tea can only cool to room temperature, after which no temperature difference is left to drive cooling. **10.** (a) linear; first differences all $+3$ (b) non-linear; differences $3, 5, 7, 9$ (c) linear; differences all -3 for each step of 2 in x (d) non-linear; differences $4, 12, 36, 108$ (the outputs triple). **11.** (a) linear; rates $6/2 = 3$, $3/1 = 3$, $9/3 = 3$ — all equal, even though the raw output differences $6, 3, 9$ are not (b) non-linear; rates $1/1 = 1$, $4/2 = 2$, $9/3 = 3$. **12.** (a) y rises 12 over the 4 from $x = 2$ to $x = 6$, so 6 for each step of 2; missing values 13 and 25 (b) $(2, 7), (4, 13), (6, 19), (8, 25), (10, 31)$. **13.** (a) $(0, 1), (1, 2), (2, 5), (3, 10), (4, 17)$; $x : 0, 1, 2, 3, 4$ and $y : 1, 2, 5, 10, 17$ (b) non-linear; first differences $1, 3, 5, 7$ are not constant. **14.** (a) $(2, 30), (4, 60), (6, 90), (8, 120)$ (b) independent h, dependent P — pay depends on hours worked, not the reverse (c) linear; P rises 30 for every 2-hour step (d) $30 \div 2 = \$15$/h, so 11 hours pays \$165. **15.** (a) $5 \times 20 = 100$ L; no — 20 L short of the 120 L capacity (b) draining takes $100 \div 40 = 2.5$ min from $t = 9$, so the tank is empty at $t = 11.5$ min (c) a line $(0, 0)$ to $(5, 100)$, a horizontal segment to $(9, 100)$, then a line down to $(11.5, 0)$ (d) the draining section; 40 L/min is twice the fill rate, so the volume changes twice as fast. **16.** (a) 300 m (b) Amir $1200 \div 8 = 150$ m/min, Bea $(1100 - 300) \div 8 = 100$ m/min; Amir is faster (c) at $t = 6$ min, both at 900 m — where the lines cross (d) Bea, by 100 m (700 vs 600); her line is still the higher one at $t = 4$. **17.** (a) Table 1: V falls 50 L every 2 min, i.e. 25 L/min, from 500 L (b) Table 2 rises 25 L/min — the same pool being *filled*; Table 3 falls 100 L every 2 min, i.e. 50 L/min — the pool draining through a wider pipe, twice as fast. **18.** A rider rises to the top, comes back down, and repeats, so the height oscillates between a minimum and a maximum — a repeating wave, never a line that keeps increasing. **19.** No. A vertical segment would put the object at many different distances at the *same* instant — infinite speed.

Lesson 4 — Extra Practice

Where Graphs Meet the Axes: x- and y-Intercepts

Part A — Intercepts from an Equation

Until now the intercepts have been read off a picture. They can also be calculated exactly: substitute $y = 0$ and solve for x to get the x-intercept, and substitute $x = 0$ and solve for y to get the y-intercept. Write every answer as an **ordered pair**.

1. Determine the x-intercept and the y-intercept of each line. Show the substitution.

(a) $y = 3x - 12$ (b) $y = -2x + 10$ (c) $y = \frac{1}{2}x - 3$ (d) $y = -\frac{3}{4}x + 6$

2. Determine both intercepts. Here the x-intercepts are **not** whole numbers — leave each one as an exact fraction, not a rounded decimal.

(a) $y = 4x - 6$ (b) $y = -5x + 2$ (c) $y = \frac{2}{3}x + 5$ (d) $y = -6x - 9$

3. An equation written in **general form**, $Ax + By = C$, is especially quick: covering up one term leaves a one-step equation. Determine both intercepts of each line.

(a) $3x + 4y = 24$ (b) $5x - 2y = 20$ (c) $2x + 7y = -14$ (d) $6x - 5y = 9$

4. (a) Determine both intercepts of $4x + 6y = 15$.

(b) Determine both intercepts of $8x - 3y = 12$.

(c) In one or two sentences, explain why finding intercepts from $Ax + By = C$ takes fewer steps than finding them from $y = mx + b$.

Part B — Intercepts from Tables, Graphs, and Special Lines

5. State the x-intercept and the y-intercept of each relation below. In part (b) no row of the table is an x-intercept, so you will have to continue the pattern.

(a) The points are listed out of order.

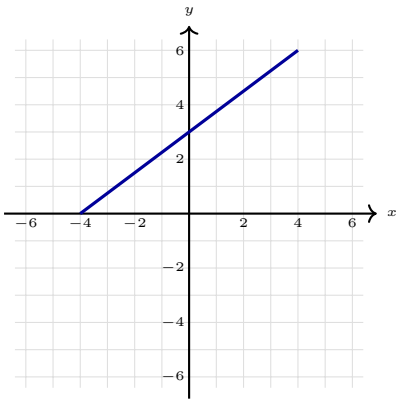
x	3	0	-1	5	2
y	0	-6	-8	4	-2

(b) The relation is linear.

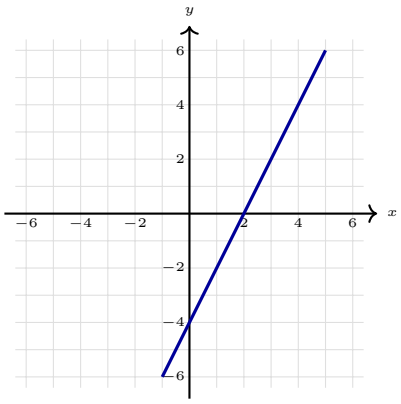
x	-2	0	2	4	6
y	-8	-5	-2	1	4

6. State both intercepts of each line below, then write the equation of the line in the form $y = mx + b$ and use it to check your intercepts by substitution. (*Method:* b is the y-intercept; put it in, then substitute the x-intercept point to find m.)

(a)



(b)



7. Each equation below describes a horizontal or vertical line, or a line through the origin. State every intercept it has — and if one kind of intercept is missing, **explain why** in terms of where the line sits on the grid.

(a) $y = 5$ (b) $x = -3$ (c) $y = -2x$ (d) $y = 0$

Part C — Intercepts That Mean Something

8. A water tank holds 60 L. A tap at the bottom is opened, and the volume of water remaining, V litres, after t minutes is given by $V = 60 - 4t$.

(a) Determine the V-intercept (the intercept on the vertical axis). What does it represent in this situation?

(b) Determine the t-intercept. What does it represent?

(c) The equation gives $V = -20$ when $t = 20$. Explain why that answer is meaningless here, and state the largest value of t that still makes sense.

(d) Which intercept tells you the size of the tank, and which tells you how long the draining takes?

9. A drone takes off with a full battery. Its charge, C percent, after t minutes of flight is $C = 100 - 8t$.

(a) Determine both intercepts.

(b) Explain what each intercept means for the pilot, and say why the t-intercept is not a whole number of minutes.

10. A taxi charges a flat pick-up fee plus a rate per kilometre. The fare, C dollars, for a ride of d kilometres is $C = 4 + 2.5d$.

(a) Determine both intercepts.

(b) One of these intercepts is meaningful and one is not. State which is which, and explain what makes the other one impossible.

Part D — Errors, Working Backwards, and Stretch

11. To find the x-intercept of $y = 2x - 8$, a student substitutes $x = 0$ and writes "the x-intercept is -8." Identify the error, explain the rule the student has mixed up, and give both correct intercepts as ordered pairs.

12. Sam says the line $3x + 5y = 15$ has a y-intercept of 5, "because the 5 is the number beside the y." Show by substitution that Sam is wrong, and state the correct y-intercept. Then state the x-intercept.

13. Work backwards. In each part, determine the equation of the line described. Give your answer in the form $y = mx + b$, and check it by substituting. (*Method:* b is the y-intercept; substitute the other point to find m.)

(a) The line has x-intercept 4 and y-intercept -2.

(b) The line has x-intercept -3 and y-intercept 5.

(c) The line has x-intercept 6 and y-intercept 6.

14. The line $y = mx+8$ has an x-intercept of 4. Determine the value of m, then state the equation of the line.

15. The line $y = 3x + b$ has an x-intercept of -2. Determine the value of b, then state the equation of the line.

16. **Stretch.** Consider the family of lines $2x + ky = 12$, where k can be any non-zero number.

(a) For what value of k does the line have a y-intercept of 3?

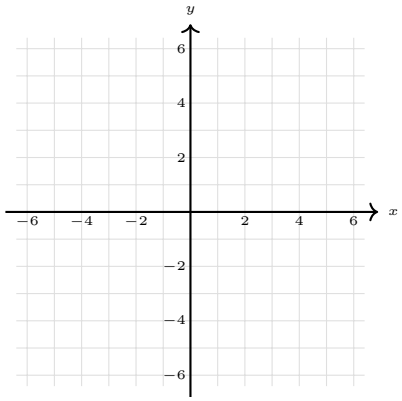
(b) Determine the x-intercept of $2x + ky = 12$ in terms of k. What do you notice?

(c) For what value of k is the x-intercept equal to 3? Explain your answer carefully.

17. **Stretch.** A line crosses the x-axis at $(6, 0)$ and the y-axis at $(0, b)$. Together with the two axes it forms a right triangle whose area is 12 square units.

(a) Determine the two possible values of b, and give the equation of each line.

(b) Sketch both lines on one set of axes on the grid at right, and explain why the two triangles have the same area.



18. Is it possible for a **straight line** to have no intercepts at all? Consider vertical, horizontal, and slanted lines in your answer, and explain why every straight line must meet at least one axis.

19. The line $y = 0$ is the x-axis itself. Explain why saying "its x-intercept is $(0, 0)$" is not the whole truth, and describe all of its x-intercepts.

Answer Key. 1. (a) x-int $(4, 0)$, y-int $(0, -12)$ (b) $(5, 0)$, $(0, 10)$ (c) $(6, 0)$, $(0, -3)$ (d) $(8, 0)$, $(0, 6)$. **2.** (a) $\left(\frac{3}{2}, 0\right)$, $(0, -6)$ (b) $\left(\frac{2}{5}, 0\right)$, $(0, 2)$ (c) $\left(-\frac{15}{2}, 0\right)$, $(0, 5)$ (d) $\left(-\frac{3}{2}, 0\right)$, $(0, -9)$. **3.** (a) $(8, 0)$, $(0, 6)$ (b) $(4, 0)$, $(0, -10)$ (c) $(-7, 0)$, $(0, -2)$ (d) $\left(\frac{3}{2}, 0\right)$, $\left(0, -\frac{9}{5}\right)$. **4.** (a) $\left(\frac{15}{4}, 0\right)$, $\left(0, \frac{5}{2}\right)$ (b) $\left(\frac{3}{2}, 0\right)$, $(0, -4)$ (c) In $Ax + By = C$, setting $y = 0$ deletes a whole term and leaves $Ax = C$, one division away from the answer. In $y = mx + b$ you must first move the b across and then divide, so each intercept takes an extra step. **5.** (a) x-int $(3, 0)$, y-int $(0, -6)$ (b) y-int $(0, -5)$; y rises 3 for every step of 2 in x, so $y = \frac{3}{2}x - 5$ and the x-int is $\left(\frac{10}{3}, 0\right)$. **6.** (a) x-int $(-4, 0)$, y-int $(0, 3)$; $y = \frac{3}{4}x + 3$ (b) x-int $(2, 0)$, y-int $(0, -4)$; $y = 2x - 4$. **7.** (a) y-int $(0, 5)$; no x-intercept — the line is horizontal 5 units above the x-axis and stays there, so it never touches it (b) x-int $(-3, 0)$; no y-intercept — the line is vertical, 3 units to the left of the y-axis, always parallel to it (c) both intercepts are $(0, 0)$; the line passes through the origin (d) y-int $(0, 0)$; the line *is* the x-axis, so every point on it is an x-intercept. **8.** (a) $(0, 60)$; the tank starts with 60 L (b) $(15, 0)$; the tank is empty after 15 min (c) a tank cannot hold a negative volume — draining stops at $t = 15$, so the largest sensible value is $t = 15$ (d) the V-intercept gives the tank's size (60 L); the t-intercept gives the draining time (15 min). **9.** (a) C-int $(0, 100)$, t-int $(12.5, 0)$ (b) the battery starts at 100%, and it is flat after 12.5 min of flight; 100 is not a multiple of 8, so the battery runs out partway through the 13th minute. **10.** (a) C-int $(0, 4)$, d-int $(-1.6, 0)$ (b) $(0, 4)$ is meaningful: a ride of 0 km still costs the \$4 pick-up fee. $(-1.6, 0)$ is not: a taxi cannot drive -1.6 km, and the fare never actually reaches \$0. **11.** The student used the y-intercept rule. An x-intercept sits on the x-axis, where $y = 0$, so substitute $y = 0$: $0 = 2x - 8$ gives $x = 4$. Intercepts: x-int $(4, 0)$, y-int $(0, -8)$. **12.** Substituting $x = 0$ gives $5y = 15$, so $y = 3$ and the y-intercept is $(0, 3)$, not 5. The x-intercept: $3x = 15$, so $(5, 0)$. **13.** (a) $y = \frac{1}{2}x - 2$ (b) $y = \frac{5}{3}x + 5$ (c) $y = -x + 6$. **14.** $0 = 4m + 8$, so $m = -2$ and the line is $y = -2x + 8$. **15.** $0 = 3(-2) + b$, so $b = 6$ and the line is $y = 3x + 6$. **16.** (a) $x = 0$ gives $ky = 12$, so $y = \frac{12}{k}$; setting $\frac{12}{k} = 3$ gives $k = 4$ (b) $y = 0$ gives $2x = 12$, so the x-intercept is $(6, 0)$ for *every* value of k — changing k tilts the line about the fixed point $(6, 0)$ (c) no value of k works; by part (b) the x-intercept is always 6, so it can never equal 3. **17.** (a) The legs are 6 and $|b|$, so $\frac{1}{2}(6)|b| = 12$ gives $|b| = 4$ and $b = 4$ or $b = -4$; the lines are $y = -\frac{2}{3}x + 4$ and $y = \frac{2}{3}x - 4$ (b) the two lines are reflections of each other in the x-axis, so their triangles have the same base 6 and the same height 4 — only the sign of b, not its size, changes. **18.** No. A vertical line $x = c$ always crosses the x-axis, and a horizontal line $y = c$ always crosses the y-axis; a slanted line is never parallel to either axis, so it must cross both. A line that missed both axes would have to be parallel to both at once, which is impossible. **19.** Every point on the x-axis has $y = 0$, so every point of $y = 0$ is an x-intercept — there are infinitely many of them, one at $(a, 0)$ for each number a. The point $(0, 0)$ is only the single one that also happens to be the y-intercept.

Lesson 5 — Extra Practice

Domain and Range of a Relation, Part 1: Reading From a Graph

Notation used in this set (a step beyond Lesson 5; Lesson 6 explains it further). For a **discrete** relation (separate dots), list the values inside braces: $D = \{-3, 0, 4\}$. For a **continuous** relation (an unbroken line or curve), use an inequality: $-3 \le x \le 4$. A **solid** dot at the end of a graph means that value *is* included, so use $\le$ or $\ge$; an **open** circle means the value is *not* included, so use $<$ or $>$; an **arrow** means the graph never ends on that side.

Part A — Solid Dots, Open Circles, and Arrows

1. Complete each sentence.

(a) A **solid** dot at the end of a graph means that x-value _____ included, so the inequality uses the symbol _____ or _____.

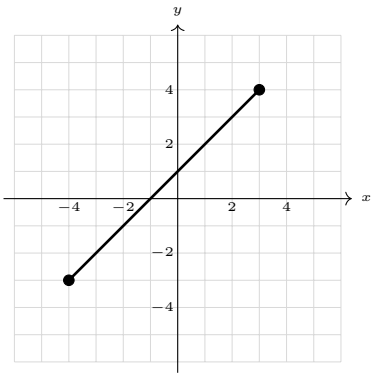
(b) An **open** circle at the end of a graph means that x-value _____ included, so the inequality uses _____ or _____.

(c) An **arrow** at the end of a graph means the graph _____, so there is no endpoint on that side.

(d) A relation drawn as separate dots is _____, and its domain is written as a _____ in braces.

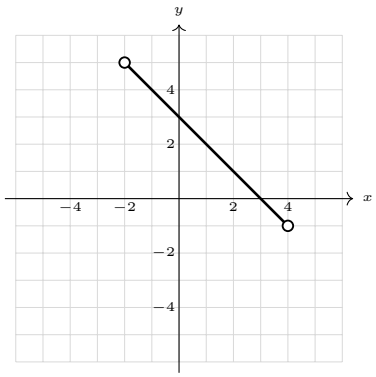
2. Each graph below is a line segment with **both** endpoints marked (solid or open). Read the endpoints carefully and state the domain and range.

(a)



D: _____
R: _____

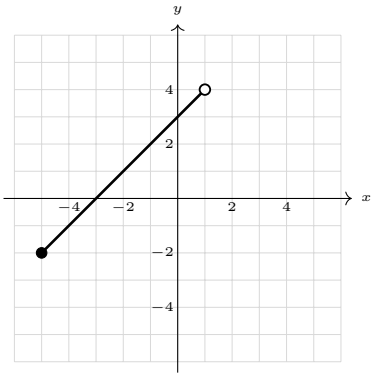
(b)



D: _____
R: _____

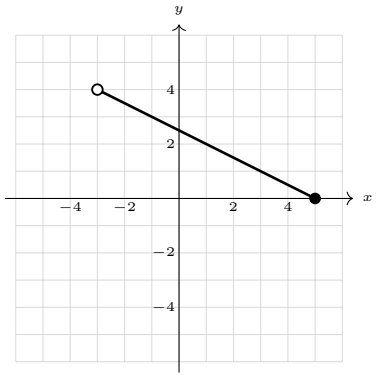
3. Each segment below has **one** solid end and **one** open end. State the domain and range, using $\le$ and $<$ in the right places.

(a)



D: _____
R: _____

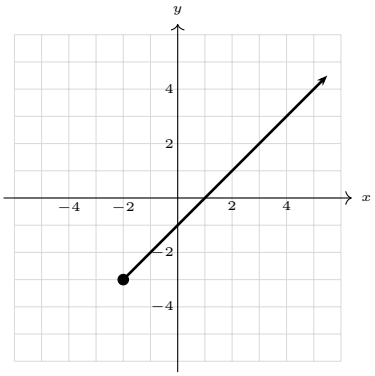
(b)



D: _____
R: _____

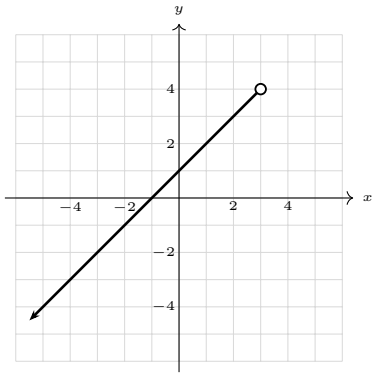
4. Each graph below is a **ray**: it starts at one endpoint and continues forever in the direction of the arrow. State the domain and range.

(a)



D: _____
R: _____

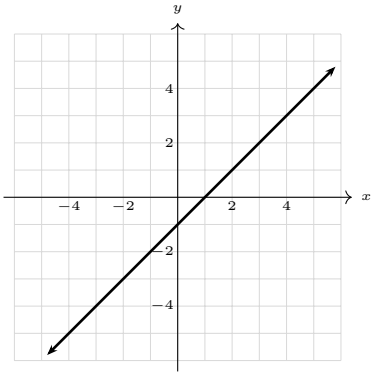
(b)



D: _____
R: _____

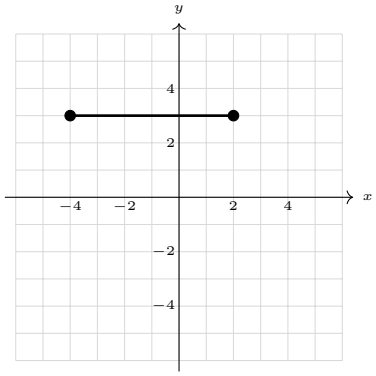
5. State the domain and range of each graph. Graph (b) is a horizontal segment — think hard about how many different y-values it actually uses.

(a)



D: _____
R: _____

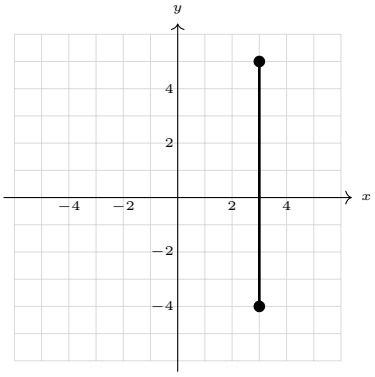
(b)



D: _____
R: _____

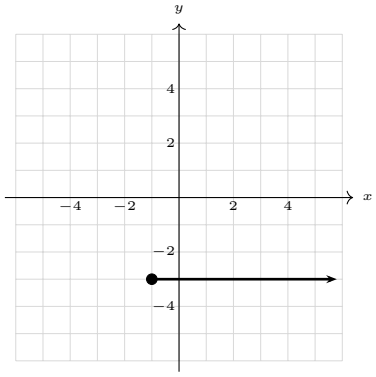
6. Graph (a) is a vertical segment and graph (b) is a horizontal ray. One of these has a domain that is a single value; the other has a range that is a single value. State both.

(a)



D: _____
R: _____

(b)

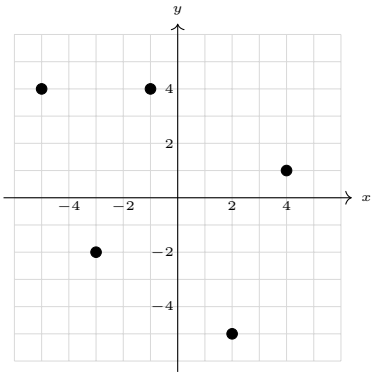


D: _____
R: _____

Part B — A List or an Interval? Discrete, Continuous, and Gaps

7. The graph in (a) is a set of five separate dots. State its domain and range as **lists** in braces, and watch for a repeated y-value. Then answer (b).

(a)



D: _____

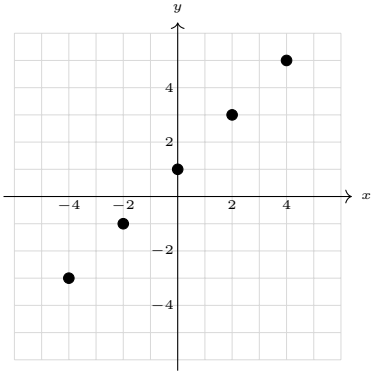
R: _____

(b) The domain of this relation has five elements but the range does not. Explain why, and state how many elements the range has.

(c) Is $x = 0$ in the domain of this relation? Explain.

8. Graphs (a) and (b) below use the *same* five lattice points, but (b) joins them into an unbroken segment. State the domain and range of each, then answer (c).

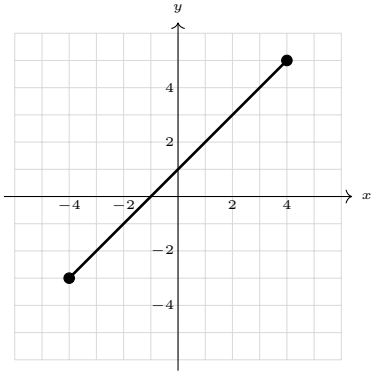
(a) Five separate dots.



D: _____

R: _____

(b) The same points, joined.



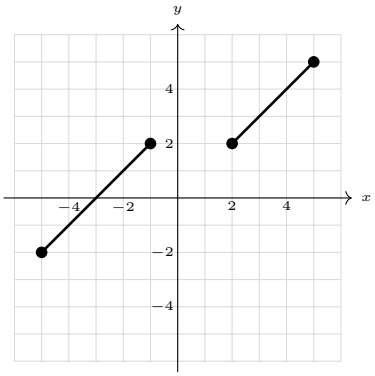
D: _____

R: _____

(c) Name two x-values that belong to the domain of graph (b) but **not** to the domain of graph (a). Then explain, in one sentence, why a discrete relation gets a list and a continuous one gets an inequality.

9. This graph is made of two separate pieces, all four endpoints solid. State the domain and the range. 10. This graph is also made of two pieces — notice the open circle at the right end of the lower piece. State the domain and the range.

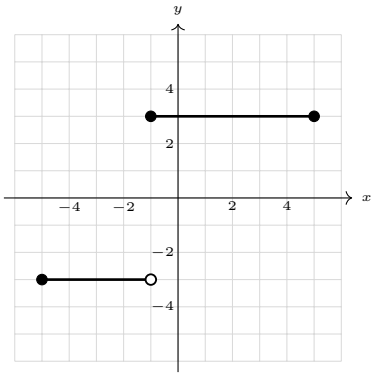
9.



D: _____

R: _____

10.



D: _____

R: _____

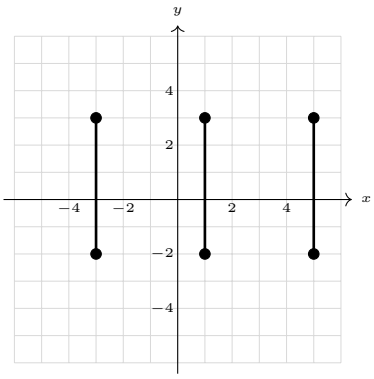
9(b) One of the two answers in question 9 has a gap in it and the other does not. Which one, and why? 10(b) In question 10 the domain is an unbroken interval but the range is a short list. Explain how that can happen.

9(b)

10(b)

11. This relation is three separate vertical segments, each with solid ends. State the domain and range. 12. This relation is one segment *plus* one lone dot off to the right. State the domain and range; each answer will need an interval **and** a single value.

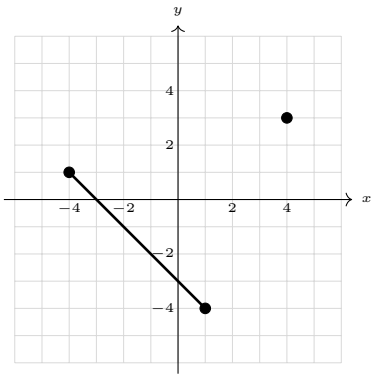
11.



D: _____

R: _____

12.



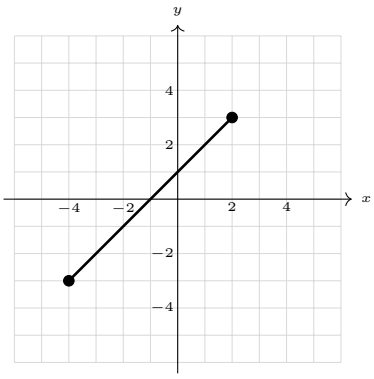
D: _____

R: _____

Part C — Error Analysis, Sketching, and Reasoning

13. **Error analysis.** Kayla read graph 13 and wrote "D: $-4 \le x \le 2$ R: $-3 \le x \le 3$." One of her answers is right and one has a mistake — name it and correct it. 14. **Error analysis.** Owen read graph 14 and wrote "D: $-3 \le x \le 4$ R: $-4 \le y \le 3$." His numbers are right but two of his symbols are wrong. Correct both answers.

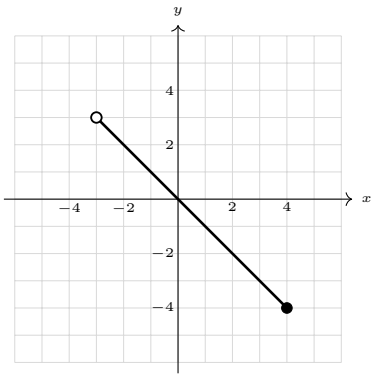
13.



Mistake: _____

Corrected: _____

14.



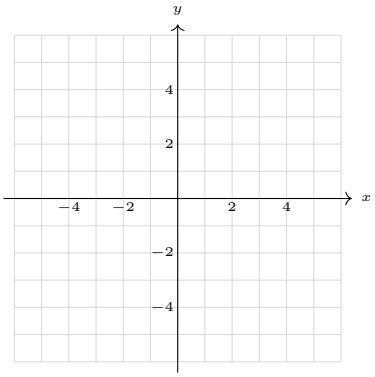
D: _____

R: _____

Why does one endpoint change *both*?

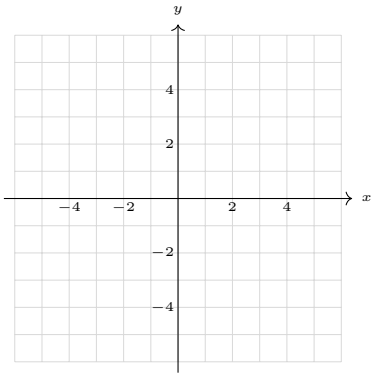
15. **Same domain, different range.** On the two grids below, sketch two *different* graphs that both have domain $-2 \le x \le 4$. Make their ranges different, and state each range underneath.

Graph 1



R: _____

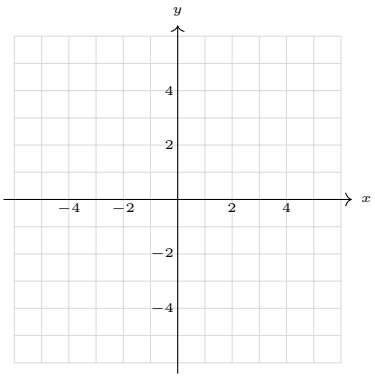
Graph 2



R: _____

Answer in a sentence: can two different graphs have the same domain but different ranges?

16. **Work backwards.** On the grid below, sketch a single line segment whose domain is $-5 \le x < 2$ and whose range is $-2 < y \le 5$. Label both endpoints with their coordinates, and make sure the solid and open ends are drawn correctly.



(a) Which endpoint is solid, and how do you know?

(b) Is your segment increasing or decreasing? Explain why it *has* to be.

17. **Reasoning.** Is it possible for an **unbroken** (continuous) graph to have a range of exactly two values, $R = \{2, 5\}$, and nothing in between? Explain your thinking. If it is impossible, describe the kind of graph that *does* have $R = \{2, 5\}$.

18. **Stretch.** A relation is one single line segment. Its domain is $-4 \le x \le 2$ and its range is $1 \le y \le 7$. There are exactly **two** different segments this could be. Give the two endpoints of each, and explain why there are exactly two and not more.

Answer Key. 1. (a) *is* included; $\le$ or $\ge$ (b) is *not* included; $<$ or $>$ (c) continues forever in that direction (d) discrete; list. **2.** (a) D: $-4 \le x \le 3$; R: $-3 \le y \le 4$ (b) D: $-2 < x < 4$; R: $-1 < y < 5$. **3.** (a) D: $-5 \le x < 1$; R: $-2 \le y < 4$ (b) D: $-3 < x \le 5$; R: $0 \le y < 4$. **4.** (a) D: $x \ge -2$; R: $y \ge -3$ (b) D: $x < 3$; R: $y < 4$. **5.** (a) D: all real numbers; R: all real numbers (b) D: $-4 \le x \le 2$; $R = \{3\}$ (every point on the segment has $y = 3$). **6.** (a) $D = \{3\}$; R: $-4 \le y \le 5$ (b) D: $x \ge -1$; $R = \{-3\}$. **7.** (a) $D = \{-5, -3, -1, 2, 4\}$; $R = \{-5, -2, 1, 4\}$ (b) the y-value 4 occurs twice (at $x = -5$ and $x = -1$) and a set lists each value once, so the range has 4 elements (c) no; no dot is plotted at $x = 0$, and only plotted dots belong to a discrete relation. **8.** (a) $D = \{-4, -2, 0, 2, 4\}$; $R = \{-3, -1, 1, 3, 5\}$ (b) D: $-4 \le x \le 4$; R: $-3 \le y \le 5$ (c) any two of -3, -1, 1, 3, 2.5, $-0.4, \ldots$; a discrete graph owns only the values it plots, so they can be listed, while a continuous graph owns every value between its ends. **9.** D: $-5 \le x \le -1$ or $2 \le x \le 5$; R: $-2 \le y \le 5$. **9(b)** the domain has the gap (no points between $x = -1$ and $x = 2$); the range has none because the first piece supplies y from -2 to 2 and the second picks up at $y = 2$ and runs to 5, so the two stretches meet at $y = 2$ with nothing missing. **10.** D: $-5 \le x \le 5$; $R = \{-3, 3\}$. **10(b)** the open circle at $(-1, -3)$ is filled in by the solid point at $(-1, 3)$, so every x from -5 to 5 is used, but the two flat pieces give only the heights -3 and 3. **11.** $D = \{-3, 1, 5\}$; R: $-2 \le y \le 3$. **12.** D: $-4 \le x \le 1$ or $x = 4$; R: $-4 \le y \le 1$ or $y = 3$. **13.** The range is wrong: it must use y, not x. Corrected R: $-3 \le y \le 3$. (Her domain, $-4 \le x \le 2$, is correct.) **14.** D: $-3 < x \le 4$; R: $-4 \le y < 3$. The open circle at $(-3, 3)$ is the culprit: that one endpoint supplies both an x-value and a y-value, so excluding the point excludes $x = -3$ from the domain *and* $y = 3$ from the range. **15.** Many answers. Example: Graph 1 is the segment $(-2, 0)$ to $(4, 3)$, R: $0 \le y \le 3$; Graph 2 is the horizontal segment $(-2, -5)$ to $(4, -5)$, $R = \{-5\}$. Yes — the domain only says how far left and right a graph reaches, not how high or low it goes. **16.** The segment from $(-5, 5)$ to $(2, -2)$, solid at $(-5, 5)$ and open at $(2, -2)$. (a) The left end is solid, since $-5 \le x$ includes -5 and $y \le 5$ includes 5. (b) Decreasing: the included x (-5) must pair with the included y (5), so the left end is high and the right end is low. **17.** Impossible. An unbroken graph cannot jump from a height of 2 to a height of 5 without passing through every height in between, so its range would have to include all of $2 \le y \le 5$. A graph with $R = \{2, 5\}$ must be broken — for example two separate horizontal segments, one at $y = 2$ and one at $y = 5$, which is exactly how question 10 is built. **18.** Endpoints $(-4, 1)$ and $(2, 7)$ (increasing), or $(-4, 7)$ and $(2, 1)$ (decreasing). The two ends must sit at $x = -4$ and $x = 2$ with heights 1 and 7; the only choice left is which height goes with which end, so there are exactly two.

Lesson 6 — Extra Practice

Domain and Range of a Relation, Part 2

Notation used in this set. When a relation uses only a few separate values, list them in braces: $D = \{2, 4, 6\}$. When it uses every value across a stretch, use an inequality: $x \geq 0$, or $2 \leq x \leq 9$ when the stretch has two ends. Use $\leq$ and $\geq$ when the end value *is* allowed and $<$ and $>$ when it is not.

Part A — Challenge: Domain and Range From an Equation

1. The equation $y = 2x + 1$ has no picture attached to it, so the domain must be decided by asking: *which numbers am I allowed to substitute for x?*

(a) Complete the table.

x	-4	-1	0	3.5	10
y					

(b) Is there any number at all that you are *not* allowed to substitute for x? Explain.

(c) State the domain of $y = 2x + 1$.

(d) Find the x-value that produces $y = 100$. Use your answer to explain why the range of $y = 2x + 1$ is also every real number.

2. State the domain and range of each linear equation. You may substitute a few numbers first if it helps.

(a) $y = -4x + 9$ (b) $y = \frac{1}{3}x - 2$ (c) $y = 7 - x$

3. The equation $y = x^2$ behaves differently. Work through the parts below carefully — this is the one to get right.

(a) Complete the table.

x	-3	-2	-1	0	1	2	3
y							

(b) Is there any number you are *not* allowed to square? State the domain of $y = x^2$.

(c) Look down your y-row. State the range of $y = x^2$, then **justify** it: explain why no choice of x could ever make y negative.

(d) Does the ordered pair $(-6, 36)$ belong to $y = x^2$? Does $(5, -25)$? Explain each.

4. Each equation below is $y = x^2$ with something done to it. State the domain and the range of each, and give a one-line reason for the range.

(a) $y = x^2 + 3$ (b) $y = x^2 - 5$ (c) $y = -x^2$

5. The equation $y = \sqrt{x}$ restricts the **domain** instead of the range.

(a) Complete the table.

x	0	1	4	9	25
y					

(b) Try $x = -4$. What goes wrong? Explain in one sentence.

(c) State the domain of $y = \sqrt{x}$.

(d) State the range of $y = \sqrt{x}$, and explain why a square root is never negative.

6. State the domain and range of each relation below.

(a) $y = \sqrt{x - 2}$ (b) $y = \sqrt{x} + 6$ (c) $y = 5$ (d) $x = -2$

Part B — From a Table and From Ordered Pairs

7. Write the domain and range of each set of ordered pairs using set notation. Watch for repeats.

(a) $(-8, 3),\ (-4, 3),\ (0, -5),\ (4, 3),\ (8, -5)$ (b) $(6, -2),\ (6, 0),\ (6, 7)$ (c) $(-1, -1),\ (0, 0),\ (1, 1),\ (2, 2)$

8. Write the domain and range of each table using set notation.

(a)

x	-3	-1	1	3	5
y	12	4	0	4	12

(b)

x	0.5	1.5	2.5	3.5
y	-7	-7	-7	-7

9. A relation has domain $D = \{2, 4, 6, 8\}$ and range $R = \{5\}$.

(a) Write every ordered pair in this relation.

(b) Could this relation have five *different* ordered pairs? Explain.

(c) A second relation has $D = \{1, 2\}$ and $R = \{3, 4\}$ and exactly two ordered pairs. Write two different relations that fit.

Part C — Restricted Domains in Context

10. A school play sells tickets for \$8 each. The theatre holds 240 seats, and the show is not oversold. The revenue, R dollars, from selling n tickets is $R = 8n$.

(a) Explain why n must be a whole number, then state the domain in set notation.

(b) State the range in set notation. (Write the first three values, then dots, then the last value.)

(c) How many values are in the domain? Be careful — do not forget $n = 0$.

(d) Does (150, 1200) belong to this relation? Justify with *both* the equation and the restriction.

(e) Does (37.5, 300) belong to this relation? The arithmetic works — so what is the problem?

11. A water tank holds 600 L. A valve is opened and the tank drains steadily at 40 L per minute until it is empty. The volume left after t minutes is $V = 600 - 40t$.

(a) How many minutes does the tank take to empty? Show your calculation.

(b) State the domain, with units, and say why it is an inequality rather than a list.

(c) State the range, with units.

(d) Substitute $t = 20$ into $V = 600 - 40t$. Explain why the answer proves $t = 20$ is outside the domain.

(e) The domain in question 10 was a list and the domain here is an inequality. Explain the difference between the two situations that causes this.

12. A rectangle has a perimeter of 40 cm. One side is x cm long and the side next to it is y cm long.

(a) Use the perimeter to show that $y = 20 - x$.

(b) Explain why the domain must use $x > 0$ and not $x \geq 0$.

(c) Explain why x must also stay below 20.

(d) State the domain and the range of this relation.

(e) Give one value of x that is in the domain and one that is not, and justify each choice.

13. A taxi charges a \$3.50 flat fee plus \$1.25 for each kilometre driven. The longest trip the company will take is 30 km. The cost of a d-kilometre trip is $C = 3.50 + 1.25d$.

(a) State the domain, with units.

(b) State the range, with units. Show the calculation for the largest value.

(c) Is this relation discrete or continuous? Explain.

(d) Why does the range not start at \$0?

Part D — Words, Set Notation, and Number Lines

14. Write each description using an inequality or a list in braces, whichever fits.

(a) All numbers greater than -3.

(b) All numbers from 2 to 9, including both ends.

(c) All numbers less than or equal to 0.

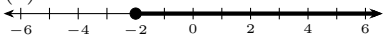
(d) All numbers between -5 and 1, including neither end.

(e) Every whole number from 1 to 6.

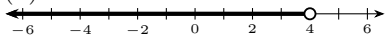
(f) All numbers that are at least -1 but under 7.

15. Write the inequality shown by each number line. Look closely at whether each dot is solid or open.

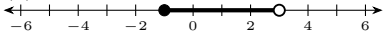
(a)



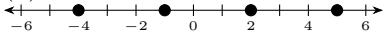
(b)



(c)

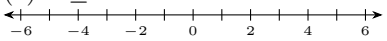


(d)

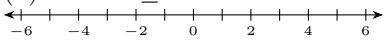


16. Draw each set on the number line provided. Use a solid dot for an included end, an open circle for an excluded end, and an arrow where the set never stops.

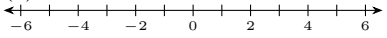
(a) $x \leq -4$



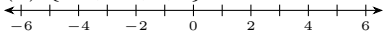
(b) $-2 < x \leq 5$



(c) $x > 0$



(d) $\{-5, -2, 1, 4\}$

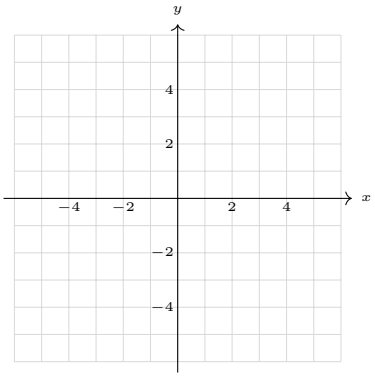


Part E — Error Analysis, Working Backwards, and a Stretch

17. **Error analysis.** A bakery sells muffins for \$2 each and bakes at most 50 per day. Marcus writes the domain as "$0 \leq n \leq 50$." Test his answer with $n = 12.7$, explain why his notation is wrong for this situation, and write the correct domain.

18. **Error analysis.** Tia says: "The domain of $y = x^2$ is $x \geq 0$, because the range is $y \geq 0$." Use the ordered pair $(-3, 9)$ to show she is wrong, name the idea she has mixed up, and write the correct domain and range.

19. **Work backwards.** Describe a relation whose domain is $x \geq 3$ and whose range is $y \leq 0$. Sketch it on the grid, label the starting point, and then write an equation for it.



(a) Where must the graph start, and should that point be solid or open? Explain.

(b) Which way must it travel from there, and why?

(c) Equation: _____

20. **Stretch: when the domain and the range are the same set.**

(a) Write three ordered pairs whose domain *and* range are both $\{1, 2, 3\}$, without using $(1, 1)$, $(2, 2)$ or $(3, 3)$.

(b) Give a linear equation whose domain and range are both every real number.

(c) State the domain and the range of $y = \sqrt{x}$ in words. Why is it fair to say they are the same set even though one is written with x and one with y?

(d) Is there a relation whose domain is $\{5\}$ and whose range is also $\{5\}$? If so, write it.

Answer Key. 1. (a) -7, -1, 1, 8, 21 (b) no — doubling and adding 1 works on every number (c) D: all real numbers (d) $2x + 1 = 100$ gives $x = 49.5$; any target y can be hit by $x = \frac{y-1}{2}$, so R: all real numbers. **2.** (a)–(c) all three have D: all real numbers and R: all real numbers (every slanted line stretches forever left, right, up and down). **3.** (a) 9, 4, 1, 0, 1, 4, 9 (b) every number can be squared, so D: all real numbers (c) R: $y \geq 0$; a positive times itself is positive and a negative times itself is also positive, so x^2 is never negative, and $x = 0$ gives the lowest value $y = 0$ (d) $(-6, 36)$ yes, since $(-6)^2 = 36$; $(5, -25)$ no, since $5^2 = 25$, not -25. **4.** All three have D: all real numbers. (a) R: $y \geq 3$ ($x^2 \geq 0$, so adding 3 gives at least 3) (b) R: $y \geq -5$ ($x^2 \geq 0$, so subtracting 5 gives at least -5) (c) R: $y \leq 0$ ($x^2 \geq 0$, so its opposite is never above 0). **5.** (a) 0, 1, 2, 3, 5 (b) no number times itself gives -4, so $\sqrt{-4}$ is not a real number (c) D: $x \geq 0$ (d) R: $y \geq 0$; the square root symbol asks for the non-negative root, and 0 is reached at $x = 0$. **6.** (a) D: $x \geq 2$ (you need $x - 2 \geq 0$); R: $y \geq 0$ (b) D: $x \geq 0$; R: $y \geq 6$ (c) D: all real numbers; $R = \{5\}$ (d) $D = \{-2\}$; R: all real numbers. **7.** (a) $D = \{-8, -4, 0, 4, 8\}$, $R = \{-5, 3\}$ (b) $D = \{6\}$, $R = \{-2, 0, 7\}$ (c) $D = \{-1, 0, 1, 2\}$, $R = \{-1, 0, 1, 2\}$. **8.** (a) $D = \{-3, -1, 1, 3, 5\}$, $R = \{0, 4, 12\}$ (b) $D = \{0.5, 1.5, 2.5, 3.5\}$, $R = \{-7\}$. **9.** (a) $(2, 5)$, $(4, 5)$, $(6, 5)$, $(8, 5)$ (b) no — every y must be 5 and there are only four x-values, so only four different pairs exist; a repeated pair is not a new one (c) e.g. $(1, 3), (2, 4)$ or $(1, 4), (2, 3)$. **10.** (a) tickets cannot be split, so n counts whole tickets; $D = \{0, 1, 2, 3, \ldots, 240\}$ (b) $R = \{0, 8, 16, \ldots, 1920\}$, since $8 \times 240 = 1920$ (c) 241 values (the numbers 1 to 240, plus 0) (d) yes — $8 \times 150 = 1200$ *and* 150 is a whole number no bigger than 240 (e) no — the arithmetic $8 \times 37.5 = 300$ is fine, but 37.5 is not a whole number of tickets, so it is not in the domain. **11.** (a) $600 \div 40 = 15$ minutes (b) D: $0 \leq t \leq 15$ (minutes); time runs smoothly, so every value in between, such as 7.3 min, really happens — there is nothing to list (c) R: $0 \leq V \leq 600$ (litres) (d) $600 - 40(20) = -200$; a tank cannot hold -200 L, and it was already empty at $t = 15$, so $t = 20$ is outside the domain (e) tickets come in whole units and can only be counted, while time (and the volume it produces) can take any value in a stretch. **12.** (a) $2x + 2y = 40$, so $x + y = 20$ and $y = 20 - x$ (b) a side of length 0 gives no rectangle, so x must be strictly greater than 0 (c) if $x = 20$ then $y = 0$, and if $x > 20$ then y is negative — neither is a rectangle (d) D: $0 < x < 20$; R: $0 < y < 20$ (e) e.g. $x = 6.5$ is in the domain (it gives $y = 13.5$, a real rectangle 6.5 cm by 13.5 cm), and $x = 20$ is not (it gives $y = 0$). **13.** (a) D: $0 \leq d \leq 30$ (km) (b) $C = 3.50 + 1.25(30) = 3.50 + 37.50 = 41.00$, so R: $3.50 \leq C \leq 41.00$ (dollars) (c) continuous — a trip can be any distance, such as 12.4 km, so the values cannot be listed (d) the \$3.50 flat fee is charged before any distance is driven, so the cheapest possible trip still costs \$3.50. **14.** (a) $x > -3$ (b) $2 \leq x \leq 9$ (c) $x \leq 0$ (d) $-5 < x < 1$ (e) $\{1, 2, 3, 4, 5, 6\}$ (f) $-1 \leq x < 7$. **15.** (a) $x \geq -2$ (b) $x < 4$ (c) $-1 \leq x < 3$ (d) $\{-4, -1, 2, 5\}$. **16.** (a) solid dot at -4, shaded left with an arrow (b) open circle at -2, solid dot at 5, shaded between them (c) open circle at 0, shaded right with an arrow (d) four solid dots, at -5, -2, 1 and 4, with no shading between them. **17.** $0 \leq n \leq 50$ allows $n = 12.7$, and the bakery cannot sell 12.7 muffins; the situation is discrete, so it needs a list, not an inequality. Correct domain: $D = \{0, 1, 2, \ldots, 50\}$ — the whole numbers from 0 to 50. **18.** $(-3)^2 = 9$, so $(-3, 9)$ is on the graph and -3 is in the domain, which breaks her rule. She has mixed up domain (inputs) with range (outputs): squaring restricts what comes *out*, not what may go *in*. Correct: D: all real numbers; R: $y \geq 0$. **19.** A ray beginning at the solid point $(3, 0)$ and travelling down and to the right forever. (a) It starts at $(3, 0)$ with a **solid** dot, because $x \geq 3$ includes 3 and $y \leq 0$ includes 0. (b) It must go right (so the x-values grow past 3) and down (so the y-values drop below 0). (c) e.g. $y = 3 - x$ for $x \geq 3$; check $x = 3 \Rightarrow y = 0$ and $x = 10 \Rightarrow y = -7$. ($y = -\sqrt{x - 3}$ also works.) **20.** (a) e.g. $(1, 2)$, $(2, 3)$, $(3, 1)$ — the x's are $1, 2, 3$ and the y's are $2, 3, 1$, and both sets are $\{1, 2, 3\}$ (b) any slanted line, e.g. $y = x$ or $y = 2x - 7$ (c) domain: all numbers that are zero or greater; range: all numbers that are zero or greater — the letter only records whether a value is an input or an output, so both describe the identical collection of numbers (d) yes — the single ordered pair $(5, 5)$, whose graph is one dot.

Lesson 7 — Extra Practice

Using Technology to Graph and Explore Relations

A word used in this set: the **slope** of a line $y = \square x + \square$ is the number multiplying x — it tells how steep the line is and whether it rises or falls. (Slope gets its own lessons in Unit 6; here you only need that idea.)

Part A — Choosing a Window Before You Graph

Every question in this set can be answered with pencil and paper. Technology draws the picture, but *you* decide what the picture should contain — and whether to trust what it shows.

1. Chantelle wants to graph $y = 2x + 30$ and see **both** intercepts on one screen. Four windows have been proposed.

W1: $x : [-10, 10]$, $y : [-10, 10]$

W2: $x : [-20, 5]$, $y : [-10, 40]$

W3: $x : [-20, 20]$, $y : [-5, 15]$

W4: $x : [0, 20]$, $y : [0, 60]$

(a) Determine both intercepts algebraically first, before looking at any window.

(b) Which window is the only one that shows both intercepts? Explain what makes it work.

(c) For each of the other three windows, name the intercept that is missing and explain, using the window numbers, why it is missing.

(d) W1 is a special case: almost none of the line appears at all. Find the one point of the line that lies on the edge of W1, and use it to explain why the screen looks nearly blank.

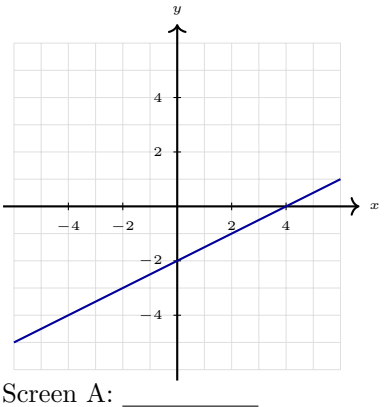
2. For each relation, determine both intercepts **without graphing**, then state a window that shows both with room to spare on every side.

(a) $y = 4x - 36$ (b) $y = -5x - 60$ (c) $y = 0.2x + 3$ (d) $y = 100 - 20x$

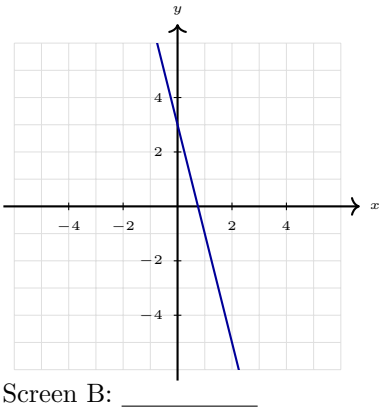
3. **Predict first.** For each equation, state three things *before* any graph is drawn: the sign of the slope (rising or falling left to right), whether the line is steeper or shallower than $y = x$, and the y-intercept.

(a) $y = -4x + 3$ (b) $y = 0.5x - 2$ (c) $y = -0.5x + 4$

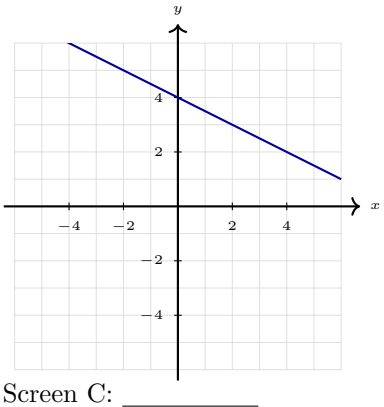
4. Now check your predictions. Each screen below uses the window $x : [-6, 6]$, $y : [-6, 6]$. Match each equation from Question 3 to Screen A, B, or C, and write down the one feature that settled each match.



Screen A: _____
Reason:



Screen B: _____
Reason:



Screen C: _____
Reason:

5. A water tank holds 200 L and drains at a steady rate. The volume V, in litres, left after m minutes is $V = 200 - 8m$ (until the tank is empty).

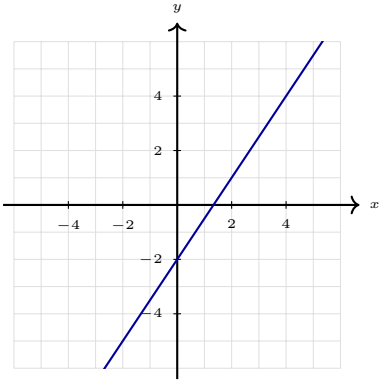
(a) Determine the V-intercept and the m-intercept.

(b) State a window suited to this situation, and say why you chose those four numbers.

(c) Explain why the standard window would show a completely blank screen here.

Part B — Reading a Screen Carefully

6. The screen below shows $y = 1.5x - 2$ in the window $x : [-6, 6]$, $y : [-6, 6]$, with gridlines 1 unit apart.



(a) Estimate the x-intercept from the screen. Between which two gridlines does it fall?

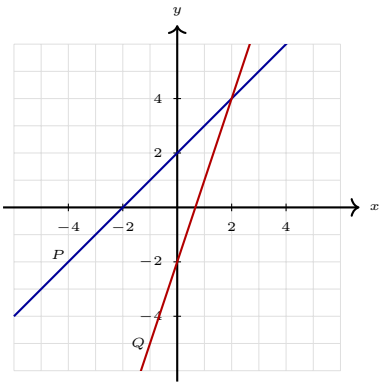
(b) Estimate the value of y when $x = 3$, and the value of x when $y = 5$.

(c) Determine the exact x-intercept algebraically. How close was your estimate in (a)?

(d) With gridlines 1 unit apart, how precise can a reading off this screen honestly be? Answer with a number, and justify it.

(e) Of all the readings in parts (a) and (b), one can be stated with far more confidence than the others. Which one, and what makes it different?

7. Two relations, P: $y = x + 2$ and Q: $y = 3x - 2$, are graphed on the same axes below.



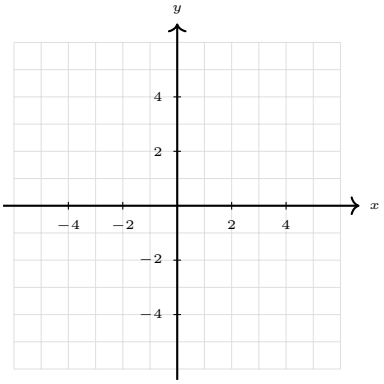
(a) Read the point where the two lines cross. Then verify it by substituting into both equations.

(b) Which relation grows faster as x increases? Name the number in the equation that tells you.

(c) Which relation has the greater y-value at $x = 5$? At $x = -2$? Show the arithmetic.

(d) For which values of x is Q above P? Answer with an inequality, and explain how the crossing point gives you the boundary.

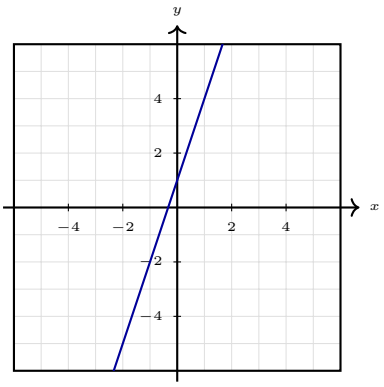
8. On the grid below, graph both $y = 2x - 1$ and $y = -x + 5$ by hand, labelling each line.



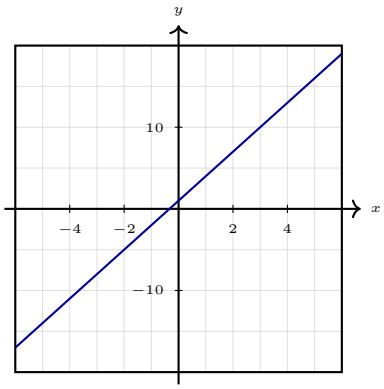
(a) State the coordinates of the point where the two lines cross, then check the point in both equations.

(b) One of these relations grows as x increases and the other shrinks. Which is which, and how does the equation show it before you graph anything?

9. **Same line, two screens.** Both screens below show the relation $y = 3x + 1$. Screen 1 uses $x : [-6, 6]$, $y : [-6, 6]$. Screen 2 uses $x : [-6, 6]$, $y : [-20, 20]$.



Screen 1



Screen 2

(a) The two pictures look different. Give two pieces of evidence from the screens that they really are the same relation.

(b) Which screen makes the line look steeper? Explain what changed, given that the slope is 3 on both screens.

(c) On Screen 2, one gridline square is 1 unit wide but 5 units tall. Explain in your own words why that squashes the picture of the line.

(d) A classmate says "a slope of 3 always looks like a 60° climb." Use these screens to explain why that claim cannot be relied on.

10. Build a table of values for each relation, then examine the **differences** between consecutive y-values.

(a) $y = 3x - 4$ for $x = -2, -1, 0, 1, 2, 3$. List the y-values, then the five differences.

(b) $y = -2.5x + 6$ for $x = 0, 2, 4, 6, 8$. List the y-values, then the four differences.

(c) In each table the differences are constant. Explain the connection between that constant, the number multiplying x in the equation, and the size of the step taken in x.

11. State whether each sentence is **True** or **False**.

(a) Changing the window changes where a relation's intercepts actually are.

(b) A line can look steeper on one screen than on another without its slope changing.

(c) If nothing appears in the standard window, the relation has no graph at all.

(d) Tracing along a line usually reports approximate x-values rather than exact ones.

(e) Two lines with different slopes must cross somewhere, even if the crossing is off screen.

(f) A window that shows both intercepts must show every point of the relation.

Part C — Error Analysis, Reasoning, and a Stretch

12. Rowan graphs $y = 0.5x + 9$ and $y = x + 2$ in the standard window, $x : [-10, 10]$, $y : [-10, 10]$. Both lines appear, but they do not meet anywhere on the screen. Rowan concludes: "these two lines never meet."

(a) Explain precisely what is wrong with Rowan's reasoning.

(b) Determine algebraically where the two lines actually meet.

(c) State a window that would show the meeting point comfortably.

(d) What feature of the two *equations* could have told Rowan, before graphing, that these lines were certain to meet somewhere?

13. Priya uses the trace feature on $y = 3x - 5$ and stops as close to the x-axis as the cursor will land. The calculator reports $x = 1.7021277$, $y = 0.1063830$. Priya decides the calculator is broken.

(a) Determine the exact x-intercept of $y = 3x - 5$.

(b) Substitute $x = 1.7021277$ into the equation. Is the reported y-value correct?

(c) Explain why trace lands on such an awkward x-value.

(d) Name two better ways to get the intercept.

14. Marcus says: "If I cannot see an x-intercept in the standard window, the relation does not have one."

(a) Give an equation of a line that is a counterexample, and state its x-intercept to prove the point.

(b) Is it true that *every* line has exactly one x-intercept? Consider $y = 4$, $y = 0$, and $y = 2x + 7$, and state the full rule you arrive at.

15. Dominic graphs $y = 12x$ in the window $x : [0, 10]$, $y : [0, 10]$.

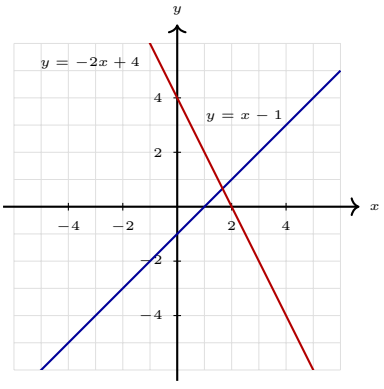
(a) Determine the exact x-value at which the line leaves the top of this window.

(b) Describe what Dominic sees, and explain why the line looks almost vertical.

(c) Give a window that shows the relation usefully for x between 0 and 10.

(d) In the window you chose in part (c), would the line still look almost vertical? Explain what happened to the appearance of the slope, and whether the slope itself changed.

16. **Stretch.** The screen below shows $y = x - 1$ and $y = -2x + 4$ in the window $x : [-6, 6]$, $y : [-6, 6]$.



(a) Estimate the coordinates of the crossing point from the screen, to the nearest half unit.

(b) Determine the crossing point algebraically by setting the two expressions equal. Leave your answer in exact fraction form.

(c) Verify your exact answer by substituting it into *both* equations.

(d) Explain why no amount of careful reading of this screen could have produced the exact answer in part (b), and describe one thing you could do with technology to improve the estimate.

Answer Key. 1. (a) x-int: $0 = 2x + 30$ so $x = -15$, giving $(-15, 0)$; y-int: $x = 0$ gives $y = 30$, i.e. $(0, 30)$. (b) W2 — -15 lies inside $[-20, 5]$ and 30 lies inside $[-10, 40]$, so both intercepts are on screen. (c) W1 misses both ($-15 < -10$ and $30 > 10$); W3 misses the y-intercept ($30 > 15$); W4 misses the x-intercept ($-15 < 0$). (d) At $x = -10$, $y = 2(-10) + 30 = 10$, so the line passes exactly through the corner $(-10, 10)$; every other point of the line has $x < -10$ or $y > 10$, so nothing else falls inside W1. **2.** (a) x-int 9, y-int -36; e.g. $x : [-2, 12]$, $y : [-45, 10]$. (b) x-int -12, y-int -60; e.g. $x : [-15, 2]$, $y : [-70, 10]$. (c) x-int -15, y-int 3; e.g. $x : [-20, 5]$, $y : [-5, 10]$. (d) x-int 5, y-int 100; e.g. $x : [-2, 8]$, $y : [-20, 120]$. (Windows vary.) **3.** (a) falling; steeper than $y = x$ (slope -4, size $4 > 1$); y-int 3. (b) rising; shallower (slope size $0.5 < 1$); y-int -2. (c) falling; shallower; y-int 4. **4.** Screen A = (b) $y = 0.5x - 2$ — rises gently and cuts the y-axis at -2. Screen B = (a) $y = -4x + 3$ — falls very steeply and crosses the y-axis at 3. Screen C = (c) $y = -0.5x + 4$ — falls gently and crosses the y-axis at 4. (B and C both fall, so steepness is what separates them.) **5.** (a) V-int 200 (the full tank at $m = 0$); m-int: $0 = 200 - 8m$ gives $m = 25$ min (the tank is empty). (b) e.g. $x : [0, 30]$, $y : [0, 220]$ — time starts at 0 and the tank is empty at $m = 25$, and V runs from 0 up to 200. (c) In the standard window only $0 \le m \le 10$ is shown, where V runs from 200 down to 120; every one of those V-values is far above Ymax = 10, so no part of the line falls inside the screen. **6.** (a) Between 1 and 2, closer to 1 — about 1.3. (b) At $x = 3$, $y = 2.5$; when $y = 5$, x is about 4.7 (between 4 and 5). (c) $0 = 1.5x - 2$ gives $x = \frac{4}{3} \approx 1.33$. (d) About half a gridline, so roughly ± 0.5 — you can see which square a point is in and whether it is nearer one edge or the middle, but not finer than that. (e) The reading at $x = 3$: the point sits on a gridline, and 2.5 lands exactly halfway between two gridlines, so it can be read off confidently; the other two readings fall between gridlines and must be estimated. **7.** (a) $(2, 4)$; check: $2 + 2 = 4$ and $3(2) - 2 = 4$. (b) Q, because its slope 3 is larger than P's slope 1. (c) At $x = 5$: P gives 7, Q gives 13, so Q is greater. At $x = -2$: P gives 0, Q gives -8, so P is greater. (d) $x > 2$. The lines swap positions only where they meet, so the crossing point $x = 2$ is the boundary; to the right of it the steeper line Q is on top. **8.** (a) $(2, 3)$; check: $2(2) - 1 = 3$ and $-(2) + 5 = 3$. (b) $y = 2x - 1$ grows (slope $+2$); $y = -x + 5$ shrinks (slope -1). The sign of the number multiplying x tells you, before graphing. **9.** (a) Both cross the y-axis at 1, and both pass through $(1, 4)$ and $(-1, -2)$; the equation is the same, only the Ymin/Ymax changed. (b) Screen 1. The slope did not change; Screen 2 fits 40 units of y into the same height, so each y-unit is drawn much shorter and the same rise takes up far less of the picture. (c) A rise of 3 crosses only part of one tall square, while a run of 1 crosses a whole square — so vertically the picture is compressed and the line tilts closer to horizontal. (d) Slope 3 looked nearly vertical on Screen 1 and close to $45°$ on Screen 2, so the angle you see depends on the window, not on the slope alone. Only a square window (equal spacing on both axes) shows a trustworthy angle. **10.** (a) y: $-10, -7, -4, -1, 2, 5$; differences $+3, +3, +3, +3, +3$. (b) y: $6, 1, -4, -9, -14$; differences $-5, -5, -5, -5$. (c) The constant difference equals the number multiplying x times the step taken in x: in (a) $3 \times 1 = 3$; in (b) $-2.5 \times 2 = -5$. Equal steps in x always give equal steps in y for a linear relation. **11.** (a) F (b) T (c) F (d) T (e) T (f) F. **12.** (a) The window shows only part of the plane; "not meeting on this screen" is not the same as "never meeting." The meeting point is simply outside the window. (b) $0.5x + 9 = x + 2$ gives $7 = 0.5x$, so $x = 14$ and $y = 16$: they meet at $(14, 16)$. (c) e.g. $x : [-5, 20]$, $y : [-5, 25]$. (d) The slopes are different (0.5 and 1), so the lines are not parallel — two non-parallel lines always cross exactly once. **13.** (a) $0 = 3x - 5$ gives $x = \frac{5}{3} \approx 1.667$. (b) $3(1.7021277) - 5 = 0.1063831$, so yes, the calculator's y-value is right for that x — it is just not zero. (c) Trace steps the cursor one pixel at a time, and pixel positions are spaced by the window width divided by the number of pixels; an exact intercept almost never lands on a pixel. (d) Use the zero (root) feature, or solve $3x - 5 = 0$ algebraically. (Zooming in repeatedly also improves the estimate but never makes it exact.) **14.** (a) Any line whose x-intercept is beyond ± 10, e.g. $y = x - 200$, whose x-intercept is 200 — far off screen but certainly there. (b) No. $y = 2x + 7$ has exactly one ($x = -3.5$); $y = 4$ is horizontal and never reaches the x-axis, so it has none; $y = 0$ *is* the x-axis, so every point is an x-intercept. Rule: every non-horizontal line has exactly one x-intercept; a horizontal line has none unless it is $y = 0$, which has infinitely many. **15.** (a) $12x = 10$ gives $x = \frac{10}{12} = \frac{5}{6} \approx 0.83$. (b) A very short, almost vertical sliver in the bottom-left corner, running from $(0, 0)$ to $\left(\frac{5}{6}, 10\right)$; the line uses all 10 units of height in less than one unit of width, so it appears nearly vertical. (c) e.g. $x : [0, 10]$, $y : [0, 120]$. (d) No — with 120 units of height it would look like a moderate climb. The *appearance* changed because the vertical scale changed; the slope is still 12. **16.** (a) About $(1.5, 0.5)$ — the crossing sits a little right of 1.5 and a little above 0.5. (b) $x - 1 = -2x + 4$ gives $3x = 5$, so $x = \frac{5}{3}$ and $y = \frac{5}{3} - 1 = \frac{2}{3}$: the point is $\left(\frac{5}{3}, \frac{2}{3}\right)$. (c) Check: $\frac{5}{3} - 1 = \frac{2}{3}$ ✓ and $-2\left(\frac{5}{3}\right) + 4 = -\frac{10}{3} + \frac{12}{3} = \frac{2}{3}$ ✓. (d) The crossing falls between gridlines and $\frac{5}{3} = 1.6\overline{6}$ is a non-terminating decimal, so no reading off a 1-unit grid can ever land on it exactly — reading a graph gives an approximation, algebra gives the exact value. Zooming in on the crossing (or using an intersect feature) would sharpen the estimate.

Lesson 8 — Extra Practice

Introducing Functions: The Vertical Line Test

Part A — The Rule, Stated Precisely

One test settles every question in this set: **each input is paired with exactly one output**. A repeated *output* is never a problem. A repeated *input* is a problem only when it comes with two *different* outputs. Watch for questions where those two ideas are easy to mix up.

1. Decide whether each set of ordered pairs is a function. Give the reason — and if it fails, name the input that breaks the rule.

(a) $(-2,7)$, $(0,7)$, $(3,7)$, $(5,7)$

(b) $(4,1)$, $(5,2)$, $(4,1)$, $(6,3)$

(c) $(-1,3)$, $(2,5)$, $(-1,-3)$, $(4,8)$

(d) $(0,0)$, $(1,-1)$, $(-1,1)$, $(1,1)$

2. Decide whether each table represents a function. Two of these are designed to catch a reader who only checks whether an x-value repeats.

(a)

x	-3	-1	1	3
y	2	2	2	2

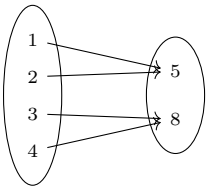
(b)

x	5	6	5	7
y	1	2	1	4

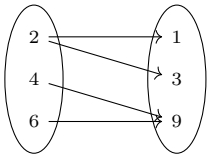
(c)

x	0	1	2	1
y	4	5	6	7

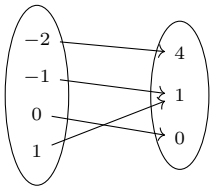
3. Decide whether each mapping diagram is a function. Justify by counting the arrows **leaving** each input.



(a) _____
Reason:



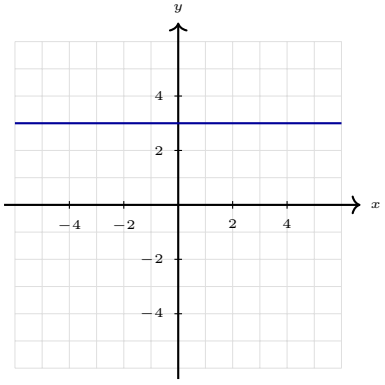
(b) _____
Reason:



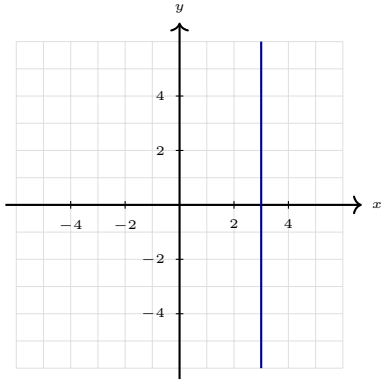
(c) _____
Reason:

4. **Two lines that look equally simple.** One of these is a function and the other is not.

$y = 3$



$x = 3$



(a) Which one is a function? Apply the vertical line test to each.

(b) For the one that fails, write two ordered pairs from it that share an input.

(c) Use the words *input* and *output* — not the words *vertical line* — to explain the difference between these two relations.

5. **Repeated outputs versus repeated inputs.**

(a) Write a set of four ordered pairs that **is** a function and in which the same y-value appears three times.

(b) Change exactly one pair in your answer to (a) so that the set is **no longer** a function. State which pair you changed and why the change matters.

(c) Complete the sentence and explain it in one line: "A repeated y-value is _____, but a repeated x-value is a problem whenever _____."

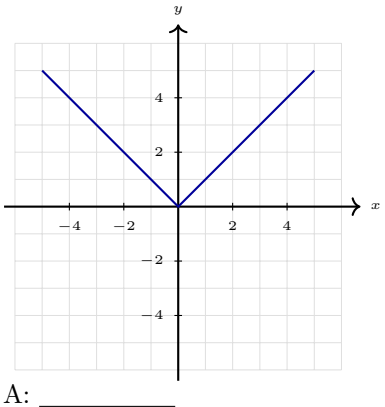
6. Jordan looks at $\{(1,5),(2,5),(3,8)\}$ and says: "This is not a function, because 1 and 2 both give 5 — the output 5 is used twice."

(a) Is Jordan's verdict right or wrong? State the correct verdict and quote the definition of a function in your answer.

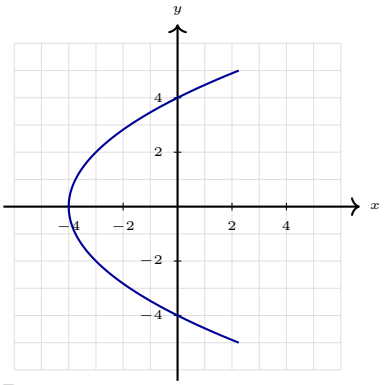
(b) Jordan has reversed the roles of input and output. Write the set Jordan *would* have been right about, by swapping the coordinates of each pair, and explain why that new set fails.

Part B — The Vertical Line Test on Graphs

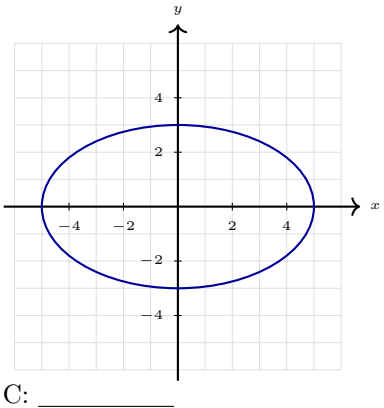
7. Classify each graph as a function or not a function. If it fails, describe a vertical line that crosses it more than once.



A: _____

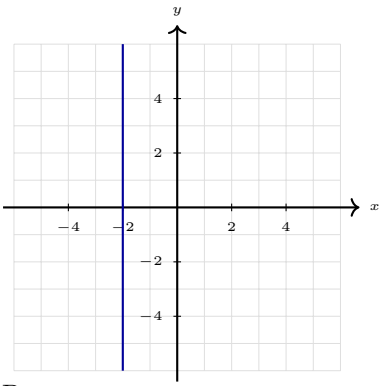


B: _____

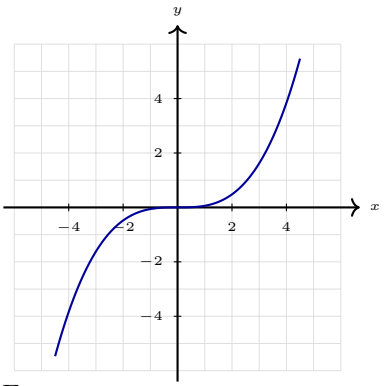


C: _____

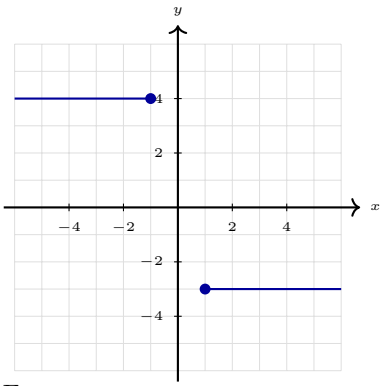
8. Classify each graph. Graph F is drawn in two separate pieces — decide carefully.



D: _____

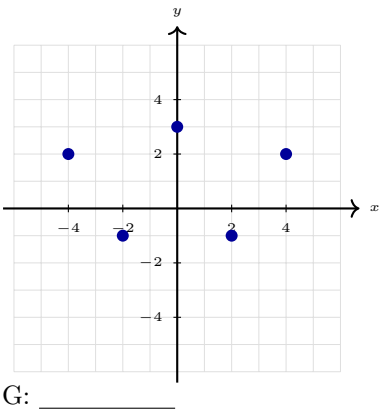


E: _____

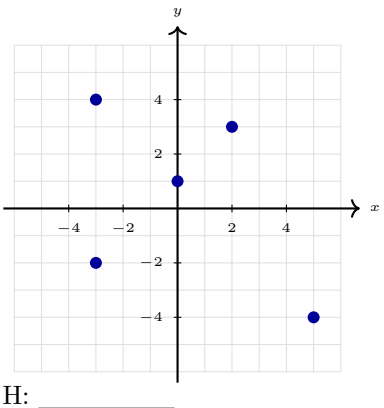


F: _____

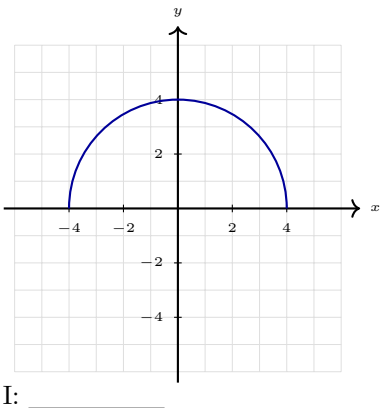
9. Graphs G and H are made of separate plotted points; graph I is half of a circle. Classify each one.



G: _____



H: _____



I: _____

10. For each graph below that is **not** a function, write two ordered pairs taken from the graph that share the same x-value but have different y-values. Read the coordinates off the grid.

(a) Graph B (sideways curve)

(b) Graph C (ellipse)

(c) Graph D (vertical line)

(d) Graph H (plotted points)

11. Amara looks at Graph D, the vertical line $x = -2$, and says: "It is a perfectly straight line, and straight lines are functions, so this is a function."

(a) Correct Amara's rule. Write the accurate version of "straight lines are functions."

(b) Explain, using the input -2, exactly how many outputs this relation gives that one input — and why a single counterexample is enough to decide the question.

Part C — Working Backwards, Context, and Reasoning

12. **Work backwards.**
(a) The set $\{(1,3),(2,5),(4,9)\}$ is a function. Add *one* ordered pair so that it is no longer a function, and add a *different* one that leaves it a function. State both.

(b) How many different first coordinates could the "breaking" pair in (a) use? List them, and explain why no other first coordinate can break the set.

(c) The set $\{(0,1),(2,4),(0,7),(5,3)\}$ is not a function. Remove one pair so that it becomes one. Is there more than one correct removal? Explain.

(d) Is it possible to *add* a pair to that same four-pair set and turn it into a function? Explain what that tells you about the difference between adding and removing.

13. **Both directions.** Every student at a school has exactly one student number, and no two students share a number.
(a) Is "student → student number" a function? Justify using the one-output rule.

(b) Is "student number → student" a function? Justify.

(c) Is "student → every course they are enrolled in" a function? Justify.

(d) Is "person → their birthday" a function? Is "birthday → the people born on it" a function? Answer both and justify each.

(e) Parts (a) and (b) were both functions, but (d) reversed gave a different answer. Explain what is special about student numbers that makes *both* directions work.

14. **Reasoning about intercepts.**
(a) Can the graph of a function have two x-intercepts? If yes, give an example (a set of ordered pairs or a sketch description) and explain why the rule is not broken.

(b) Can the graph of a function have two y-intercepts? Decide, then justify your answer twice — once using the input/output rule, and once using the vertical line test.

(c) A relation has x-intercepts at $(-3,0)$ and $(4,0)$, and y-intercepts at $(0,2)$ and $(0,-2)$. Can this relation be a function? Name the exact feature that decides it.

15. State whether each statement is true (T) or false (F). For every false statement, write a short correction.

(a) If two different inputs give the same output, the relation is not a function.

(b) A horizontal line is a function, but a vertical line is not.

(c) Every function is a relation, and every relation is a function.

(d) A graph made of separate, disconnected pieces cannot be a function.

(e) If a vertical line misses the graph completely, the relation is not a function.

(f) Listing the same ordered pair twice can turn a function into a non-function.

16. **Stretch.** Consider the relation $\{(1,2),(3,4),(k,5)\}$, where k is a number you get to choose.

(a) Test $k = 1$. Write out the three pairs and decide whether the relation is a function.

(b) Test $k = 3$, then test $k = 5$. Decide each case.

(c) State every value of k for which the relation **is** a function.

(d) Now consider $\{(1,2),(3,4),(k,4)\}$. Does changing the last output from 5 to 4 change your answer to (c)? Explain what that shows about the role of outputs.

(e) Explain why no choice of k can make the original set contain the same ordered pair listed twice.

Answer Key. 1. (a) Function — the inputs $-2, 0, 3, 5$ are all different; the repeated output 7 does not matter. (b) Function — 4 appears twice but with the *same* output 1 both times, so the input 4 still has exactly one output. (c) Not a function — the input -1 is paired with both 3 and -3. (d) Not a function — the input 1 is paired with both -1 and 1. **2.** (a) Function — four different inputs, each with one output (all outputs equal 2, which is allowed). (b) Function — $x = 5$ is listed twice but gives $y = 1$ both times, so the pair $(5, 1)$ is simply repeated. (c) Not a function — $x = 1$ gives 5 and also 7. **3.** (a) Function — exactly one arrow leaves each of $1, 2, 3, 4$; two arrows arriving at 5 and two at 8 is fine. (b) Not a function — two arrows leave the input 2 (to 1 and to 3). (c) Function — one arrow leaves each input; the output 1 receives two arrows, which is allowed. **4.** (a) $y = 3$ is a function: every vertical line meets it exactly once. $x = 3$ is not: the vertical line $x = 3$ lies right on top of the graph and meets it infinitely many times. (b) e.g. $(3, 0)$ and $(3, 5)$ — any two points with $x = 3$. (c) For $y = 3$, every input has exactly one output, and that output happens to be 3 every time. For $x = 3$, the single input 3 has every possible output at once, and all other inputs have none — so it fails the one-output rule. **5.** (a) e.g. $\{(1, 6), (2, 6), (3, 6), (4, 9)\}$ — answers vary; inputs must all differ. (b) e.g. change $(4, 9)$ to $(1, 9)$, giving $\{(1, 6), (2, 6), (3, 6), (1, 9)\}$: the input 1 now has two different outputs, 6 and 9. (c) A repeated y-value is *always allowed*; a repeated x-value is a problem whenever *the two outputs attached to it are different*. **6.** (a) Wrong — the set *is* a function. A function requires each *input* to have exactly one output; nothing forbids two inputs from sharing an output. Here $1 \to 5$, $2 \to 5$, $3 \to 8$: one output each. (b) Swapping gives $\{(5, 1), (5, 2), (8, 3)\}$, which is not a function, because now the input 5 has the two different outputs 1 and 2. **7.** A: Function — the V has exactly one point above or below each x-value. B: Not a function — e.g. the vertical line $x = 0$ meets it at $(0, 4)$ and $(0, -4)$; the line $x = -3$ meets it at $(-3, 2)$ and $(-3, -2)$. C: Not a function — e.g. $x = 0$ meets the ellipse at $(0, 3)$ and $(0, -3)$; any vertical line strictly between $x = -5$ and $x = 5$ meets it twice. **8.** D: Not a function — the vertical line $x = -2$ is the graph itself, so it crosses it everywhere. E: Function — the curve always moves left to right, so every vertical line meets it at most once. F: Function — the two pieces sit over different x-values (one covers $-6 \le x \le -1$, the other $1 \le x \le 6$), so no vertical line meets both; the gap between -1 and 1 just means those inputs have no output. **9.** G: Function — the inputs $-4, -2, 0, 2, 4$ are all different (the repeated outputs 2 and -1 are fine). H: Not a function — $(-3, 4)$ and $(-3, -2)$ share the input -3. I: Function — the upper half of the circle gives exactly one y for each x from -4 to 4. **10.** (a) $(0, 4)$ and $(0, -4)$ (or $(-3, 2)$ and $(-3, -2)$). (b) $(0, 3)$ and $(0, -3)$ (or $(-4, 1.8)$ and $(-4, -1.8)$; any vertical pair). (c) $(-2, 1)$ and $(-2, 5)$ — any two points with $x = -2$. (d) $(-3, 4)$ and $(-3, -2)$. **11.** (a) Every *non-vertical* straight line is a function; a vertical line is the one exception. (b) The input -2 is paired with every y-value on the line — infinitely many outputs, when a function allows exactly one. One input with two different outputs is already enough to disqualify a relation, so a single counterexample settles it; you never need to check the rest. **12.** (a) Breaking pair: e.g. $(2, 7)$ — the input 2 would then have outputs 5 and 7. Safe pair: e.g. $(6, 11)$ — a brand-new input. (Answers vary.) (b) Three: 1, 2, or 4. A breaking pair must reuse an input that is already there with a *different* output, and those are the only inputs present; any other first coordinate is new, so it brings its own single output. (c) Remove $(0, 1)$ or remove $(0, 7)$ — either works, so the removal is not unique. Removing $(2, 4)$ or $(5, 3)$ does not help, since 0 would still have two outputs. (d) No. Adding pairs can only add more inputs and outputs; the conflict at $x = 0$ is still there no matter what you add. Removing can repair a relation, adding never can. **13.** (a) Function — each student has exactly one student number. (b) Function — because no two students share a number, each number leads back to exactly one student. (c) Not a function — a student enrolled in several courses gives one input several outputs. (d) "Person $\to$ their birthday" is a function (one birthday each); "birthday $\to$ people born on it" is not, because a single date is shared by many people. (e) The pairing is one-to-one: every student gets a number and every number belongs to exactly one student, with none shared and none left over. Birthdays are not one-to-one — many people share a date — so that relation only works in one direction. **14.** (a) Yes. For example $\{(-3, 0), (4, 0), (0, -5)\}$, or any parabola such as $y = x^2 - 4$ with x-intercepts $(-2, 0)$ and $(2, 0)$. Two *inputs* share the *output* 0, which the rule allows. (b) No. A y-intercept has $x = 0$; two different y-intercepts would mean the single input 0 has two different outputs, which the definition forbids. By the vertical line test, the y-axis *is* a vertical line, so it may meet the graph at most once. (c) No, it cannot be a function. The two

x-intercepts are harmless, but the two y-intercepts $(0, 2)$ and $(0, -2)$ give the input 0 two different outputs — that alone decides it. **15.** (a) F — different inputs are allowed to share an output; the relation can still be a function. (b) T. (c) F — every function is a relation, but not every relation is a function. (d) F — disconnected pieces are fine as long as no x-value is used twice (see Graph F). (e) F — a vertical line that misses the graph just means that input has no output; the test only fails when a vertical line crosses *more than once*. (f) F — a repeated identical pair adds nothing new; the input still has exactly one output. **16.** (a) $k = 1$: $(1, 2)$, $(3, 4)$, $(1, 5)$ — not a function, since the input 1 has outputs 2 and 5. (b) $k = 3$: $(1, 2)$, $(3, 4)$, $(3, 5)$ — not a function, since 3 has outputs 4 and 5. $k = 5$: $(1, 2)$, $(3, 4)$, $(5, 5)$ — a function; a point whose coordinates match each other is perfectly ordinary. (c) Every value except 1 and 3; that is, $k \neq 1$ and $k \neq 3$. (d) Yes: now only $k = 1$ fails. With $k = 3$ the pair $(3, 4)$ is just written twice — the same pair repeated, which is allowed — so the relation is a function for every $k \neq 1$. A clash of inputs breaks a function only when the outputs are *different*. (e) A repeated pair would need the same first *and* second coordinate. The third pair has output 5, while the others have outputs 2 and 4, so it can never match either one no matter what k is.

Lesson 9 — Extra Practice

Function Notation: Evaluating and Interpreting $f(x)$

Part A — Evaluating at Negatives, Zero, and Fractions

A function machine does not care whether the input is friendly. Substitute the number exactly as given — brackets around every negative and every fraction — then simplify with order of operations. Leave fractional answers as exact fractions, not decimals.

1. If $f(x) = 6x - 4$, determine each value.
(a) $f(-3)$ (b) $f(0)$ (c) $f\left(\frac{1}{2}\right)$ (d) $f\left(-\frac{1}{2}\right)$

2. If $g(x) = 8 - 4x$, determine each value.
(a) $g(-2)$ (b) $g(0)$ (c) $g\left(\frac{3}{4}\right)$ (d) $g\left(-\frac{1}{4}\right)$

3. If $h(x) = x^2 - 2x$, determine each value. Watch the sign carefully in parts (a) and (d): squaring a negative gives a positive, but the $-2x$ term does not.
(a) $h(-3)$ (b) $h(0)$ (c) $h\left(\frac{1}{2}\right)$ (d) $h\left(-\frac{1}{2}\right)$

4. If $p(x) = \dfrac{x+7}{2}$, determine each value.
(a) $p(-3)$ (b) $p(0)$ (c) $p\left(\frac{1}{2}\right)$ (d) $p(-7)$

Part B — Running the Machine Backwards: Solving $f(x) = k$

So far every question has handed you the *input* and asked for the output. A statement such as $f(x) = 19$ does the opposite: it hands you the *output* and asks which input produced it. To answer it, replace $f(x)$ with the rule and solve the equation. For $f(x) = 4x + 3$:

$$f(x) = 19 \implies 4x + 3 = 19 \implies 4x = 16 \implies x = 4,$$

so the input 4 is the one that produces the output 19. Check it: $f(4) = 4(4) + 3 = 19$. ✓

5. Let $f(x) = 3x - 7$. Solve each equation for x, and check each answer by evaluating.
(a) $f(x) = 8$ (b) $f(x) = -1$

(c) $f(x) = 0$ *(your answer will be a fraction)* (d) $f(x) = -13$

6. Let $g(x) = 12 - 5x$. Solve each equation for x.
(a) $g(x) = 2$

(b) $g(x) = 27$

(c) $g(x) = 0$

(d) $g(x) = 12$ *(what does your answer say about where the graph crosses an axis?)*

7. Let $f(x) = 4x + 9$. The number -2 appears in both parts below, but in different roles.
(a) Determine $f(-2)$.

(b) Solve $f(x) = -2$.

(c) The two answers are completely different numbers. Explain, in a sentence or two, what job the -2 is doing in part (a) and what job it is doing in part (b).

8. **Stretch.** Let $h(x) = x^2 - 9$. Solving $h(x) = k$ here means solving a squared equation, so an output can come from *two* different inputs — or from none at all. Solve each, and state how many inputs work.
(a) $h(x) = 0$

(b) $h(x) = 16$

(c) $h(x) = -9$

(d) $h(x) = -25$ *(think before you write — can x^2 ever be negative?)*

Part C — Expressions, Variable Inputs, and Two Functions at Once

9. Let $f(x) = 2x + 3$. Evaluate each expression. In parts (a) and (b), find each function value *first*, then combine.
(a) $f(2) + f(3)$

(b) $2f(1)$

(c) Evaluate $f(5)$. Is $f(2) + f(3) = f(5)$?

10. The input does not have to be a number. To find $f(a)$ you replace every x with a; to find $f(x+1)$ you replace every x with the whole expression $(x+1)$, brackets included, then expand. For $f(x) = 3x - 2$:

$$f(x+1) = 3(x+1) - 2 = 3x + 3 - 2 = 3x + 1.$$

Now let $f(x) = 5x - 1$ and determine each expression, simplified.
(a) $f(a)$ (b) $f(a+2)$ (c) $f(x+1)$

11. Let $f(x) = 3x + 4$ and $g(x) = x^2 - 1$. Evaluate each of the following. Show the two function values separately before combining them.
(a) $f(2) - g(2)$ (b) $g(-3) - f(-3)$ (c) $f(0) + g(0)$

12. Let $f(x) = 2x + 1$ and $g(x) = 5x - 8$. An equation such as $f(x) = g(x)$ asks: *is there an input that both machines send to the same output?*
(a) Verify that $f(2) \neq g(2)$ by evaluating both.

(b) Solve the equation $f(x) = g(x)$.

(c) Substitute your answer from (b) into *both* rules to find the output they share.

(d) If the graphs of f and g were drawn on one grid, what would your answers to (b) and (c) tell you about the two lines?

Part D — Interpreting, Error Analysis, and Working Backwards

13. **Error analysis.** Dmitri believes $f(x)$ means "f times x." Given $f(x) = 2x + 6$, he writes

$$f(3) = f \times 3 = (2x + 6)(3) = 6x + 18.$$

(a) Explain what the brackets in $f(3)$ actually mean, and give the correct value of $f(3)$.

(b) Dmitri then argues that $f(2) + f(3)$ must equal $f(5)$ "because $2 + 3 = 5$." Use $f(x) = 2x + 6$ to show that this is false.

14. **Stretch: working backwards to the rule.** A function is linear, so it has the form $f(x) = mx + b$, but you are given only two of its values: $f(0) = 4$ and $f(3) = 13$.

(a) Explain why $f(0) = 4$ immediately gives you the value of b.

(b) Use $f(3) = 13$ together with your value of b to find m, then write the complete rule.

(c) Use your rule to determine $f(-2)$.

(d) Use your rule to solve $f(x) = 25$.

15. **Stretch.** A second linear function g satisfies $g(-1) = 9$ and $g(3) = -3$. This time neither input is 0, so you must find m first.

(a) Find m by comparing how much the output changed with how much the input changed.

(b) Substitute one known pair into $g(x) = mx + b$ to find b, then write the rule.

(c) Solve $g(x) = 0$, then determine $g(5)$.

16. **From a table.** The table shows values of a linear function f. Three entries are missing.

(a) The outputs change by the same amount each time the input goes up by 1. Find that amount and complete the table.

x	-2	-1	0	1	2	3	4
$f(x)$	-8		-2	1		7	

(b) State the value of $f(4)$.

(c) Solve $f(x) = 7$.

(d) Write the rule for f in the form $f(x) = mx + b$, and use it to check your answer to (b).

17. **Reading the notation in context.** At a theatre, $C(n)$ represents the total cost, in dollars, of buying n tickets.

(a) Explain in words what the statement $C(5) = 42$ means in this situation.

(b) Explain in words what the *question* "solve $C(n) = 100$" is asking someone to find.

(c) Write "eight tickets cost \$63" as a statement in function notation.

(d) One of $C(3)$ and "solve $C(n) = 3$" asks for a number of dollars; the other asks for a number of tickets. Say which is which, and explain how you can tell.

18. A veterinarian records a puppy's mass with the function $W(t)$, the mass in kilograms t weeks after birth.

(a) Interpret $W(6) = 4.2$ in words, with units.

(b) The vet solves $W(t) = 10$. Describe what she is trying to find out.

(c) $W(0)$ has a sensible meaning here, but $W(-2)$ does not. Explain both.

19. **Stretch: when input equals output.** Some functions have an input that the machine leaves unchanged — a value of x for which $f(x) = x$. To find it, set the rule equal to x and solve.

(a) If $f(x) = 2x - 5$, find the value of x for which $f(x) = x$, then check it.

(b) If $g(x) = 4x + 9$, find the value of x for which $g(x) = x$.

(c) Try the same thing with $h(x) = x + 7$. Explain what goes wrong and what it tells you about h.

Answer Key. 1. (a) -22 (b) -4 (c) -1 (d) -7. **2.** (a) 16 (b) 8 (c) 5 (d) 9. **3.** (a) $9 + 6 = 15$ (b) 0 (c) $\frac{1}{4} - 1 = -\frac{3}{4}$ (d) $\frac{1}{4} + 1 = \frac{5}{4}$. **4.** (a) 2 (b) $\frac{7}{2}$ (c) $\frac{15}{4}$ (d) 0. **5.** (a) $x = 5$ (b) $x = 2$ (c) $x = \frac{7}{3}$ (d) $x = -2$. **6.** (a) $x = 2$ (b) $x = -3$ (c) $x = \frac{12}{5}$ (d) $x = 0$; the output 12 comes from the input 0, so the graph crosses the vertical axis at $(0, 12)$. **7.** (a) $f(-2) = 1$ (b) $x = -\frac{11}{4}$ (c) In (a) the -2 is the *input* — it is substituted into the rule. In (b) the -2 is the *output* — it is the answer the machine gave, and the unknown input must be solved for. **8.** (a) $x^2 = 9$, so $x = 3$ or $x = -3$; two inputs (b) $x^2 = 25$, so $x = 5$ or $x = -5$; two inputs (c) $x^2 = 0$, so $x = 0$; one input (d) $x^2 = -16$ is impossible, since a square is never negative; no input works. **9.** (a) $f(2) = 7$, $f(3) = 9$, total 16 (b) $f(1) = 5$, so $2f(1) = 10$ (c) $f(5) = 13$, and $16 \neq 13$, so no. **10.** (a) $5a - 1$ (b) $5(a + 2) - 1 = 5a + 9$ (c) $5(x + 1) - 1 = 5x + 4$. **11.** (a) $10 - 3 = 7$ (b) $8 - (-5) = 13$ (c) $4 + (-1) = 3$. **12.** (a) $f(2) = 5$, $g(2) = 2$, so they differ (b) $2x + 1 = 5x - 8$ gives $x = 3$ (c) $f(3) = 7$ and $g(3) = 7$ (d) the two lines cross at the single point $(3, 7)$. **13.** (a) The brackets hold an *input*, not a multiplication: $f(3)$ means substitute 3 for x, so $f(3) = 2(3) + 6 = 12$ (b) $f(2) = 10$ and $f(3) = 12$, so $f(2) + f(3) = 22$, but $f(5) = 16$; since $22 \neq 16$, the claim fails. **14.** (a) Substituting $x = 0$ gives $f(0) = m(0) + b = b$, so $b = 4$ (b) $3m + 4 = 13$ gives $m = 3$, so $f(x) = 3x + 4$ (c) $f(-2) = -2$ (d) $3x + 4 = 25$ gives $x = 7$. **15.** (a) The input rises from -1 to 3 (a change of 4) while the output falls from 9 to -3 (a change of -12), so $m = \frac{-12}{4} = -3$ (b) $-3(-1) + b = 9$ gives $b = 6$, so $g(x) = -3x + 6$ (c) $-3x + 6 = 0$ gives $x = 2$; $g(5) = -9$. **16.** (a) The outputs go up by 3 each step: $-8, -5, -2, 1, 4, 7, 10$ (b) $f(4) = 10$ (c) $f(x) = 7$ when $x = 3$ (d) $f(x) = 3x - 2$, and $3(4) - 2 = 10$ agrees with (b). **17.** (a) Buying 5 tickets costs \$42 (b) It asks how many tickets can be bought for a total cost of \$100 (c) $C(8) = 63$ (d) $C(3)$ asks for dollars, because the 3 sits in the input position (tickets) and the answer is the output (cost); solving $C(n) = 3$ asks for tickets, because 3 is the output (cost) and the unknown n is the input. **18.** (a) Six weeks after birth the puppy has a mass of 4.2 kg (b) She wants the number of weeks after birth at which the puppy's mass reaches 10 kg (c) $W(0)$ is the puppy's mass at birth, which makes sense; $W(-2)$ would be its mass two weeks before it was born, which has no meaning. **19.** (a) $2x - 5 = x$ gives $x = 5$; check $f(5) = 5$ ✓ (b) $4x + 9 = x$ gives $3x = -9$, so $x = -3$ (c) $x + 7 = x$ simplifies to $7 = 0$, which is never true, so no input is left unchanged — h shifts every input up by 7.

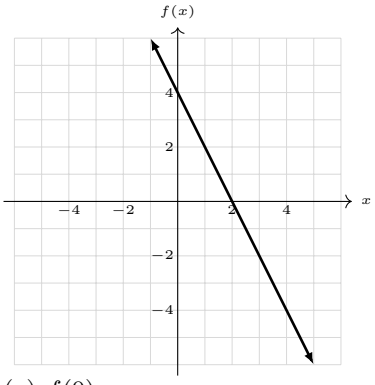
Lesson 10 — Extra Practice

Function Notation: Graphs, Conversions, and Real-World Models

Part A — Both Directions Off One Graph

Every graph below answers two different kinds of question. To find $f(a)$, start at a on the horizontal axis and read *up or down* to the graph, then across. To solve $f(x) = b$, start at b on the vertical axis and read *across* to the graph, then up or down. One question travels $x \to y$; the other travels $y \to x$.

1. The graph of the function f is shown.

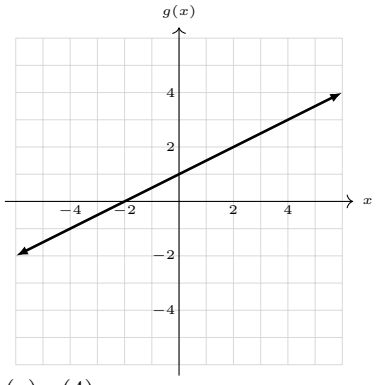


(a) $f(0) =$ _____
(b) $f(3) =$ _____
(c) Solve $f(x) = 2$.

(d) Solve $f(x) = 0$.

(e) Write your answer to (b) as an ordered pair on the graph.

2. The graph of the function g is shown.

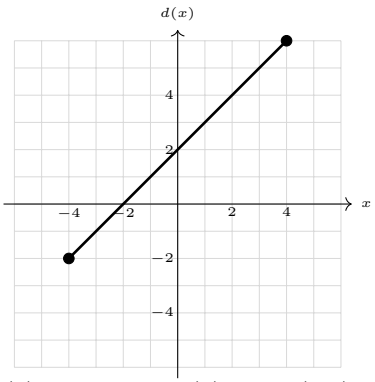


(a) $g(4) =$ _____
(b) $g(-2) =$ _____
(c) Solve $g(x) = -1$.

(d) Solve $g(x) = 3$.

(e) Parts (a) and (d) both involve the numbers 4 and 3, but in opposite roles. Explain.

3. The graph of a function d is shown. The solid dots mark where the graph begins and ends.



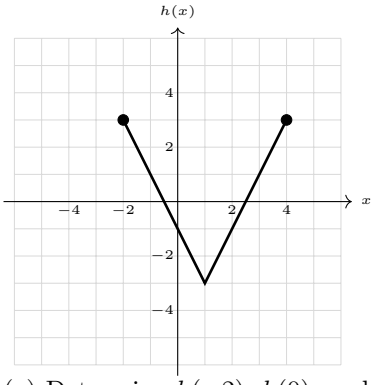
(a) Determine $d(1)$ and $d(-4)$.

3. *(continued)*
(b) Solve $d(x) = 0$, then solve $d(x) = 5$.

(c) State the domain and range of d in set notation.
D: _____
R: _____

(d) Explain why $d(6)$ has no value, and why this is a fact about the *domain*.

4. The graph of a function h is shown, drawn only between its two endpoints.



(a) Determine $h(-2)$, $h(0)$, and $h(3)$.

(b) Solve $h(x) = 1$. *(Careful — read across at* 1 *and see how many times you meet the graph.)*

(c) Solve $h(x) = -3$, then explain why this output is different from the one in (b).

4. *(continued)*
(d) Solve $h(x) = -5$, or explain why no solution exists.

(e) State the domain and range of h in set notation.
D: _____
R: _____

(f) A student claims $h(5) = 5$ because "the graph keeps going up." Explain why this claim cannot be made from this graph.

Part B — Moving Between Equations, Tables, and Ordered Pairs

5. The same rule can wear four different costumes: an equation in y, an equation in function notation, a table, and a set of ordered pairs. Convert each as directed.

(a) Write $f(x) = 3x - 4$ in x-y notation.

(b) Write $C(n) = 8n + 12$ in x-y notation.

(c) Write $y = 7 - 2x$ in function notation, naming the function f.

(d) Write $y = 4w - 1$ in function notation, naming the function P with input w.

6. A point (a, b) on the graph of f and the statement $f(a) = b$ carry exactly the same information — the input is always written first.

(a) The point $(3, 11)$ lies on the graph of f. Write this in function notation.

(b) It is known that $g(-2) = 7$. Write the matching ordered pair.

(c) The graph of h passes through $(0, -5)$. Write this in function notation, and name the feature of the graph it describes.

(d) A student says $(4, 9)$ on the graph of f means $f(9) = 4$. Correct the student.

7. The table below lists some values of a linear function f.

x	-3	-1	0	2	5
$f(x)$	7	3	1	-3	-9

(a) Write three statements in function notation that the table tells you.

(b) Use the table to solve $f(x) = -3$.

7. *(continued)*
(c) The inputs do not go up in equal steps, so compare a pair of columns carefully to find how much $f(x)$ changes per unit of x. Then write the rule in the form $f(x) = mx + b$.

(d) Use your rule to determine $f(10)$, a value the table does not show.

8. Let $f(x) = -2x + 5$.
(a) Complete the table of values.

x	-2	-1	0	1	2	3
$f(x)$						

(b) Write the same rule in x-y notation.

(c) Write the six ordered pairs from your table.

(d) Use the table to solve $f(x) = -1$, then confirm your answer algebraically from the rule.

Part C — Real-World Models: Evaluate, Interpret, Solve, Restrict

In a model, every number means something. The constant term is the starting value, the coefficient is the rate of change, and the domain and range are limited by what the situation physically allows. Write domain and range in set notation, as in $0 \le t \le 20$.

9. A shop rents kayaks. The total charge, in dollars, for a rental of t hours is $C(t) = 25 + 15t$. The shop is open for 8 hours a day, and a kayak must be returned the same day.
(a) Determine $C(0)$ and $C(4)$, and explain in words what $C(4)$ tells a customer.

(b) Explain what the 25 and the 15 each represent in this situation, with units.

(c) Solve $C(t) = 145$, and state what the answer means.

(d) State a sensible domain and range for C in set notation.
D: _____
R: _____

10. A tank holding 600 litres is drained at a steady rate. The volume remaining, in litres, t minutes after the valve is opened is $V(t) = 600 - 40t$.

(a) Determine $V(0)$ and $V(5)$, and interpret each value in words.

(b) Solve $V(t) = 240$, and explain what the answer means.

(c) How long does the tank take to empty completely? Write the equation you solved using function notation.

(d) State a sensible domain and range for V in set notation.

D: _____

R: _____

11. As a weather balloon rises, the air temperature drops steadily. The temperature, in degrees Celsius, at an altitude of h kilometres is $T(h) = 20 - 6h$. The balloon rises to a maximum altitude of 10 km.

(a) Determine $T(0)$ and $T(3)$, and interpret each in words.

(b) Solve $T(h) = -16$, and state what the answer means.

(c) Determine the temperature at the balloon's highest point.

(d) State a sensible domain and range for T in set notation.

D: _____

R: _____

12. **Intercepts in context.** An intercept is just a function value at a boundary: the vertical intercept is the output when the input is 0, and a horizontal intercept is an input whose output is 0. Use the models from questions 9 and 10.

(a) The graph of C meets the vertical axis at $(0, 25)$. Explain what this point means to a customer.

(b) Solving $C(t) = 0$ gives $t = -\frac{5}{3}$. Explain why this number is real mathematics but meaningless in this situation.

(c) The graph of V has intercepts $(0, 600)$ and $(15, 0)$. Interpret both in words.

13. **Error analysis.** Asked for the domain of $V(t) = 600 - 40t$ from question 10, a student writes: "The domain is all real numbers, because you can substitute any number into $600 - 40t$."

(a) The student is right about the algebra and wrong about the model. Give *two* separate reasons why the domain must be restricted — one about small inputs, one about large ones.

(b) Write the correct domain and range in set notation, and state the value of $V(20)$ that the student's domain would wrongly allow.

14. **Build the rule, then use it.** A gym charges a one-time joining fee of \$40 plus \$22 for each month of membership.
(a) Write a function $M(t)$ for the total amount paid, in dollars, after t months.

(b) Determine $M(9)$ and explain what it represents.

(c) Solve $M(t) = 370$, and state what the answer means.

(d) A member signs a two-year contract (24 months). State the domain and range in set notation.

15. **Build the rule, then use it.** A water tank starts with 900 litres and is pumped out at a constant 45 litres per minute.
(a) Write a function $W(t)$ for the volume remaining, in litres, after t minutes of pumping.

(b) Determine $W(8)$, and write its meaning as a sentence.

(c) Solve $W(t) = 450$. Then explain, without further calculation, why the answer is exactly half the time needed to empty the tank.

(d) Determine how long emptying takes, then state the domain and range in set notation.

Part D — Stretch: Comparing Two Models

16. Two courier companies price a delivery of k kilometres differently. Company P charges $P(k) = 45 + 4k$ dollars; Company Q charges $Q(k) = 9k$ dollars.
(a) Determine $P(5)$ and $Q(5)$. Which company is cheaper for a 5 km delivery?

(b) Solve $P(k) = Q(k)$, and determine the cost that both companies charge at that distance.

(c) Determine $P(15)$ and $Q(15)$. Which company is cheaper now?

(d) Using your answers, describe completely the distances for which each company is the cheaper choice. Explain why the answer switches over at exactly one distance.

Answer Key. 1. (a) 4 (b) -2 (c) $x = 1$ (d) $x = 2$ (e) $(3, -2)$. **2.** (a) 3 (b) 0 (c) $x = -4$ (d) $x = 4$ (e) In (a) the 4 is the input and the 3 is the output; in (d) the 3 is the output given and the 4 is the input found. Same pair, opposite direction. **3.** (a) $d(1) = 3$, $d(-4) = -2$ (b) $d(x) = 0$ at $x = -2$; $d(x) = 5$ at $x = 3$ (c) D: $-4 \le x \le 4$, R: $-2 \le d(x) \le 6$ (d) the graph ends at $x = 4$, so 6 is not an allowed input — the set of allowed inputs is exactly what "domain" means. **4.** (a) $h(-2) = 3$, $h(0) = -1$, $h(3) = 1$ (b) the line $y = 1$ meets the graph twice, at $x = -1$ and $x = 3$ (c) $x = 1$ only; -3 is the lowest output on the graph, so only the corner point produces it (d) no solution — no point of the graph is as low as -5 (e) D: $-2 \le x \le 4$, R: $-3 \le h(x) \le 3$ (f) 5 is outside the domain: the graph stops at $x = 4$, so nothing can be read there. **5.** (a) $y = 3x - 4$ (b) $y = 8n + 12$ (c) $f(x) = 7 - 2x$ (d) $P(w) = 4w - 1$. **6.** (a) $f(3) = 11$ (b) $(-2, 7)$ (c) $h(0) = -5$; this is the y-intercept (d) the input is written first, so the correct statement is $f(4) = 9$. **7.** (a) e.g. $f(-3) = 7$, $f(0) = 1$, $f(5) = -9$ (b) $x = 2$ (c) from $x = 0$ to $x = 2$ the output falls 4 over 2 steps, so $m = -2$; $b = f(0) = 1$ and $f(x) = -2x + 1$ (d) $f(10) = -19$. **8.** (a) 9, 7, 5, 3, 1, -1 (b) $y = -2x + 5$ (c) $(-2, 9), (-1, 7), (0, 5), (1, 3), (2, 1), (3, -1)$ (d) $x = 3$, and $-2(3) + 5 = -1$ ✓. **9.** (a) $C(0) = 25$, $C(4) = 85$; a 4-hour rental costs \$85 (b) 25 is the fixed charge in dollars just to take a kayak out, 15 is the cost in dollars for each additional hour (c) $t = 8$; a charge of \$145 means the kayak was out for 8 hours (d) D: $0 \le t \le 8$, R: $25 \le C(t) \le 145$. **10.** (a) $V(0) = 600$, the tank held 600 L when the valve was opened; $V(5) = 400$, after 5 min 400 L remain (b) $t = 9$; 240 L are left after 9 min (c) solve $V(t) = 0$: $t = 15$ min (d) D: $0 \le t \le 15$, R: $0 \le V(t) \le 600$. **11.** (a) $T(0) = 20$, the ground-level temperature is $20\,^\circ$C; $T(3) = 2$, at 3 km the temperature is $2\,^\circ$C (b) $h = 6$; the temperature is $-16\,^\circ$C at an altitude of 6 km (c) $T(10) = -40\,^\circ$C (d) D: $0 \le h \le 10$, R: $-40 \le T(h) \le 20$. **12.** (a) Renting for 0 hours still costs \$25 — it is the fixed fee (b) a negative t would be time before the rental began, which cannot happen, and the charge never actually reaches \$0 (c) $(0, 600)$: the tank started with 600 L; $(15, 0)$: the tank is empty after 15 min. **13.** (a) Inputs below 0 describe time before the valve was opened, and inputs above 15 give volumes that do not exist, since the tank is already empty (b) D: $0 \le t \le 15$, R: $0 \le V(t) \le 600$; the student's domain would allow $V(20) = -200$ L, a negative volume. **14.** (a) $M(t) = 22t + 40$ (b) $M(9) = 238$; after 9 months the member has paid \$238 in total (c) $t = 15$; a total of \$370 has been paid after 15 months (d) D: $0 \le t \le 24$, R: $40 \le M(t) \le 568$. **15.** (a) $W(t) = 900 - 45t$ (b) $W(8) = 540$; after 8 minutes of pumping, 540 L remain (c) $t = 10$; 450 L is half of the starting 900 L, and since the rate is constant, half the water leaves in half the total time (d) $W(t) = 0$ at $t = 20$ min; D: $0 \le t \le 20$, R: $0 \le W(t) \le 900$. **16.** (a) $P(5) = 65$, $Q(5) = 45$; Q is cheaper (b) $45 + 4k = 9k$ gives $k = 9$ km, and both charge $P(9) = Q(9) = \$81$ (c) $P(15) = 105$, $Q(15) = 135$; P is cheaper (d) Q is cheaper for $0 \le k < 9$, the two are equal at $k = 9$, and P is cheaper for $k > 9$. Each cost is a straight line, and two lines with different rates of change cross exactly once, so the cheaper company can change only at that single crossing distance.

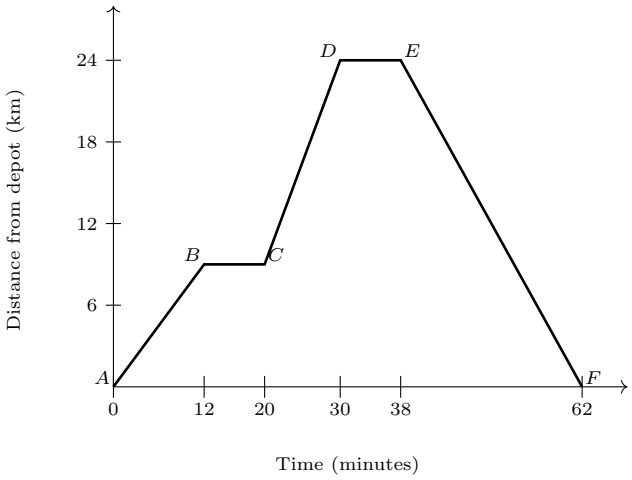
Lesson 11 — Extra Practice

Interpreting and Sketching Real-World Graphs

How to read this set. Every graph here tells a story, and every story hides a **rate of change**. A *rising* segment means the quantity is growing, a *falling* segment means it is shrinking, and a *flat* segment means it is not changing at all — which is never the same thing as going backwards. The *steepness* of a segment is how fast the quantity changes, found the same way every time: the change in the vertical quantity divided by the change in time.

Part A — Reading a Multi-Segment Story Graph

1. A delivery van leaves its depot, makes two drop-offs, and returns. The graph shows the van's **distance from the depot** over the whole trip, made of five straight segments meeting at A–F.



(a) Describe, with reference to the van, what is happening during each interval.

(i) A to B:

(ii) C to D:

(iii) E to F:

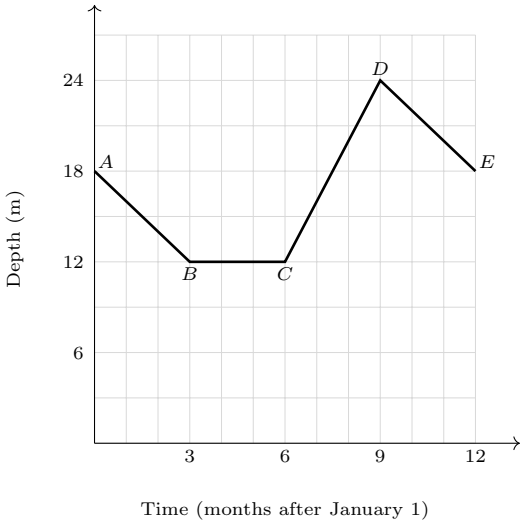
(b) Determine the rate of change from C to D, in km per minute, and then convert it to km/h.

(c) Which segment is the steepest? State what that steepness means for the van at that moment.

(d) For how many minutes in total was the van not moving? Which segments tell you this?

(e) How many kilometres did the van *drive* altogether? Explain why the answer is not 0 km even though it ends where it started.

2. The graph shows the water depth in a reservoir over one year, measured in months after January 1.



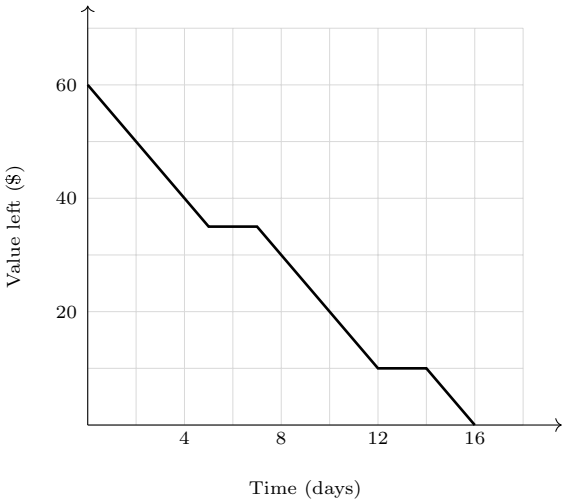
(a) Complete the table.

	A–B	B–C	C–D	D–E
Change in time (months)	3			
Change in depth (m)	-6			
Rate (m per month)	-2			

(b) Which interval is the steepest, and what does that represent for the reservoir?

(c) During which interval did the depth not change, and for how long?

3. Nadia loads \$60 onto a prepaid transit card. The graph shows the value left on the card over the next 16 days. She rides only on weekdays.



(a) State the coordinates of the vertical intercept and explain what it means here.

(b) There are two flat sections. Explain what they represent, and why they are **not** days on which the card gained value.

(c) Determine the rate of change on a day that Nadia rides. Include the units and the sign.

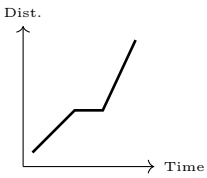
(d) State the coordinates of the horizontal intercept and explain what it means here.

(e) Suppose instead Nadia had ridden *every* day, weekends included, at the same daily cost. On what day would the card have reached \$0? Explain how the graph would change.

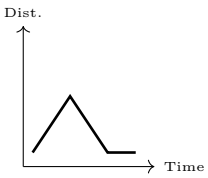
Part B — Matching Stories to Graphs

4. Each graph below shows Kai's **distance from home** against **time** on a different morning. Match each story to exactly one graph.

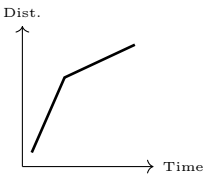
A



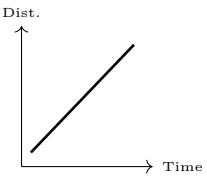
B



C



D



(i) Kai walks at one steady pace the whole way, never stopping. ____

(ii) Kai walks to the bus stop, waits there for the bus, then rides the bus — much faster than he walked — the rest of the way. ____

(iii) Kai jogs quickly at first, then tires and walks the rest of the way, never stopping. ____

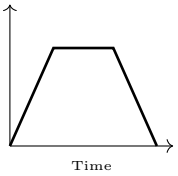
(iv) Kai starts out, realizes he forgot his lunch, walks home at the same pace, and waits inside. ____

(v) In the graph you chose for story (iv), the falling segment is exactly as steep as the rising segment. Explain what that tells you about Kai's two walks.

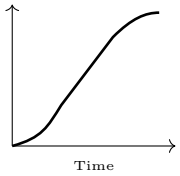
(vi) Graphs **C** and **D** both rise the whole way with no flat part. Explain precisely what makes them different stories.

5. A city bus pulls away from one stop, drives to the next stop, and halts there. The two graphs below describe *the same* trip: one shows the bus's **speed**, the other its **distance travelled**. Neither is labelled.

Graph P



Graph Q



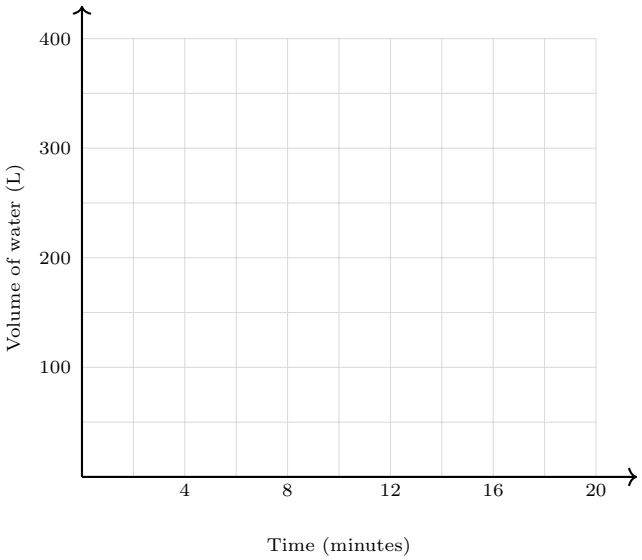
(a) Which graph shows speed and which shows distance travelled? Justify each choice.

(b) At the instant the bus is halted at the second stop, describe what each graph is doing.

(c) Explain why the distance-travelled graph can never have a falling segment, no matter what the bus does.

Part C — Sketching a Graph From a Story

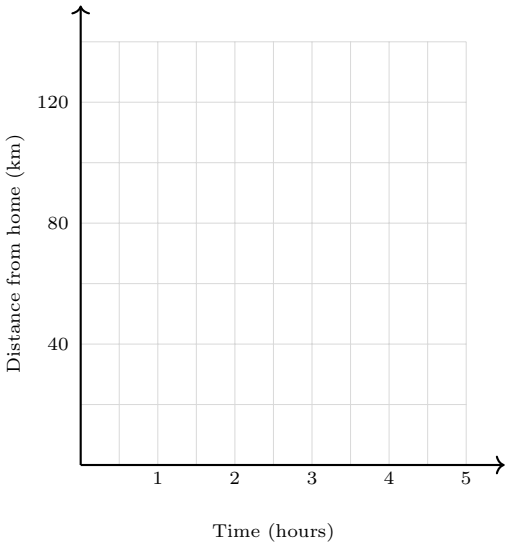
6. A tank holds 100 L of water when a pump is switched on. The pump adds water at a steady 50 L per minute for 4 minutes, then shuts off for 6 minutes. A valve is then opened and the tank drains at a steady 30 L per minute until it is empty. Sketch the volume of water against time on the grid, then answer (a) and (b).



(a) How much water is in the tank when the pump shuts off?

(b) At what time is the tank empty? Mark that point on your graph.

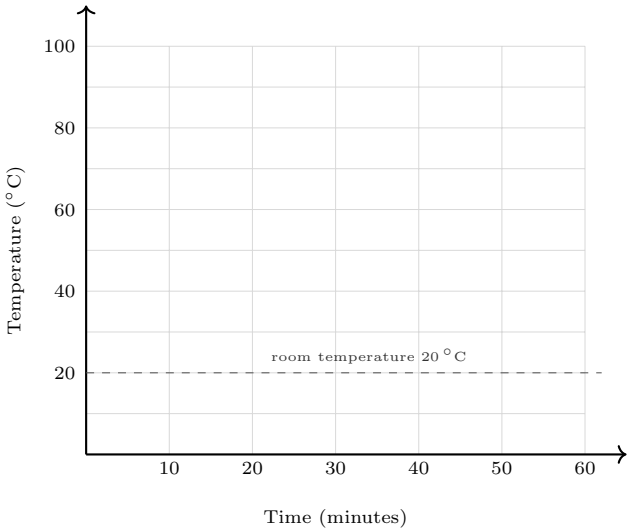
7. Amara drives from home to a cabin and back. She drives 80 km in the first hour, stops at a rest area for half an hour, then drives on for another half hour at the same speed, arriving at the cabin 120 km from home. She stays at the cabin for one hour, then drives straight home at 80 km/h. Sketch her distance from home against time.



(a) At what time does she arrive back home?

(b) Two parts of your graph are flat. Explain why they are flat for two *different* reasons.

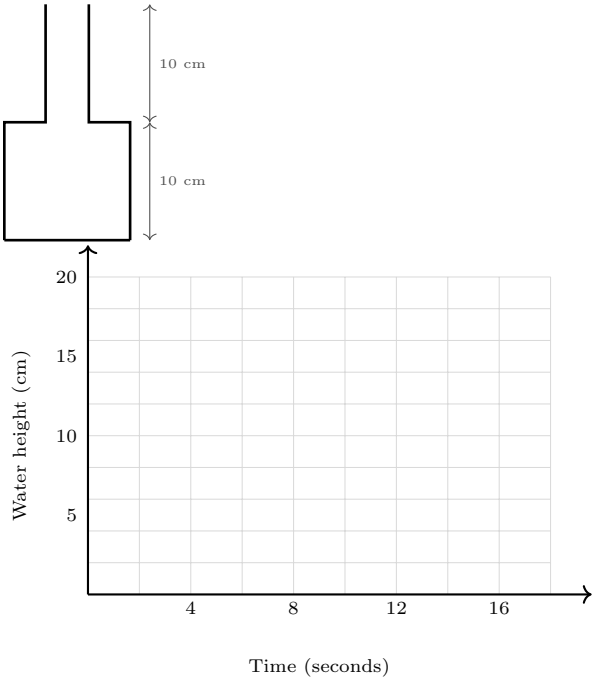
8. A mug of coffee is poured at 90 °C into a room that stays at 20 °C. The coffee cools quickly at first and then more and more slowly, getting closer and closer to room temperature without ever dropping below it. Sketch the temperature against time. The dashed line marks room temperature.



(a) Explain how the steepness of your curve changes, and what that means about how fast the coffee is losing heat.

(b) Explain why your curve must never cross the dashed line.

9. The container at the left is made of a wide cylinder with a narrow cylinder on top; each part is 10 cm tall. Water is poured in at a **steady rate** from empty until the container is full. Sketch the *height of the water* against time on the grid.



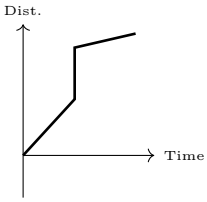
(a) Is your graph one straight line, or two straight pieces? Explain using the shape of the container.

(b) Which piece is steeper, and why?

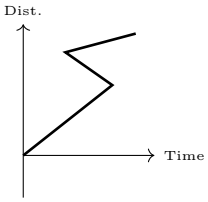
Part D — Impossible Graphs and Error Analysis

10. Each graph below is offered as a picture of a runner's **distance from the start line** during a race.

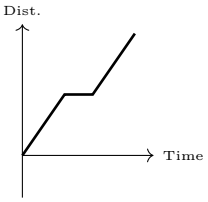
A



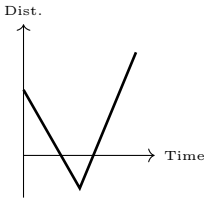
B



C



D



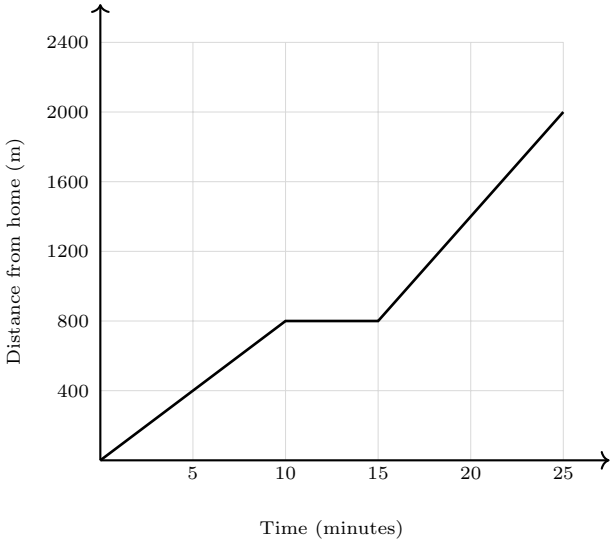
(a) Exactly one graph could really happen. Which one?

(b) Graph **A** contains a vertical segment. Explain why this is impossible, and connect your explanation to the vertical line test.

(c) Explain what is impossible about Graph **B**.

(d) Explain what is impossible about Graph **D**. (Think about what the horizontal axis means.)

11. The graph shows Dev's distance from home on a 25-minute walk. His classmate Rowan wrote: *"Between* 10 *and* 15 *minutes the graph is flat, so Dev turned around and walked back toward home during those five minutes."*



(a) Explain why Rowan is wrong, and state what actually happened between 10 and 15 minutes.

(b) Describe what the graph *would* look like if Dev really had walked back toward home for those five minutes at his original pace, and state how far from home he would then have been at 15 minutes.

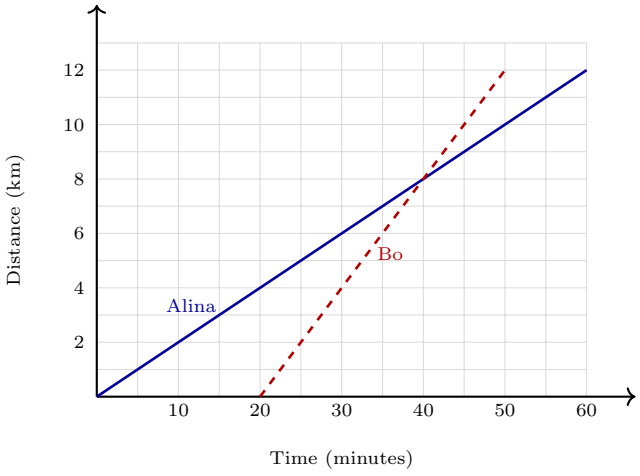
(c) Determine Dev's rate of change, in metres per minute, on each of the three segments. Which segment is fastest?

12. A graph of a car's **speed** against time falls steadily from 60 km/h to 0 km/h over the last three minutes of a trip. A student writes: *"The graph goes down, so the car was reversing."* Explain the student's mistake, and state what the falling segment really shows. Was the car moving forward at all during those three minutes? Explain.

13. On a distance–time graph, one segment is twice as steep as another. A student concludes, *"so that part of the trip is twice as long."* Explain why steepness does not measure length of trip, and state what being twice as steep actually tells you. Give a short example in which the steeper segment is in fact the *shorter* part of the trip.

Part E — Comparing Two Journeys on One Set of Axes

14. Alina and Bo cycle the same 12 km route. Both journeys are shown on one grid.



(a) Who starts first, and by how many minutes?

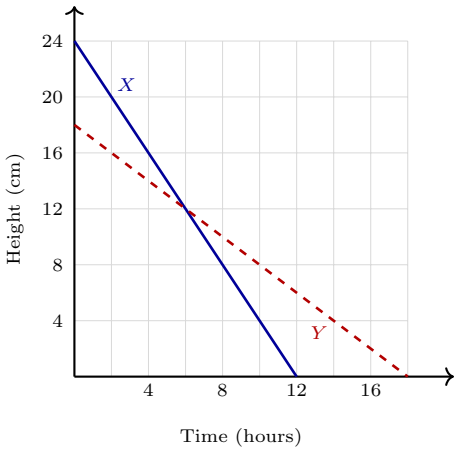
(b) Determine each cyclist's speed in km/h.

(c) At what time does Bo catch up to Alina, and how far along the route are they at that moment?

(d) Who finishes first, and by how many minutes?

(e) How far apart are the two cyclists 30 minutes after Alina starts?

15. Two candles are lit at the same moment. Candle X starts at 24 cm and Candle Y starts at 18 cm; both burn at steady rates, shown on the grid.



(a) Which candle burns faster? State both rates, with units, and say how the graph shows it.

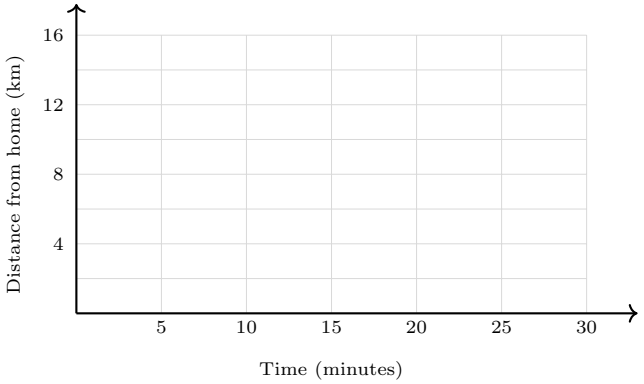
(b) At what time are the two candles the same height, and what is that height?

(c) Which candle is taller after 4 hours, and by how much?

(d) Each line has a horizontal intercept. State both, and explain what they mean here.

Part F — Stretch

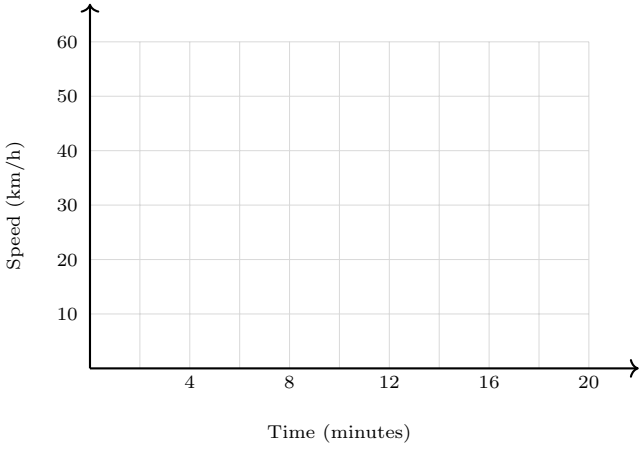
16. On the grid, sketch **one** graph of a car's distance from home against time that satisfies *all three* conditions at once. Then, beside it, describe the journey in words.



- The car is 5 km from home at $t = 0$ and again 5 km from home at $t = 30$.
- The car is stopped for exactly 10 minutes somewhere in the middle.
- The car is never more than 12 km from home.

Describe the journey:

17. On the grid, sketch **one** speed–time graph for a 20-minute bus route that satisfies *all three* conditions at once.

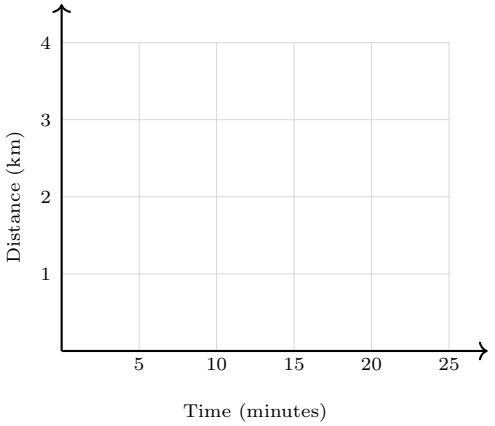


- Besides the start and the finish, the bus comes to a complete stop exactly three times.
- The fastest stretch of the route happens between the second stop and the third stop.
- The bus never exceeds 50 km/h.

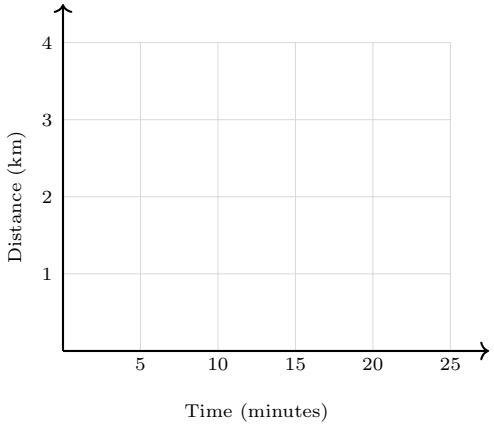
(a) How many times does your graph come down to speed 0 *between* the start and the finish?

18. Priya walks 2 km to a store in 10 minutes, shops for 5 minutes, then walks the same 2 km home in 10 minutes. Sketch **two** graphs of this one trip: her *distance from home*, and the *total distance she has walked*.

(a) Distance from home



(b) Total distance walked



(c) The two graphs agree for the first 15 minutes and then disagree completely. Explain why.

(d) One of the two graphs ends at the same height it started. Say which, and explain why that is the correct picture of the trip rather than a mistake.

Answer Key. 1. (a)(i) The van drives away from the depot at a steady speed, reaching 9 km in 12 min. (ii) After the first drop-off it drives further out, 9 km to 24 km, in 10 min — its fastest stretch. (iii) It drives all the way back to the depot, 24 km in 24 min. (b) $\frac{24-9}{30-20} = 1.5$ km/min $= 90$ km/h. (c) C–D: it is the steepest segment, so the van is driving fastest there (highway speed). (d) 16 min — B to C (8 min) and D to E (8 min), the two flat segments. (e) $9 + 15 + 24 = 48$ km. The graph shows *distance from the depot*, not distance driven; the return trip still adds 24 km even though it brings the van back to 0. **2.** (a) B–C: 3, 0, 0; C–D: 3, +12, +4; D–E: 3, −6, −2. (b) C–D, at 4 m per month — the reservoir is refilling fastest (spring melt or heavy rain). (c) B to C, a span of 3 months (months 3 to 6). **3.** (a) $(0, 60)$: she loaded \$60 onto the card on day 0. (b) The weekends, when she does not ride; the value stays the same, and staying the same is not the same as going up — a gain would be a *rising* segment. (c) −\$5 per day. (d) $(16, 0)$: the card runs out of money on day 16. (e) $60 \div 5 = 12$, so day 12. The graph would be a single straight segment from $(0, 60)$ to $(12, 0)$ with no flat parts. **4.** (i) D (ii) A (iii) C (iv) B. (v) Equal steepness means equal speed, so he walked home at exactly the pace he walked out. (vi) D has one constant steepness the whole way (one steady pace); C starts steep and then becomes gradual (fast first, then slower) — both rise, but C's *rate* changes and D's does not. **5.** (a) P is speed; Q is distance travelled. P returns to 0, which makes sense for speed (stopped) but not for distance travelled, and P is made of a rise, a flat, and a fall — speeding up, cruising, slowing down. Q only ever increases and flattens out at the end. (b) P sits at 0; Q is flat at its maximum height (no new distance is being added). (c) Distance travelled is a running total, so it can grow or stay the same, but it can never un-travel; a falling segment would mean the total distance already covered had decreased. **6.** Segments $(0, 100)$ to $(4, 300)$, flat $(4, 300)$ to $(10, 300)$, then down to $(20, 0)$. (a) $100 + 50(4) = 300$ L. (b) $300 \div 30 = 10$ min of draining after minute 10, so the tank is empty at $t = 20$ min, the point $(20, 0)$. **7.** Segments $(0, 0)$ to $(1, 80)$, flat to $(1.5, 80)$, up to $(2, 120)$, flat to $(3, 120)$, down to $(4.5, 0)$. (a) 4.5 h after leaving. (b) The first flat is the rest-area stop *on the way there*; the second is the hour spent *at the cabin*. Both mean "not moving," but at two different places — 80 km from home and 120 km from home. **8.** A curve from $(0, 90)$ falling steeply at first, then flattening as it approaches (but never reaches) the dashed line at 20. (a) It starts steep and becomes more and more gradual: the coffee loses heat quickly while it is far hotter than the room, and more slowly as the gap closes. (b) Crossing the line would mean the coffee became colder than the room it is sitting in, which heat will not do on its own. **9.** Two straight pieces: a gentle one from $(0, 0)$ up to a height of 10 cm, then a steeper one from 10 cm up to 20 cm. (a) Two pieces, because the container changes width partway up; each cylinder on its own has a constant width, so each gives a straight segment. (b) The upper piece. The narrow top holds less water per centimetre of height, so at the same pour rate the level climbs faster there. **10.** (a) C. (b) A vertical segment says the runner was at two (in fact infinitely many) different distances at the *same* instant. A vertical line meets the graph more than once, so the graph fails the vertical line test and is not a function of time — and a runner cannot jump forward instantaneously. (c) B doubles back to the left, so time runs backwards. Time only moves forward, and again one time value would give two distances. (d) D dips below the horizontal axis, giving a negative distance from the start line. A distance is never negative — if the runner went behind the start line, the distance would grow again, not turn negative. **11.** (a) Flat means *no change* in distance from home, not a change of direction: Dev stood still for five minutes, staying 800 m from home. Walking back would show as a *falling* segment. (b) A falling segment from $(10, 800)$ down toward home at 80 m/min; after five minutes he would be $800 - 5(80) = 400$ m from home, at $(15, 400)$. (c) 0–10 min: $\frac{800}{10} = 80$ m/min; 10–15 min: 0 m/min; 15–25 min: $\frac{1200}{10} = 120$ m/min. The last segment is fastest. **12.** On a *speed*–time graph a falling segment means the speed is getting smaller — the car is slowing down (braking). It would only mean reversing on a graph of distance or position. Yes, the car is still moving forward the whole time: its speed is positive right up until the instant it reaches 0, which is the moment it finally stops. **13.** Steepness is a *rate*: it compares distance to time, so twice as steep means twice as fast, not twice as far or twice as long. A trip can be fast and short. For example, driving 2 km in 1 minute is twice as steep as walking 4 km in 4 minutes, yet the steep segment covers half the distance and takes a quarter of the time. **14.** (a) Alina, by 20 minutes. (b) Alina: $12 \div 60 = 0.2$ km/min $= 12$ km/h; Bo: $12 \div 30 = 0.4$ km/min $= 24$ km/h. (c) At 40 min, 8 km along (where the lines cross). (d) Bo, at 50 min against Alina's 60 min — 10 minutes earlier. (e) At 30 min Alina is at 6 km and Bo is at 4 km, so they are 2 km apart. **15.** (a) X burns faster: X loses 24 cm in 12 h $= 2$ cm/h, while Y loses 18 cm in 18 h $= 1$ cm/h. X's line is steeper. (b) At 6 h, both are 12 cm tall (the crossing point). (c) After 4 h, X is $24 - 8 = 16$ cm and Y is $18 - 4 = 14$ cm, so X is taller by 2 cm. (d) $(12, 0)$ and $(18, 0)$: candle X burns out after 12 h and candle Y after 18 h. **16.** Many answers. One is: $(0, 5)$ up to $(10, 12)$, flat from $(10, 12)$ to $(20, 12)$, then down to $(30, 5)$. It starts and ends 5 km from home, is stopped for exactly 10 minutes, and its highest point is 12 km. Any graph that starts and ends at height 5, has exactly one flat stretch 10 minutes long, and never rises above 12 is acceptable. **17.** Many answers. One is: rise from $(0, 0)$ to $(3, 40)$, fall to $(5, 0)$, flat to $(6, 0)$; rise to $(8, 30)$, fall to $(10, 0)$, flat to $(11, 0)$; rise to $(14, 50)$, fall to $(16, 0)$, flat to $(17, 0)$; rise to $(18, 30)$, fall to $(20, 0)$. (a) Three times — the three stops between the start and the finish (the sample graph is flat at speed 0 at 5–6, 10–11

and 16–17 minutes). **18.** (a) $(0,0)$ to $(10,2)$, flat to $(15,2)$, down to $(25,0)$. (b) $(0,0)$ to $(10,2)$, flat to $(15,2)$, up to $(25,4)$. (c) For the first 15 minutes she has only walked away from home, so how far she *is* from home and how far she *has walked* are the same number. On the walk home the two numbers move in opposite directions: her distance from home shrinks back to 0 while the total she has walked keeps climbing to 4 km. (d) Graph (a). It is correct because she really does end up back home, a distance of 0 from it. Only the total-distance graph must keep rising, and it does, ending at 4 km.

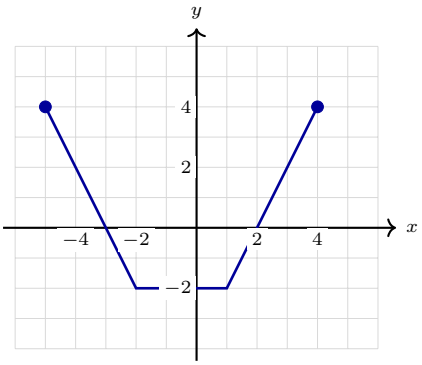
Lesson 12 — Extra Practice

Unit Review: Relations and Functions

About this set. These are the hardest questions in the unit, and none of them tests one idea on its own. A single graph may ask you for intercepts, domain, range, the vertical line test and a function value all at once; a single story may ask you to build the rule, use it forwards and backwards, and then decide what the answer means. Work carefully and state your reasons — on a test, that reasoning is most of the mark.

Part A — Reading Graphs

1. The graph of $y = f(x)$ below is made of three straight pieces, with solid endpoints.



(a) Is this relation a function? Justify your answer using the vertical line test.

(b) State the domain and the range, using set notation.

(c) State the coordinates of every x-intercept and of the y-intercept.

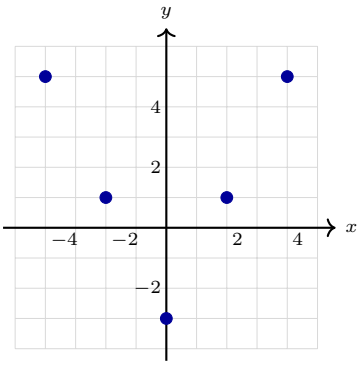
(d) Determine $f(3)$ and $f(-4)$.

(e) Determine *all* values of x for which $f(x) = 2$.

(f) In which quadrant does the point $\big(-4,\ f(-4)\big)$ lie? Justify with the signs of its coordinates.

(g) Explain why the equation $f(x) = -2$ has infinitely many solutions, and state them using an inequality.

2. The graph of $y = g(x)$ consists of the five points shown.



(a) Write the relation as a set of ordered pairs, then as a table of values.

(b) State the domain and the range in set notation. Explain why the range has only three values even though there are five points.

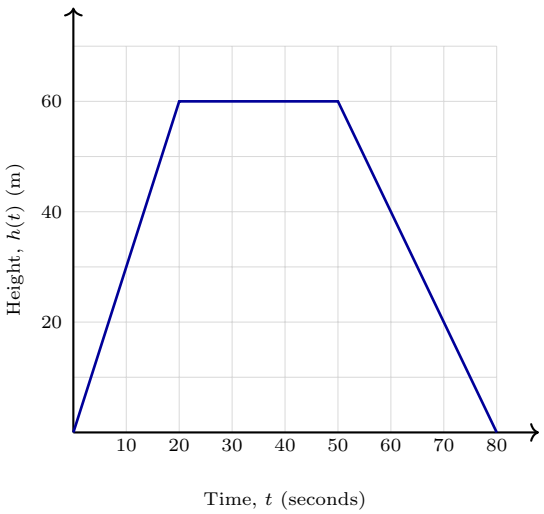
(c) Is g a function? Justify.

(d) Determine $g(-2) + g(4)$.

(e) Solve $g(x) = 1$.

(f) State the y-intercept. Then explain why this relation has **no** x-intercept.

3. A drone takes off, hovers, and lands. Its height $h(t)$, in metres, t seconds after launch is graphed below.



(a) Determine $h(35)$ and describe what the drone is doing at that moment.

(b) Determine the rate of change during the climb and during the descent. Include units and signs, and say which is steeper.

(c) Solve $h(t) = 30$. Explain why there are two answers, and why that does **not** stop h from being a function.

(d) State the domain and range of h, using set notation.

(e) State both t-intercepts and explain what each one means for the drone.

Part B — Functions and Notation

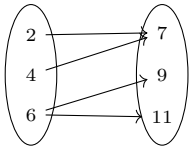
4. Decide whether each relation below is a function. In every case give the reason — and if it is not a function, name the input that breaks the rule.

(a) $(-1, 4)$, $(0, 4)$, $(1, 4)$, $(2, 4)$

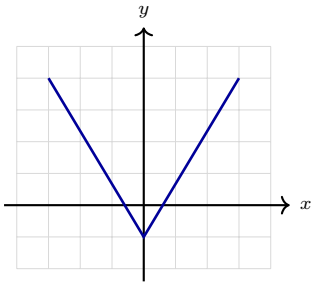
(b)

x	3	5	3	7
y	2	4	2	8

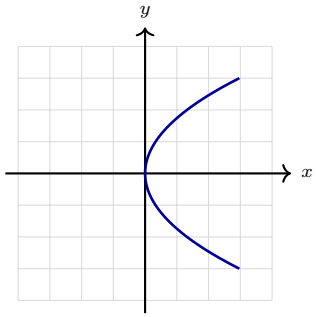
(c)



(d)



(e)



(f) Every student at a school is paired with their student number, and no two students share a number. Is "student → number" a function? Is "number → student" a function?

5. Consider the functions $f(x) = 5 - 2x$ and $g(x) = 3x + 10$.

(a) Determine $f(-3)$.

(b) Solve $f(x) = 15$.

(c) Solve $f(x) = g(x)$, then verify by evaluating both functions at your answer.

(d) Determine the x-intercept of each function. Give exact values.

(e) For which values of x is $f(x) > g(x)$? Explain how your answer to (c) helps you decide.

(f) Zeke says "$f(x) = g(x)$ has no solution, because f and g are different functions." Explain what he has confused.

6. A tiling pattern is built from square tiles. Figure 1 uses 5 tiles, and each figure after that uses 4 more tiles than the one before.

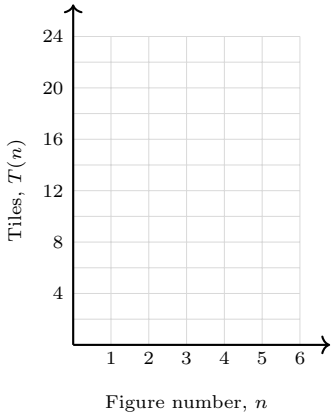
(a) Complete the table of values.

Figure, n	1	2	3	4	5
Tiles, $T(n)$	5				

(b) State the independent and the dependent variable.

(c) Write a function rule $T(n)$ for the number of tiles in Figure n, and check it against Figure 1 and Figure 5.

(d) Graph the first five figures on the grid.



(e) Determine $T(12)$.

(f) Which figure uses 101 tiles?

(g) Is your graph discrete or continuous? State the domain and range for Figures 1 to 5.

(h) The rule gives $T(0) = 1$. Explain why that answer has no meaning in this pattern.

7. A function h has a constant rate of change, and $h(2) = 11$ and $h(5) = 26$.

(a) Determine the rate of change of h.

(b) Determine $h(0)$, and use it to write a rule for $h(x)$.

(c) Determine $h(9)$.

(d) Solve $h(x) = 96$.

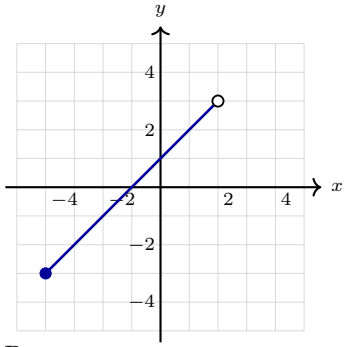
Part C — Domain, Range and Intercepts

8. State the domain and the range of each relation, using the notation that fits it.

(a) $(-6, 2)$, $(-1, 7)$, $(3, 2)$, $(8, 7)$

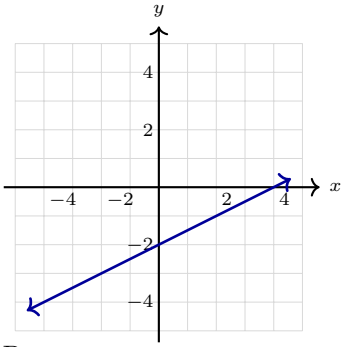
D: _____
R: _____

(b)



D: _____
R: _____

(c)



D: _____
R: _____

(d) A ferry sells tickets at \$12 each and carries at most 240 passengers, so the ticket revenue from p passengers is $C(p) = 12p$. State the domain and range of C in this context, and say what kind of numbers the domain must contain.

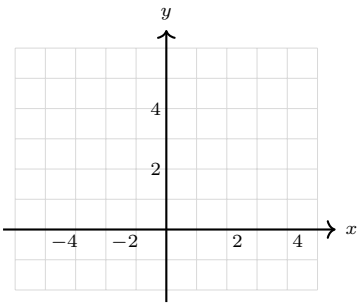
(e) Explain why the domain in part (a) is written with braces but the domain in part (b) is written with an inequality. Then explain what the open circle in (b) changes about the answer.

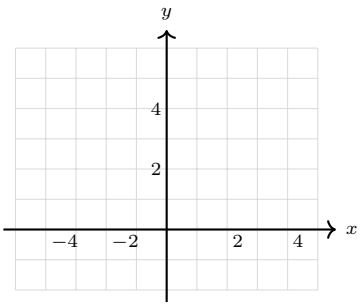
9. **Work backwards.** On each grid, draw one relation that has exactly the domain and range given. Label the endpoints of anything you draw.

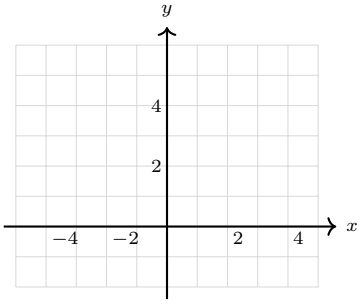
(a) D: $-4 \le x \le 2$, R: $1 \le y \le 5$, and it **is** a function.

(b) The *same* domain and range as (a), but it is **not** a function.

(c) $D = \{-3, 0, 3\}$ and $R = \{2\}$.







(d) Explain why part (c) can only be drawn one way, while parts (a) and (b) have infinitely many correct answers.

(e) Could a relation have domain $-4 \le x \le 2$ and range $\{3\}$ and be a function? Draw or describe it, or explain why not.

10. A water cooler holds 24 L at the start of a shift and empties at a steady rate. The amount left, in litres, after x hours is $f(x) = 24 - 3x$.

(a) Determine the x-intercept algebraically and explain what it means.

(b) State the y-intercept and explain what it means.

(c) Determine $f(5)$ and solve $f(x) = 6$.

(d) State the domain and range of f in this context, using set notation.

(e) A student answers "the domain is all real numbers, because the rule $24 - 3x$ works for any x." Explain why that is the wrong answer to part (d), and name two values of x that the rule accepts but the situation does not.

(f) Sketch the graph of f over its context domain, and use your sketch to explain why the range can be read off the two endpoints alone.

Part D — Reasoning and Error Analysis

11. Each statement below contains one error. Identify the error and give the correct answer.

(a) "The point $(0, -4)$ is in Quadrant III, because its y-coordinate is negative."

(b) "The relation $(1, 5)$, $(2, 5)$, $(3, 5)$ is not a function, because the outputs repeat."

(c) "For $f(x) = 2x + 7$, the expression $f(3)$ means f multiplied by 3."

(d) "The graph is a solid line segment from $(-2, -3)$ to $(4, 5)$, so the domain is $\{-2, 4\}$."

(e) "The x-intercept of $y = 4x - 20$ is $(0, -20)$."

(f) "A distance–time graph is flat from 5 to 9 minutes, so the runner ran backwards then."

12. State whether each statement is **true** or **false**. Give a one-sentence justification, or a counterexample, for every answer.

(a) Every function is a relation. _____

(b) Every relation is a function. _____

(c) A graph can have two different y-intercepts and still be a function. _____

(d) A graph can have two different x-intercepts and still be a function. _____

(e) If $f(2) = 7$ and $f(5) = 7$, then the relation is not a function. _____

(f) If $f(2) = 7$ and $f(2) = 9$, then the relation is not a function. _____

(g) The domain of a relation drawn as five separate dots should be written with an inequality. _____

(h) On a distance–time graph, a steeper segment always means the object travelled farther in total. _____

13. Mia says the vertical line $x = 3$ is a function "because it is a straight line, and straight lines are functions." Jon says it is not. Decide who is right and explain, then state the domain and range of $x = 3$.

14. Now consider the horizontal line $y = 3$. Is it a function? State its domain and range, and explain in one sentence why swapping the roles of x and y changes the answer so completely.

Part E — Applications

15. A drone may be rented for at most 6 hours. The cost, in dollars, of an h-hour rental is $C(h) = 18 + 7h$.

(a) Determine $C(4)$ and state what it means.

(b) A customer is charged \$60. Determine the length of the rental.

(c) A customer claims to have been charged \$74. Explain why this is impossible here.

(d) Determine $C(0)$ and explain what it represents.

(e) State the domain and range of C in this context, using set notation.

(f) Explain why the y-intercept of the graph of C must equal the answer to part (d), and what that number is called in the situation.

16. Two phone plans charge for m minutes of long distance: $A(m) = 30 + 0.05m$ and $B(m) = 0.20m$.

(a) Determine $A(150)$ and $B(150)$. Which plan is cheaper for 150 minutes?

(b) Solve $A(m) = B(m)$ and explain what the answer means to a customer.

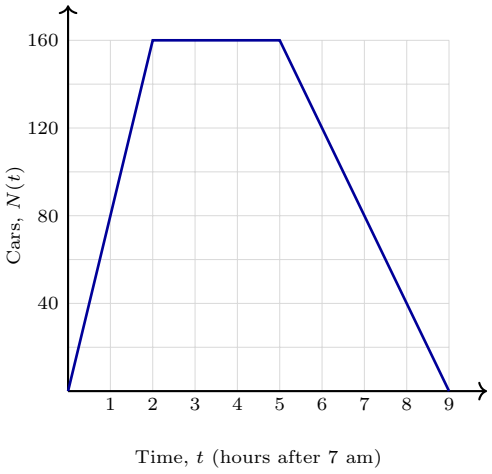
(c) Which plan is cheaper for 300 minutes? Justify using function notation.

(d) State the domain of each function in this context.

(e) You want to graph both plans on a calculator and see the point where they cross. State a window $x : [\ \ ,\ \ ]$, $y : [\ \ ,\ \ ]$ that would show it clearly, and explain why the standard window $x : [-10, 10]$, $y : [-10, 10]$ would not.

(f) Explain, without any calculation, why plan B must be cheaper for small numbers of minutes and plan A must be cheaper for large numbers of minutes.

17. The graph shows $N(t)$, the number of cars in a parking lot, t hours after 7 am.



(a) Determine $N(4)$ and describe the lot at that time.

(b) Determine the rate of change on each of the three segments, with units. Which is steepest, and what does it represent?

(c) Solve $N(t) = 80$. Explain why two different times give the same answer.

(d) State both t-intercepts, give the clock time for each, and explain what they mean.

(e) State the domain and range of N, using set notation.

Part F — Challenge

18. The graph of a function g is a single straight segment joining $(-6, 9)$ to $(6, -3)$, with both endpoints included.

(a) Determine the rate of change, then write a rule for $g(x)$.

(b) State the x-intercept and the y-intercept as ordered pairs.

(c) State the domain and range, using set notation.

(d) Determine $g(-2)$, then solve $g(x) = 7$.

(e) Solving $g(x) = 12$ with your rule gives $x = -9$. Explain why 12 is nevertheless **not** in the range of g.

(f) Give one value that *is* in the range of g but is *not* in its domain, and one value that is in both. Explain how you chose each.

19. A candle-making club can make at most 40 candles. Its profit, in dollars, from selling n candles is $P(n) = 6n - 90$.

(a) Determine $P(25)$ and explain what it means.

(b) Determine the n-intercept and explain its meaning for the club.

(c) Determine $P(0)$ and explain what it represents.

(d) How many candles must be sold for a profit of \$120?

(e) State the domain of P in this context, and explain why it should be whole numbers rather than all real numbers. State the least and the greatest value in the range.

(f) Because of your answer to (e), the graph of P is not a solid line. Describe what it looks like instead, and explain why the club still finds the rule $P(n) = 6n - 90$ useful.

Answer Key. 1. (a) Yes — no vertical line meets the graph more than once, so each x has exactly one y. (b) D: $-5 \le x \le 4$, R: $-2 \le y \le 4$. (c) x-intercepts $(-3, 0)$ and $(2, 0)$; y-intercept $(0, -2)$. (d) $f(3) = 2$ and $f(-4) = 2$. (e) $x = -4$ and $x = 3$ (one on each sloped piece). (f) $(-4, 2)$ is in Quadrant II, since $x < 0$ and $y > 0$. (g) The middle piece is the horizontal segment $y = -2$ running from $x = -2$ to $x = 1$, so every x in that stretch gives output -2: the solutions are $-2 \le x \le 1$. (A function may repeat an *output* as often as it likes.) **2.** (a) $(-4, 5)$, $(-2, 1)$, $(0, -3)$, $(2, 1)$, $(4, 5)$; table x: $-4, -2, 0, 2, 4$ with y: $5, 1, -3, 1, 5$. (b) $D = \{-4, -2, 0, 2, 4\}$, $R = \{-3, 1, 5\}$. Two pairs of points share an output (1 twice and 5 twice), and a set lists each value only once. (c) Yes — every x-value appears exactly once, so no input has two outputs. (d) $g(-2) + g(4) = 1 + 5 = 6$. (e) $x = -2$ and $x = 2$. (f) y-intercept $(0, -3)$. No point has $y = 0$, and the relation is only these five dots — there is nothing joining them — so the graph never touches the x-axis. **3.** (a) $h(35) = 60$ m: the drone is hovering at a constant height. (b) Climb $\frac{60}{20} = +3$ m/s; descent $\frac{-60}{30} = -2$ m/s. The climb is steeper, so the drone rises faster than it comes down. (c) $t = 10$ s and $t = 65$ s — it passes 30 m once going up and once coming down. A function may give the same *output* at two different *inputs*; it is only broken when one input has two outputs. (d) D: $0 \le t \le 80$, R: $0 \le h(t) \le 60$. (e) $t = 0$ (launch, still on the ground) and $t = 80$ (landing). **4.** (a) Function — each input appears once; repeated outputs are allowed. (b) Function — the pair $(3, 2)$ is simply listed twice, so the input 3 still has only the one output 2. (c) Not a function — the input 6 has two arrows, to 9 and to 11. (d) Function — the V passes the vertical line test. (e) Not a function — it opens sideways, so a vertical line crosses it twice (e.g. $x = 3$ meets both the top and the bottom branch). (f) "Student → number" is a function (one number each). "Number → student" is also a function here, because no two students share a number, so each number leads back to exactly one student. **5.** (a) $f(-3) = 5 - 2(-3) = 11$. (b) $5 - 2x = 15 \Rightarrow x = -5$. (c) $5 - 2x = 3x + 10 \Rightarrow -5 = 5x \Rightarrow x = -1$; check $f(-1) = 7$ and $g(-1) = 7$. (d) f: $x = \frac{5}{2}$, so $\left(\frac{5}{2}, 0\right)$; g: $x = -\frac{10}{3}$, so $\left(-\frac{10}{3}, 0\right)$. (e) $x < -1$. The two graphs meet only at $x = -1$; f falls while g rises, so f is above g on the left of that point and below it on the right. (f) $f(x) = g(x)$ is not a claim that the two functions are the same. It asks for the input (if any) at which they happen to give the same output — here, the single value $x = -1$. **6.** (a) 5, 9, 13, 17, 21. (b) Independent: figure number n; dependent: number of tiles. (c) $T(n) = 4n + 1$; check $T(1) = 5$ and $T(5) = 21$. (d) Five separate dots at $(1, 5)$, $(2, 9)$, $(3, 13)$, $(4, 17)$, $(5, 21)$. (e) $T(12) = 49$. (f) $4n + 1 = 101 \Rightarrow n = 25$, Figure 25. (g) Discrete — there is no "Figure 2.5." $D = \{1, 2, 3, 4, 5\}$, $R = \{5, 9, 13, 17, 21\}$. (h) There is no Figure 0; the pattern starts at Figure 1, so $n = 0$ is outside the domain even though the arithmetic works. **7.** (a) $\frac{26-11}{5-2} = 5$ per unit of x. (b) $h(0) = 11 - 2(5) = 1$, so $h(x) = 5x + 1$. (c) $h(9) = 46$. (d) $5x + 1 = 96 \Rightarrow x = 19$. **8.** (a) $D = \{-6, -1, 3, 8\}$, $R = \{2, 7\}$. (b) D: $-4 \le x < 2$, R: $-3 \le y < 3$. (c) D: all real numbers, R: all real numbers — both arrows continue forever. (d) D: $p = 0, 1, 2, \ldots, 240$ (whole numbers — you cannot carry part of a passenger); R: $C = 0, 12, 24, \ldots, 2880$, the multiples of 12 from \$0 to \$2880. (e) Part (a) is discrete — four separate points, so its domain is a *list* in braces. Part (b) is continuous — an unbroken segment containing every x-value between the endpoints, so it needs an inequality. The open circle at $(2, 3)$ means $x = 2$ and $y = 3$

are *not* included, so those ends use $<$ instead of $\leq$. **9.** (a) Many answers; simplest is the segment from $(-4, 1)$ to $(2, 5)$ with both endpoints solid. (b) Many answers; e.g. the two segments $(-4, 1)$ to $(2, 5)$ *and* $(-4, 5)$ to $(2, 1)$, forming an X. Domain and range are unchanged, but a vertical line through the crossing meets it twice. (c) The three points $(-3, 2)$, $(0, 2)$, $(3, 2)$ — and nothing else. (d) In (c) the domain and range are finite lists, and each of the three x-values must pair with the only available y-value, so the picture is forced. In (a) and (b) the domain is a whole interval, and infinitely many different curves run across that interval while still reaching every height from 1 to 5. (e) Yes — the horizontal segment from $(-4, 3)$ to $(2, 3)$. Every input has exactly one output, and the only output used is 3. **10.** (a) $24 - 3x = 0 \Rightarrow x = 8$; the cooler is empty 8 hours into the shift. (b) $(0, 24)$: the cooler holds 24 L at the start. (c) $f(5) = 9$ L; $24 - 3x = 6 \Rightarrow x = 6$ h. (d) D: $0 \leq x \leq 8$, R: $0 \leq f(x) \leq 24$. (e) The rule is only a model of a real cooler: before the shift starts and after the cooler is empty it no longer describes anything. For example $x = -2$ is time before the shift began, and $x = 10$ would give $f(10) = -6$ L, a negative volume. (f) A straight segment from $(0, 24)$ down to $(8, 0)$. Because the graph is a single unbroken segment with no turns, it never rises above its left endpoint or falls below its right one, so the largest and smallest outputs are exactly the endpoint heights 24 and 0. **11.** (a) A point with $x = 0$ lies *on the y-axis*, not in any quadrant; quadrants require both coordinates to be non-zero. (b) It *is* a function. Repeated outputs are fine — only a repeated *input* with two different outputs breaks the rule. (c) $f(3)$ means substitute 3 for x: $f(3) = 2(3) + 7 = 13$. The symbol f is the name of a function, not a number being multiplied. (d) A solid segment contains every x-value in between, so D: $-2 \leq x \leq 4$; braces would only be right for two separate dots. (e) That is the y-intercept. The x-intercept has $y = 0$: $4x - 20 = 0 \Rightarrow x = 5$, so $(5, 0)$. (f) Flat means the distance is not changing — the runner *stopped*. Running back would show as a falling segment. **12.** (a) T — a function is a relation with one extra condition. (b) F — e.g. $(1, 2)$, $(1, 5)$ is a relation but not a function. (c) F — the y-axis is itself a vertical line, so two y-intercepts means the input 0 has two outputs. (d) T — two x-intercepts are two *inputs* sharing the output 0, which is allowed (e.g. any U-shaped curve cutting the x-axis twice). (e) F — two inputs sharing the output 7 is fine. (f) T — the single input 2 would have two different outputs. (g) F — separate dots are discrete, so the domain is a list in braces. (h) F — steepness is a *rate* (distance per unit time), not a total. A short, fast sprint can be steeper than a long, slow walk that covers more ground. **13.** Jon is right. Every point on $x = 3$ has the same input 3 but a different output, so the one input 3 has infinitely many outputs; the line *is* a vertical line, so it fails the vertical line test outright. $D = \{3\}$; R: all real numbers. **14.** Yes, $y = 3$ is a function: every input has exactly one output, namely 3, and no vertical line meets it twice. D: all real numbers; $R = \{3\}$. The definition is not symmetric — it restricts what *inputs* may do, and says nothing about outputs, so repeating a y-value is harmless while repeating an x-value is fatal. **15.** (a) $C(4) = 18 + 28 = \$46$, the cost of a 4-hour rental. (b) $18 + 7h = 60 \Rightarrow h = 6$ hours. (c) $18 + 7h = 74 \Rightarrow h = 8$ hours, which is more than the 6-hour maximum, so 8 is outside the domain and \$74 is not a possible charge. (d) $C(0) = \$18$ — the booking fee, charged before any hours are used. (e) D: $0 \leq h \leq 6$, R: $18 \leq C(h) \leq 60$. (f) The y-intercept is the value of C when $h = 0$, which is exactly $C(0)$. In the situation it is the fixed booking fee. **16.** (a) $A(150) = \$37.50$, $B(150) = \$30$; plan B is cheaper. (b) $30 + 0.05m = 0.20m \Rightarrow 30 = 0.15m \Rightarrow m = 200$ minutes: at 200 minutes both plans cost the same (\$40), so that is the break-even point. (c) $A(300) = \$45$ and $B(300) = \$60$, so plan A is cheaper. (d) $m \geq 0$ for both — you cannot talk for a negative number of minutes. (e) Something like $x : [0, 400]$, $y : [0, 80]$. The standard window shows only x up to 10 and y up to 10, but the crossing is at $(200, 40)$ — far outside it. In fact plan A already costs \$30 at $m = 0$, so its graph is off the top of the standard screen immediately. (f) B starts at \$0 and A starts at \$30, so B wins while the minutes are few. But A charges only 5 cents a minute against B's 20 cents, so B's cost climbs four times as fast and must eventually overtake A's \$30 head start. **17.** (a) $N(4) = 160$ cars: the lot has been full (at its maximum) since $t = 2$. (b) 0–2 h: +80 cars/h; 2–5 h: 0 cars/h; 5–9 h: -40 cars/h. The first is steepest — the morning arrival rush, when cars fill the lot fastest. (c) $t = 1$ and $t = 7$ — 80 cars once while filling and once while emptying. The graph rises to 160 and comes back down, so it passes every height between 0 and 160 twice. (d) $t = 0$ (7 am) and $t = 9$ (4 pm): the lot is empty before the day starts and empty again once everyone has left. (e) D: $0 \leq t \leq 9$, R: $0 \leq N(t) \leq 160$. **18.** (a) $\frac{-3-9}{6-(-6)} = -1$; the segment falls 1 unit for every 1 across, and $g(0) = 3$, so $g(x) = 3 - x$ (check: $g(-6) = 9$, $g(6) = -3$). (b) x-intercept $(3, 0)$; y-intercept $(0, 3)$. (c) D: $-6 \leq x \leq 6$, R: $-3 \leq g(x) \leq 9$. (d) $g(-2) = 5$; $3 - x = 7 \Rightarrow x = -4$. (e) The *rule* extends forever, but the *graph* stops at $x = -6$. Since $x = -9$ is outside the domain, the point $(-9, 12)$ is not on the graph, and no allowed input produces 12 — the largest output is $g(-6) = 9$. (f) Many answers. In the range but not the domain: 8 (outputs reach 9, but inputs stop at 6). In both: 0 (an allowed input, and also an achieved output, since $g(3) = 0$). **19.** (a) $P(25) = 150 - 90 = \$60$ profit from selling 25 candles. (b) $6n - 90 = 0 \Rightarrow n = 15$: selling 15 candles breaks even — no profit and no loss. (c) $P(0) = -\$90$: with no candles sold the club is \$90 down, the cost of its supplies. (d) $6n - 90 = 120 \Rightarrow n = 35$ candles (within the limit of 40). (e) D: $n = 0, 1, 2, \ldots, 40$ (whole numbers). Candles are sold whole, so a value such as $n = 12.5$ has no meaning. Least range value $P(0) = -\$90$; greatest $P(40) = \$150$. (f) It is 41 separate dots, one for each whole number of candles from 0 to 40, rising in steps of \$6 — not a solid line. The rule is still useful because it gives the height of every one of those dots and lets the club answer questions such as (b) and (d) without listing them all.